Analytical Techniques and Bioinformatics

Career Avenues

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PROTEIN SEQUENCING

Frederic Sanger first time achieved complete sequence of protein (bovine insulin) in 1953. For his work, he was awarded the Nobel Prize of Chemistry in (1958).

Protein sequencing refers to the techniques employed to determine the amino acid sequence of a protein. There are several applications of protein sequencing, which are:-

a) Identification of the protein family to which a particular protein belongs and finding the evolutionary history of that protein. Function prediction.

b) Prediction of the cellular localization of the protein based on its target sequence (sequence of amino acids at the N terminal end of the protein which determines the location of the protein inside the cell).

c) Prediction of the sequence of the gene encoding the particular protein.

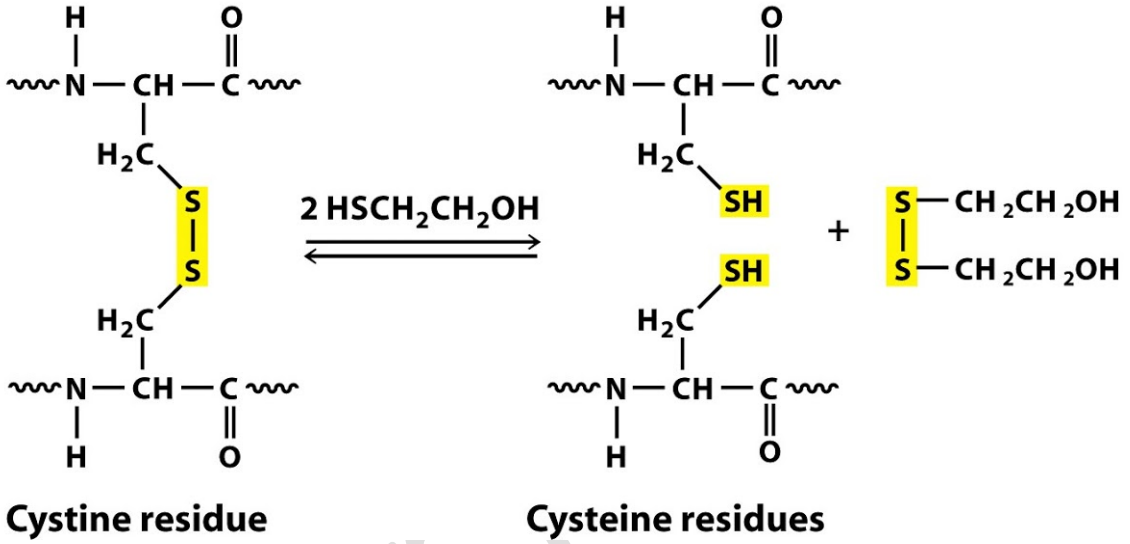
d) Discovering the structure and function of a protein through various computational methods and experimental methods.

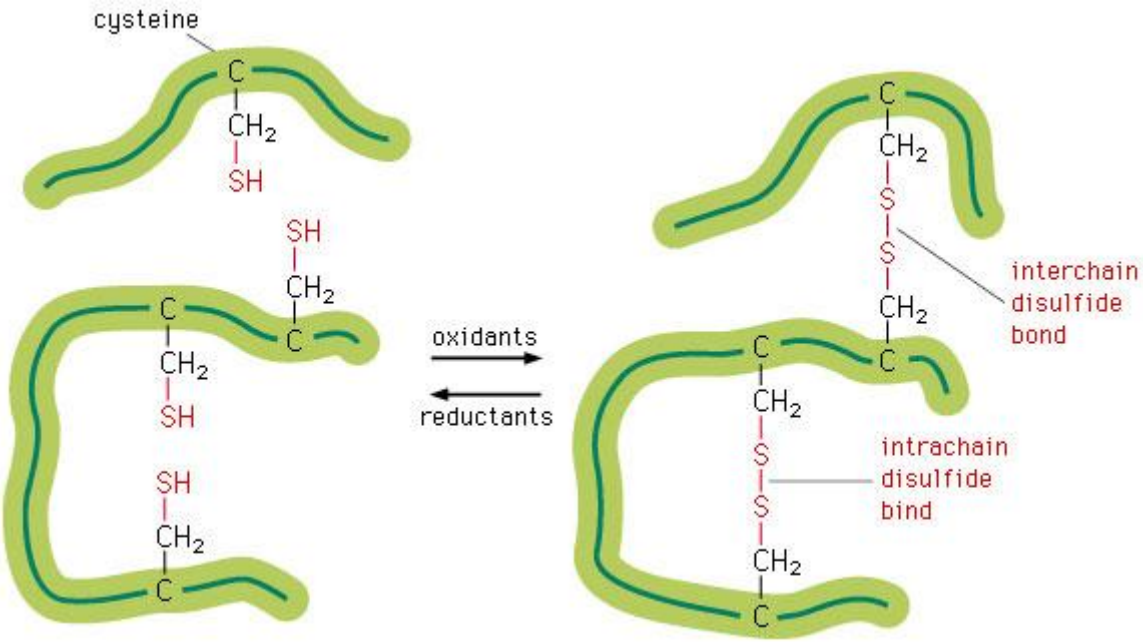
Till date several methods have been utilized for protein sequencing. Two main methods include Edman degradation and Mass Spectrometry. Protein sequence can also be generated from the DNA/mRNA sequence that codes for the protein, which has been explained in details in the recombinant DNA section. Here, we have discussed the most important methods used for protein sequencing and the pros and cons of each method.

Edman degradation

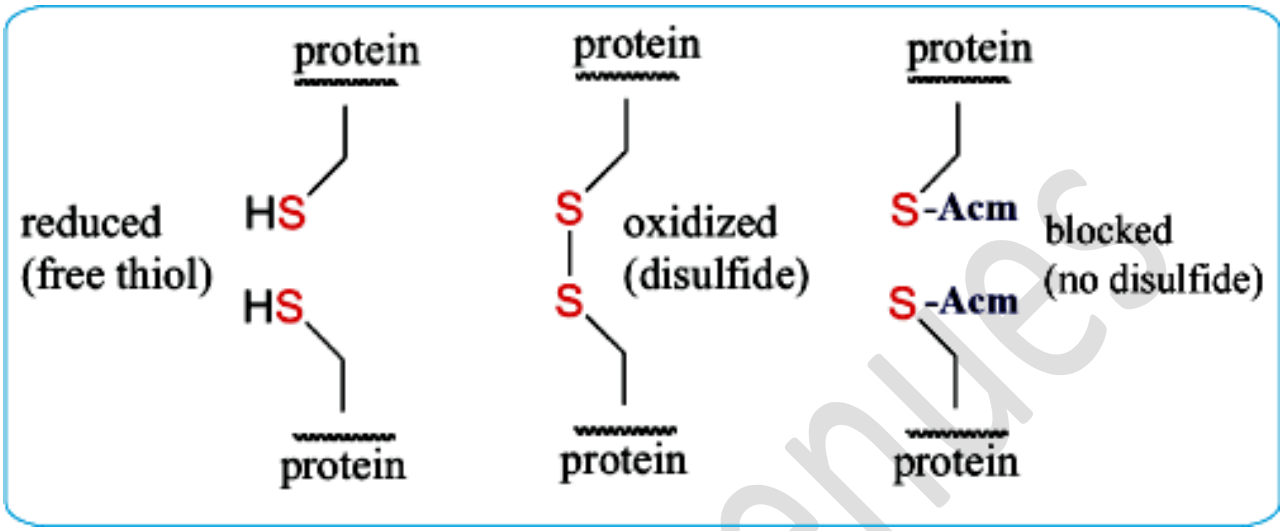
Before sequencing process is initiated, it is necessary to break all non-covalent interaction by denaturants (like high concentration of urea or GuHCl). This process will also separate subunits, in case of oligomeric proteins. Occasionally, subunits of a oligomeric protein are connected by covalent interations. In that case special treatments are required to separate subunits. The protein is treated with Edman's reagent (phenyl isothiocyanate) which reacts with the N-terminal amino acid and under mild acidic condition forms a cyclic compound Phenyl thiohydantoin derivative (PTH–amino acid) of N-terminal amino acid is released. Amino acid of PTH –amino acid derivative is identified by chromatographic property of the PTH –amino acid derivative. In this process N-terminal amino acid is identified after first cycle. *Since this method proceeds from the N terminal residue, the reaction will not work if that N-terminal of a protein is blocked*

(generally due to post-translational modification). After first cycle of the reaction, amino group of the second amino acid is free for reaction with Edman's reagent and at the end of reaction PTH derivative of second amino acid from N-terminal is released. The process continues till end of sequence or a disulfide bond is encountered in the sequence. PTH-cysteine derivative will remain attached with polypeptide and PTH-cystein will not be released. Thus, reduction of disulfide bond in the polypeptide sequence needed before sequencing process can be initiated. Reduction of free cysteine can be done by use of β-mercaptoethanol.

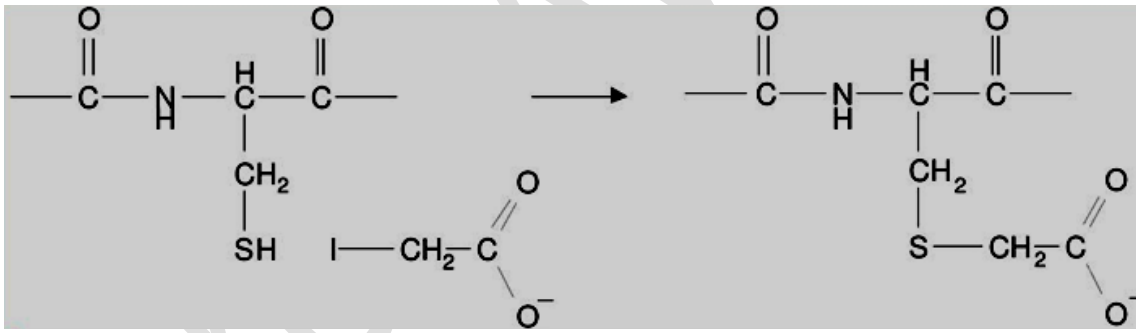




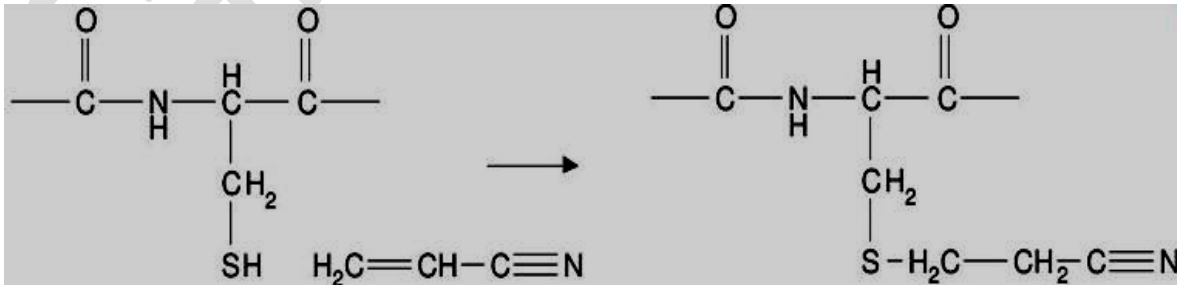
As free cysteine can re-oxidize to form disulfide it is necessary to block free cysteine. This may be done by use of iodoacetic acid or acrylonitrile (free cysteine modification) as shown in Fig. below



Diagrammatic representation of reduced state, oxidized state and blocked state of the disulfide bond



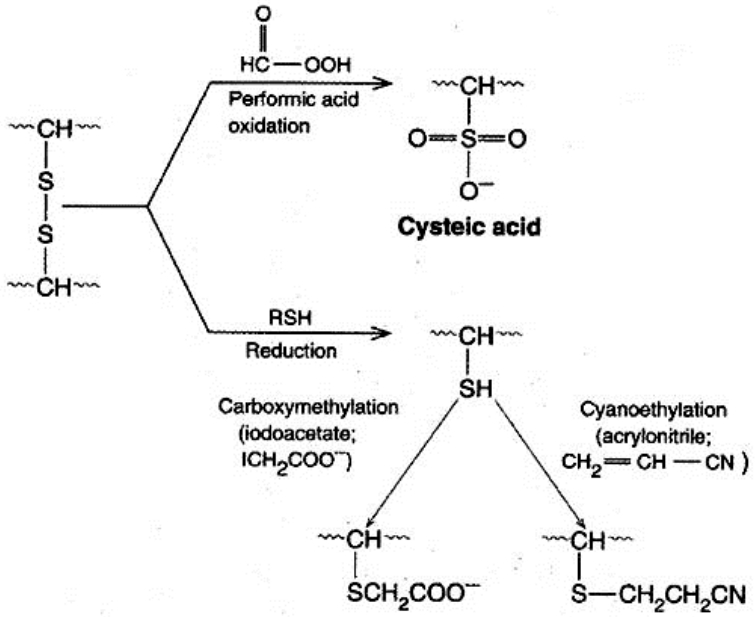
Free Cysteine blocked with the use of Idoacetic acid

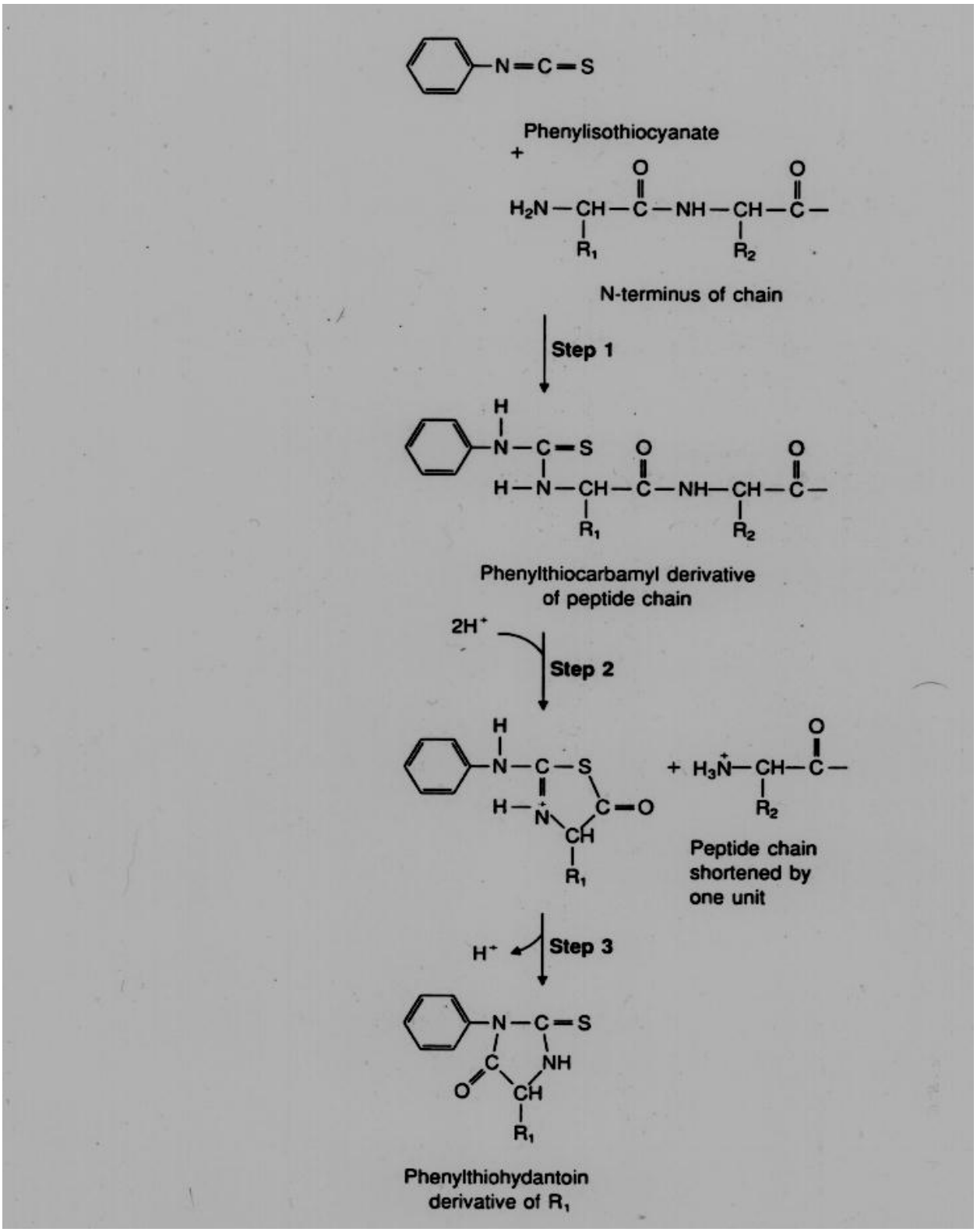


Free Cysteine blocked with the use of Acetonitrile

Other method for irreversible oxidation of disulfide bond is use of preformic acid. As shown in the figure below, performic acid oxidizes cysteine to negatively charge cysteic acid. Repulsion of

negatively charged cysteic acid group prevents re-formation of disulfide and alkylation is not required.





Edman Degradation method for protein sequencing

Further, the accuracy of each cycle is 98%. So after 60 steps the accuracy is less than 30%. Thus, this method cannot be used for sequencing of proteins larger than 50 amino acids. In case of larger proteins it has to be broken down to short peptide fragments using cleavage proteases such as trypsin (cleaves a protein at carboxyl side of lysine and arginine residues) or chymotrypsin (cleaves at carboxyl side of tyrosine, tryptophan and phenylalanine). Specific cleavage can also be achieved by chemical methods like cynogen bromide, which always cleaves at carboxyl side of methionine residue (a protein with 12 methionine will yield 13 fragment polypeptide on cleavage with cynogen bromide (CNBr).

Protein fragments after a protease (for example trypsin) will be separated and sequenced. Let us assume that the following two peptide sequences are obtained.

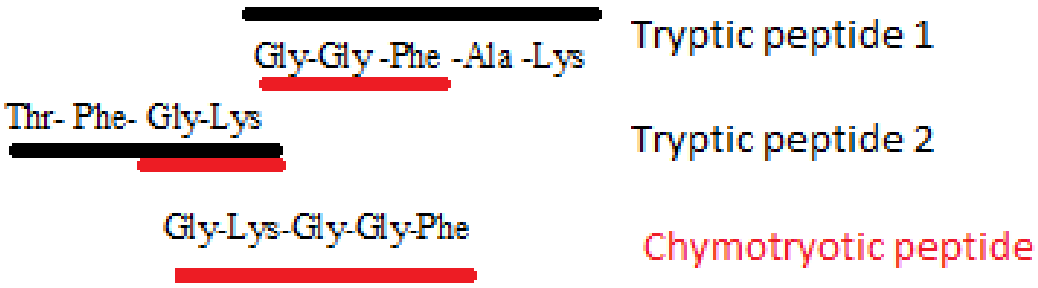
Gly-Gly -Phe -Ala -Lys
Thr- Phe- Gly-Lys

Now, a second method is used to cleave protein at other site. For example if we use chymotrypsin. Following sequence is found for one peptide sequence.

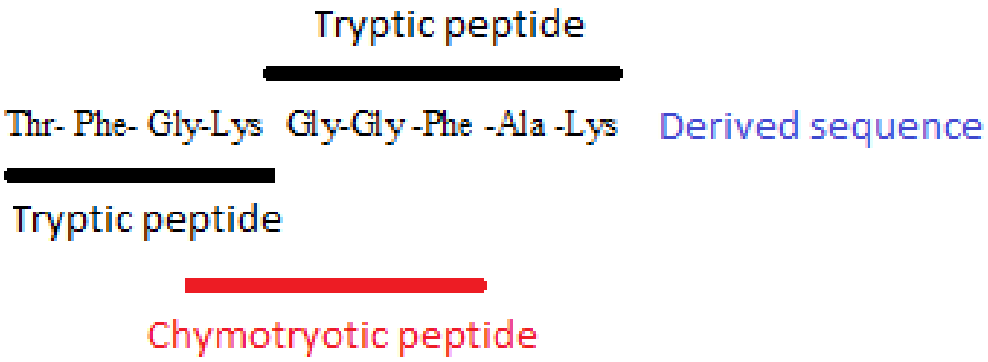
Gly-Lys-Gly-Gly-Phe

Now data from tyrosine cleavage fragment sequence and chymotrypsin cleavage fragment sequence can be analyze to get larger sequence information Sequences of smaller fragments may be overlapped and analyzed for complete sequence as depicted in the figure below

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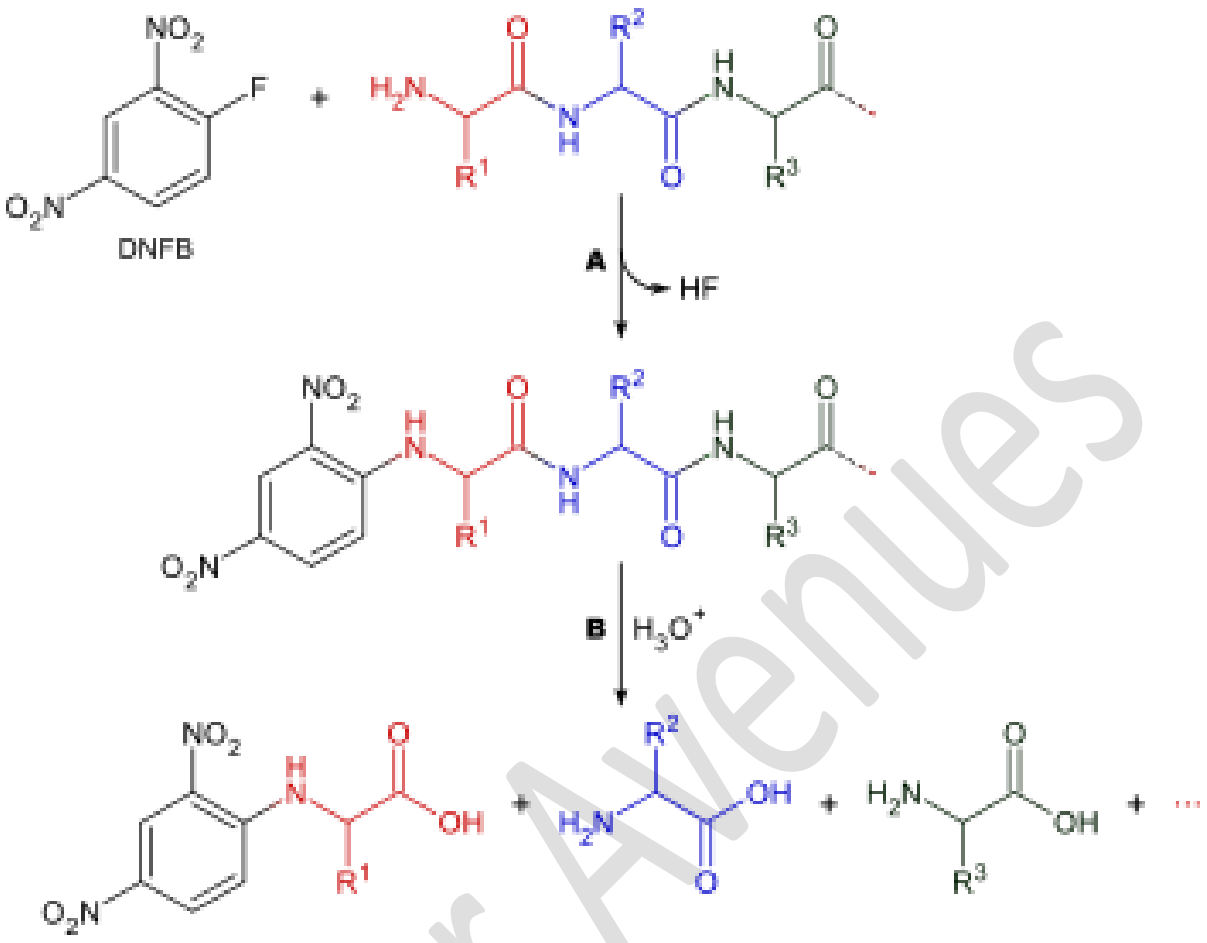


Let us see how this data can be combined to get bigger sequence



Sanger's Protein sequencing method

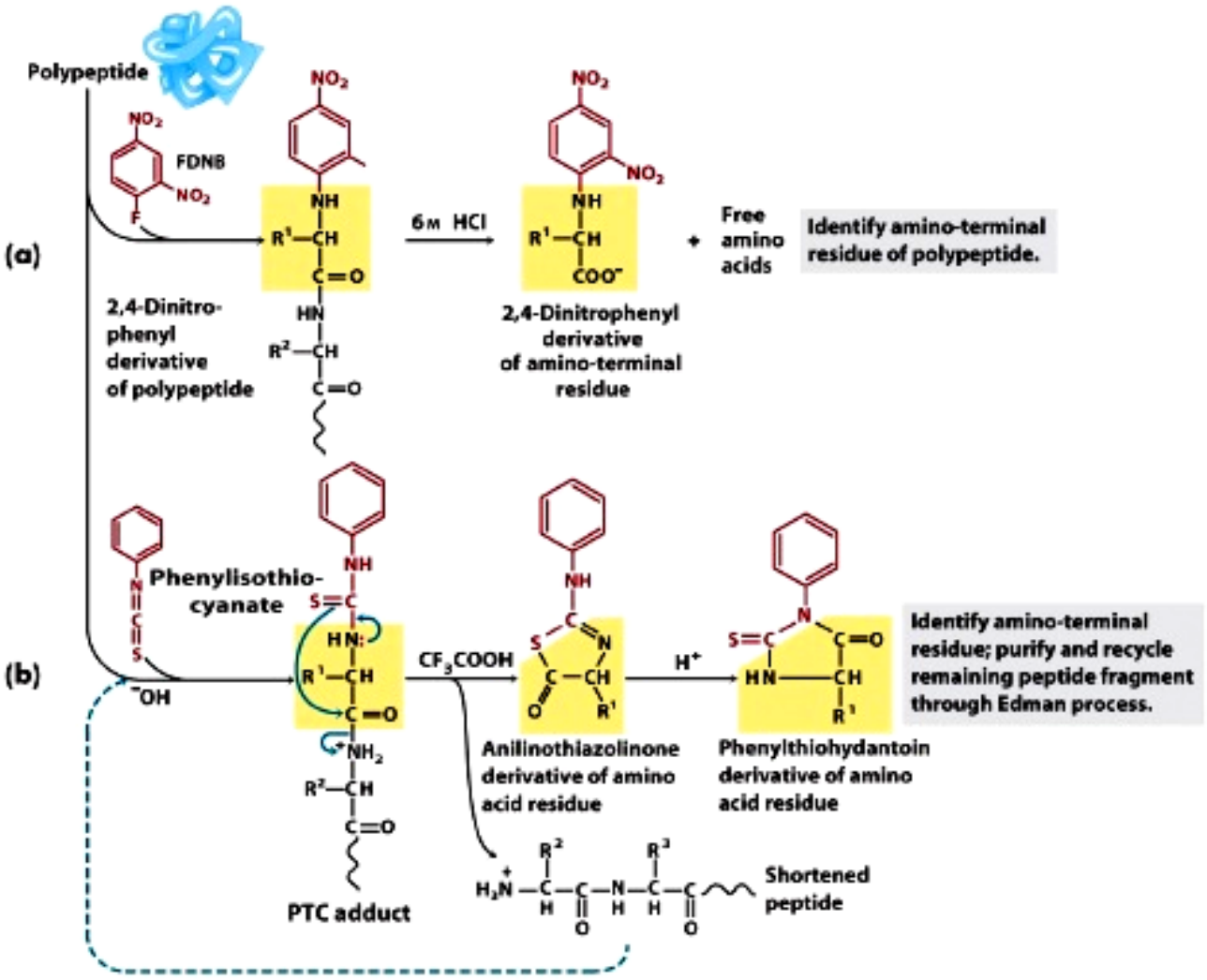
Here, the N terminal amino acid of the protein is labeled by dyes like Sanger's reagent (fluoro-dinitrobenzene) or dansyl chloride. The labeled protein is then hydrolyzed by 6M HCl at 110 ºC by the above mentioned method and loaded in Dowex 50 column and the elution profile is matched with the standard profile obtained from FNB or DNSCl derivative of all the amino acids, to obtain the N terminal amino acid. The reagents produce coloured derivatives which can be easily detected by absorbance



Protein sequencing using Sanger's reagent and dansyl chloride

One major difference between Edman and Sanger method of Protein sequencing is that Edman degradation removes only the amino-terminal amino acid from a polypepetide, leaving all the other bonds intact while Sanger method cleaves the entire polypeptide to form free amino acids. Hence, Edman's Process has been automated using a machine called Sequenator.

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Disadvantages of this method include:

1. Once we get the N terminal amino acid, the protein is already hydrolyzed in constituent amino acids. Thus we cannot repeat the cycle with same sample. For second amino acid sequencing we require new stock of protein sample and the N-terminal residue need to be cleaved from the protein using an appropriate protease such as amino peptidase. This makes the process very tedious and complicated.

2. These dyes selectively labels the amine groups present in the protein and therefore can label the amine groups present in the side chains as well, which may give erroneous results.

Protein sequencing using Molecular Biology techniques

If first few N-terminal amino acid of a protein is known, complete aminoacid sequence can be derived using Molecular Biology techniques. A simple example is as follow:

The genome sequence of *Calotropis procera*, a plant, or the sequence of procerain B, a novel cystein protease from the plant, gene is not yet known. Thus, the only information for cloning of cDNA we have is the fifteen N-terminal amino acid residues. The double stranded cDNA can be amplified with help of degenerate primer (based of N-terminal amino acid sequence) and oligo dT primer. Total RNA can be isolated from young leaf or latex of the plant and first strand of cDNA can be synthesised with oligo dT primer by reverse transcription. The second strand of cDNA can be synthesised and the subsequent amplification of double stranded cDNA can be achieved by PCR with degenerate primer as forward and oligo dT primer as reverse primmer. The amplified double stranded cDNA of expected size can be subjected to TA cloning and confirmed by sequencing. Once sequence of cDNA is available, it can be translated in protein sequence.

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Microscopy

Introduction

Microscopes constitute the very basic requirement for cell biologists, thereby facilitating the deciphering the fine details of intracellular components. The microscope is an obligatory tool for this purpose because most cellular structures are too small to be seen by the unaided eye. The beginnings of cell biology can be traced to the invention of the light microscope, which made it possible for scientists to examine enlarged images of the cells and thus to analyze cellular structure and its dynamics. The first light microscope was developed in 1590 by Z. Janssen and H. Janssen. During the next century, many microscopic observations were reported, notably those of Robert Hooke (who observed the first cells) and Antonie van Leeuwenhoek (who provided first glimpse of internal cell structure through improved microscopes). Since then, the light microscopes have undergone numerous improvements and modifications till the present time. Light microscopy has experienced a renaissance in recent years through specialized technological improvements that allow researchers to explore aspects of cell structure and behaviour. Most images produced by microscopes are now recorded electronically using digital imaging techniques, like digital cameras, digital image acquisition software, digital printing and digital display methods. Additionally, vast improvements have been made in the biological aspects of specimen preparation. These advancements have fostered many more applications of the microscope in biomedical research. These advances have involved the merging of technologies from physics, engineering, chemistry and molecular biology, that have greatly expanded the ability to study cells using light microscope. Biochemical analyses is frequently accompanied by microscopic examination of tissue, cell or organelle preparations. Such examinations are used in many different applications, for example: 1. to evaluate the integrity of samples during an experiment; 2. to map the fine details of the spatial distribution of macromolecules within cells; 3. to directly measure biochemical events within the living tissues. There are fundamentally two different types of microscopes:

1. the light microscope and
2. the electron microscope.

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Light microscopes use a series of glass lenses to focus light in order to form an image whereas electron microscopes use electromagnetic lenses to focus a beam of electrons.

Light microscopes are able to magnify to a maximum of approximately 1500 times whereas electron microscopes to a maximum of approximately 200000 times. Standard light microscopes have a lateral resolution limit of about 0.5 micrometers for routine analysis. In contrast, electron microscopes have a lateral resolution of up to 1 nanometer. Both living and dead specimens are viewed with a light microscope, and often in real colour, whereas only dead ones are viewed with an electron microscope and never in real colour. Applications of the microscope in biomedical research may be relatively simple and routine, for example, a quick check of the status of a preparation or the health of cells growing in a plastic dish in the tissue culture. The application may be more involved, for example, measuring the concentration of calcium in a living embryo over a millisecond timescale through a more advanced light microscope (often called an imaging system). Some microscopes are more suited to specific applications than others but there may be constraints imposed by the specimen. Images may be required from specimens of vastly different sizes and magnifications; for example, for imaging whole animals (metres), through tissues and embryos (micrometres) and down to cells, proteins and DNA (nm). The study of living cells may also require time resolution from days (like when imaging neuronal development or disease processes) to milliseconds (like when imaging cell signalling events).

Principles of microscopy

The principles of microscopy need to be examined through special emphasis on the factors that determine how small an object can be observed and analyzed with current technologies.

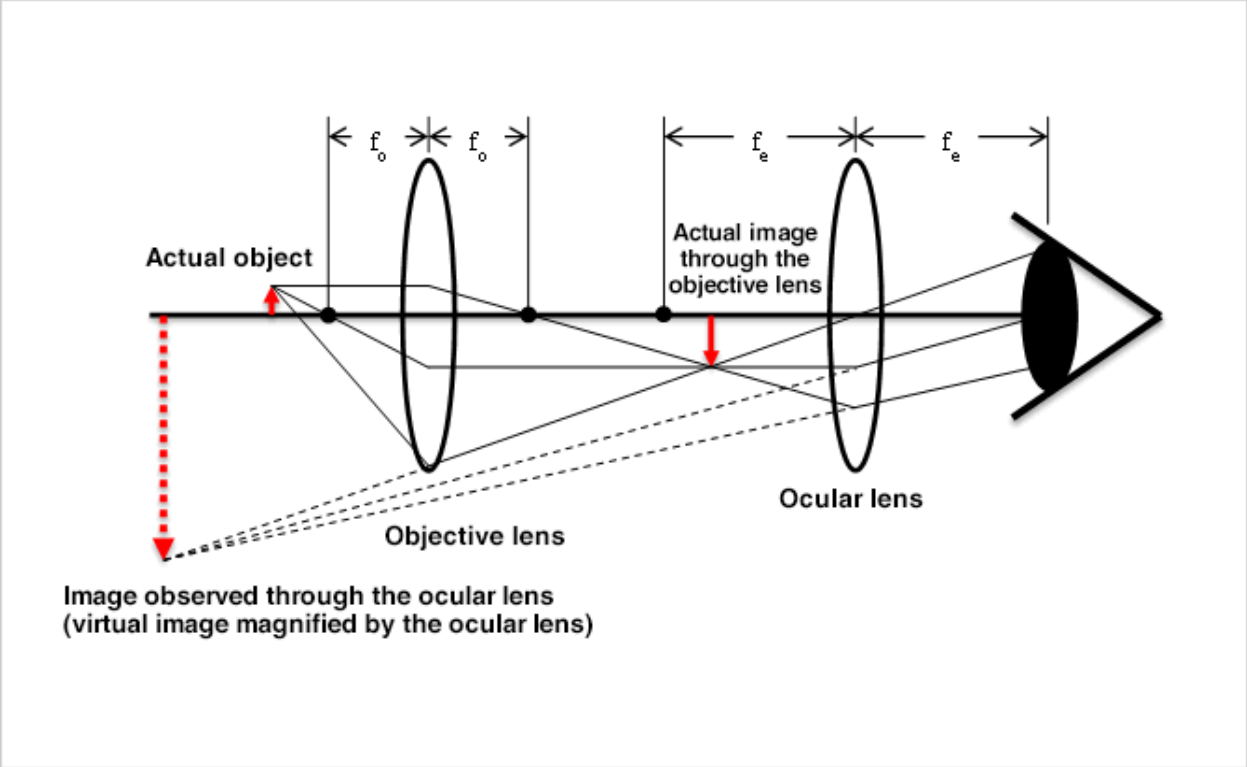
Illuminating wavelength Three elements are always needed to form an image, regardless of the type of microscope being used:

- a source of illumination
- a specimen to be examined
- a system of lenses that focuses the illumination on the specimen and forms the image.

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The source of illumination in a light microscope is visible light (in the wavelength range of 400-700 nm) and the lens system consists of a series of glass lenses. The image can either be viewed directly through the eye piece or focused on a detector, such as photographic film or an electronic camera. In an electron microscope, the illumination source is a beam of electrons emitted by a heated tungsten filament, and the lens system consists of a series of electromagnets. Despite these differences in illumination source and instrument design, both types of microscopes depend on the same principle of optics and form images in a similar manner. When a specimen is placed in the path of light or electron beam, physical characteristics of beam are changed in a way that creates an image that can be interpreted by the human eye or recorded on a photographic detector. To have a better understanding of illumination source and specimen, the concept of wavelength needs to be understood. The ability of an object to perturb a wave's motion depends significantly on the size of the object in relation to the wavelength of the motion. This principle is of great importance in microscopy because the wavelength of the illumination source sets a limit on how small an object be seen. When a beam of light (or electrons) encounters a specimen, the specimen alters the physical characteristics of illuminating beam. Since an object can be detected only by its effect on the wave, the wavelength must be comparable in size to the object that is to be detected. As we understand this relationship between wavelength and object size, one can readily appreciate that why small objects can be seen only by electron microscopy. The wavelength of electrons is very much shorter than those of photons and thus, objects such as viruses and ribosomes are too small to perturb the waveform of photons, but can readily interact with electrons.

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Principle that enables magnified observation with a biological microscope

Resolution

The image formed when waves of light or electrons pass through a lens and are focused, results from a property of waves called interference. Thus, the image seen when you look at a specimen through a series of lenses, is just a pattern of either additive or cancelling interference of the waves that went through the lenses (a pattern called diffraction). A light microscope uses glass lenses to direct photons, whereas an electron microscope uses electromagnets as lenses to direct electrons. However, both kinds of microscopes have two fundamental properties in common:

- focal length
- angular aperture

The focal length is determined by

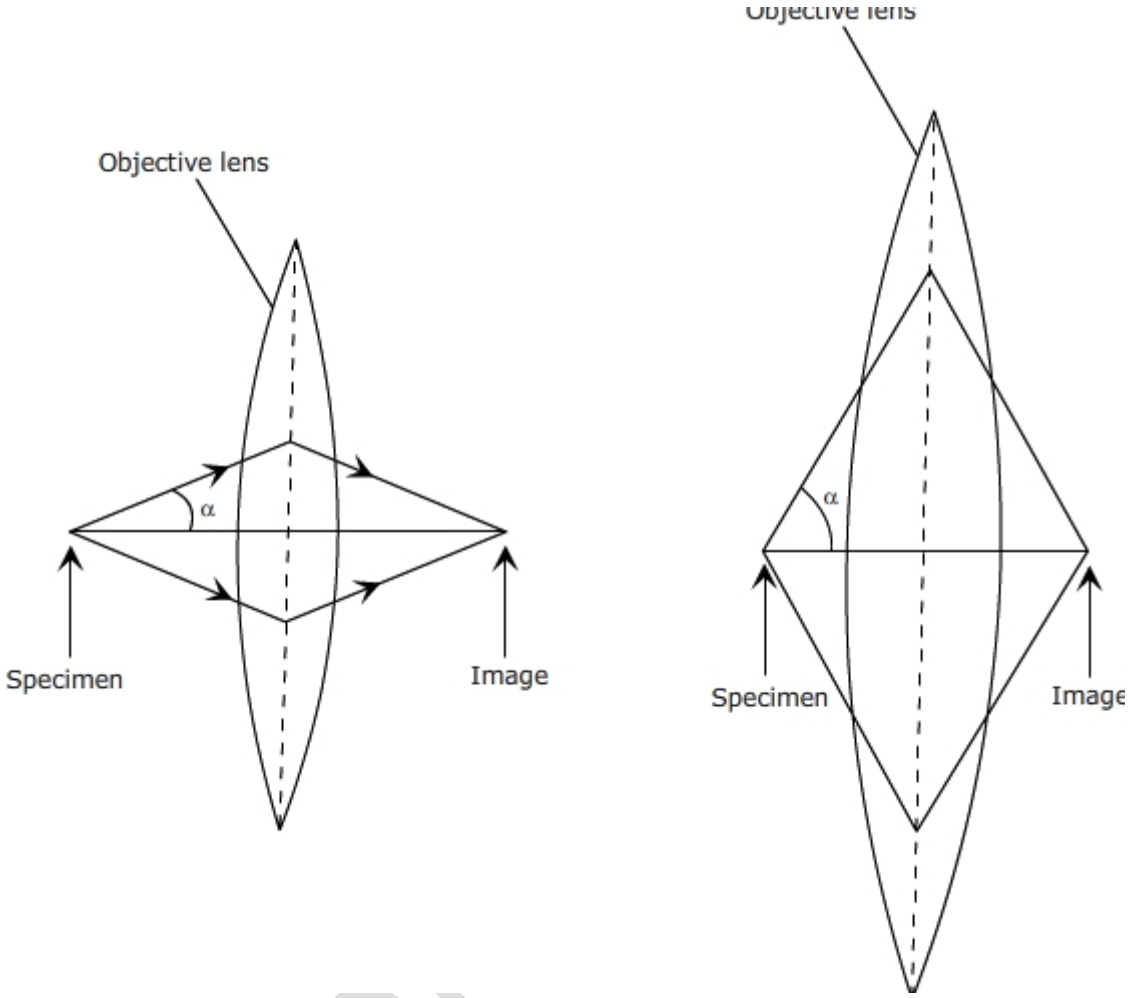
- The index of the refraction of the lens itself
- The medium in which it is immersed
- The geometry of the lens

The magnifying strength of a lens (measured in dioptres) is the inverse of focal length (measured in meters). The angular aperture refers to a measure of how much of the illumination that leaves the specimen actually passes through the lens. This, thus determines the sharpness of the interference pattern and, therefore, the ability of the lens to convey information about the specimen. The angular aperture is about 70° in the best light microscopes. Resolution, and not magnification, is a more reliable estimate of the utility of a microscope. Three factors influence a microscope's resolution:

- The wavelength of light used to illuminate the specimen
- The angular aperture
- The refractive index of the medium surrounding the specimen

The effect of these variables on resolution is described quantitatively through Abbé equation: $r = (0.61\ \lambda)/ (n \sin \alpha)$ where r = resolution λ = wavelength of light being used to illuminate the specimen n = refractive index of the medium between the specimen and the objective lens of the microscope α = angular aperture The constant 0.61 implies the degree to which image points can overlap and still be identified as separate points by an observer. Alternatively, the above equation can also be written as: $r = (0.61\ \lambda)/ NA$ where NA refers to the quantity $(n \sin \alpha)$ and is called as numerical aperture of the objective lens.

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The angular aperture of a lens is the half-angle α of the cone of light entering the objective lens of the microscope from the specimen. A low aperture lens (a) and a high aperture lens (b) is shown in the above diagram.

Numerical aperture (NA) is a measure of the ability of a lens to collect light from the specimen. It is usually better to choose the lens with higher NA if there is a choice between lenses of same magnification.

An important goal in both light and electron microscopy is to maximize resolution. Resolution improves as r becomes smaller because r is a measure of how close two points can be and still be

distinguished from each other. Therefore, for best resolution, the numerator in the above equation should be as small as possible and the denominator should be as large as possible. If we consider a glass lens that uses visible light as an illumination source, the minimum value for wavelength is set by the shortest wavelength in the range of visible light (i.e. 400-700 nm) that is practical to use for illumination that turns out to be blue light of approximately 450 nm. This would make the numerator as small as possible. In order to maximize the denominator, both the values of refractive index and sine of the angular aperture must be maximized to achieve optimal resolution. The maximum value for sine α is about 0.94 (because the angular aperture for the best objective lenses is approximately 70°). The refractive index of air is 1. Thus, resolution of a sample illuminated in air with blue light of 450 nm can be calculated as follows:

$r = (0.61\ \lambda)/\ NA$

$= (0.61)\ (450)/\ (1)\ (0.94)$

$= 292$ nm

Limit of resolution is thus, 300 nm for a glass lens in air.

In order to increase the numerical aperture, some microscopic lenses have been designed to be used with a layer of immersion oil between the lens and the specimen. Immersion oil (higher refractive index than air) allows the lens to receive more of the light transmitted through the specimen. Since the refractive index of immersion oil is about 1.5, the maximum NA for an oil immersion lens is about (1.5) X (0.94) = 1.4. Resolution of an oil immersion lens is approximately 200 nm. In actual practice, such limits of resolution (best possible resolution for a microscope) can rarely be reached because of aberrations in the lenses. The resolution can be further enhanced by using ultraviolet rays as an illumination source. Because the wavelength of UV rays is shorter (200-300 nm), the resolution can be enhanced to approximately 100 nm. However, when using ultraviolet light, special cameras (because UV rays are invisible to the human eye) and expensive quartz lenses (because ordinary glass is opaque to UV rays) must be used. The limit of resolution sets an upper boundary on useful magnification that is possible with any given lens. Generally, the greatest useful magnification that can be achieved with a light microscope is approximately 1000 times the numerical aperture of the lens being used. Since NA ranges from approximately 1.0 to 1.4, the useful magnification of a light microscope is limited to roughly 1000X in air and 1400X with oil immersion lens. The magnification for each objective lens can be increased above a point where it is impossible to resolve any more detail in the

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subject. Any magnification above this point is called empty magnification. The best way is to use a higher magnification and higher NA objective lens.

In order to achieve better magnification in an effective way it is necessary to switch from visible light to electron beam as the illumination source. The theoretical limit of resolution of the electron microscope (0.002 nm) is orders of magnitude better than that of light microscope (200 nm) because the wavelength of an electron is approximately 100,000 times shorter than that of a visible light. However, practical problems in the design of electromagnetic lenses (to focus the electron beam) prevent the electron microscope from achieving this theoretical potential. Electromagnets produce considerable distortion when the angular aperture is more than a few tenths of a degree. This angle is of several orders of magnitude less than that of a good glass lens (about 70°), providing a numerical aperture considerably smaller than that of a light microscope. The limit of resolution for an optimal electron microscope is therefore, only approximately 0.2 nm, far from the theoretical limit of 0.002 nm. Practical limit of resolution is often closer to 2 nm when viewing biological samples because of the problems with sample preparation and contrast. Resolution in an electron microscope is generally about 100 times better than that of light microscope. The useful magnification of an electron microscope is approximately 100 times than that of a light microscope or approximately 100,000X.

Light microscopy

Light microscopy can be considered under the following aspects:

Magnifying lens

The simplest form of light microscope is a magnifying lens which consists of a glass lens mounted in a metal frame. The specimen hardly requires any sample preparation and is usually held close to the eye. Focusing of the region of interest is achieved by moving the lens and specimen relative to one another. The source of light is usually the sun or indoor light. The detector is human eye and the recording device is a hand drawing.

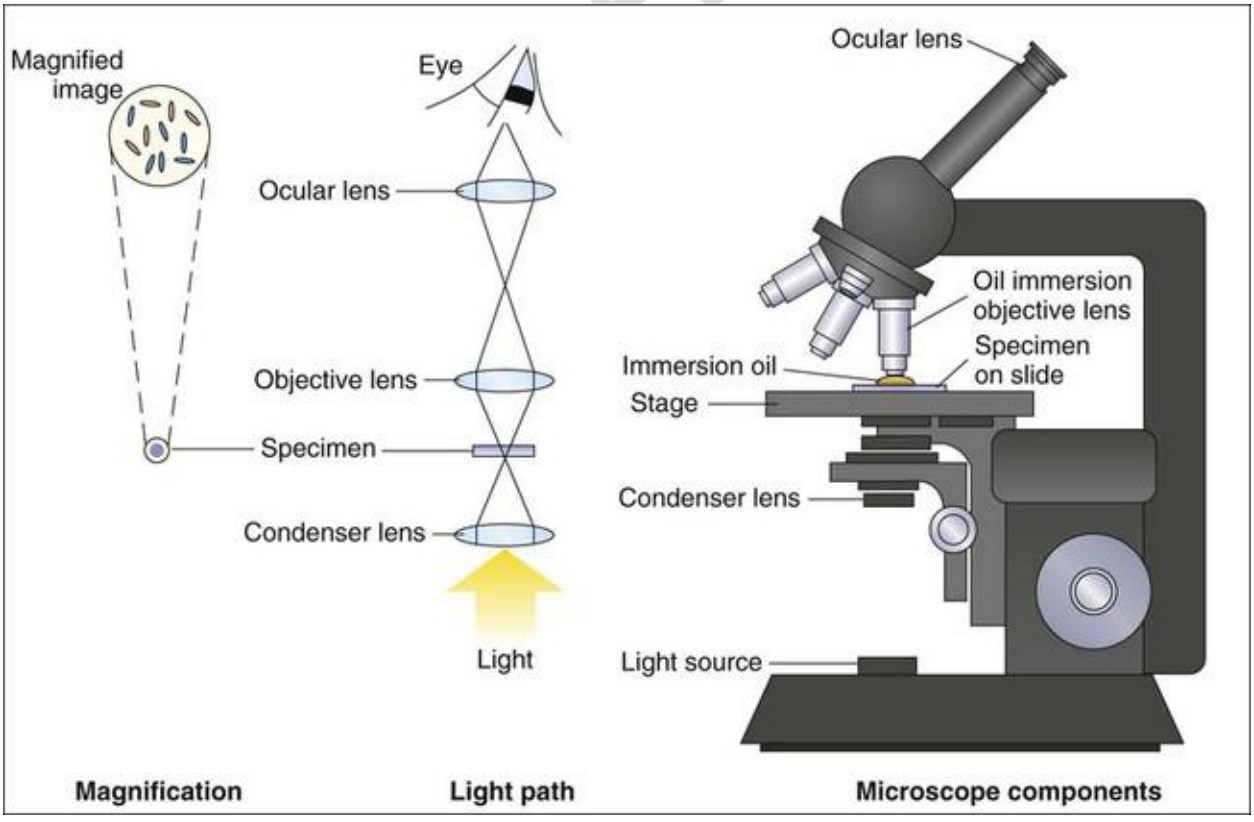
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Bright field microscopy

The most basic mode of light microscope is called brightfield (bright background) which can be achieved with the minimum optical elements. All light microscopes include a system of three glass lenses:

- The condenser lens that focuses light from the source onto the specimen
- The objective lenses which magnify the specimen to different degrees
- The eyepiece lenses which further magnify the image.

Upon comparison with other microscopes, these are inexpensive and simple to align and use. Contrast in images obtained from such microscopes is usually produced by the colour of the object itself or by exploiting some other property that affects the amount of light that can pass through. This kind of microscope is, therefore, used to collect images from pigmented tissues or histological sections, or from tissue culture cells that have been stained with dyes. Major components of all modern light microscopes include a condenser lens, the objective lens and the eyepiece lens.



Light from a lamp is focussed at the specimen by a glass condenser lens. The image is magnified with an objective lens and projected onto a detector with an eyepiece lens. The detector can be the eye or a digital camera.

The main components of the compound light microscope include:

Light source

A proper illumination of the specimen is important for achieving high-quality images and photomicrographs and this is achieved using a light source. Light sources typically include mercury lamps, xenon lamps, lasers or light-emitting diodes (LEDs). Light from the light source passes into the condenser lens (mounted beneath the microscopic stage in an upright microscope and above the stage in an inverted microscope) in a bracket that can be raised and lowered for focusing. The specimen is illuminated with parallel beams of light through the condenser focused light from the light source. Correct position of condenser lens produces illumination that is uniformly bright and free from glare across the viewing area of the specimen. Light that passes through the specimen (transmitted light) or is reflected back from the specimen (reflected light) is focused by the objective lens into the eyepiece lens. The image is either directly viewed by the eye in the eyepiece or often projected onto a detector (like photographic film or more likely a digital camera). The images are stored in a digital format and reproduced using digital methods. Condenser misalignment and an improperly adjusted condenser aperture diaphragm contribute to major sources of poor images in a light microscope.

Stand

This part of microscope holds all the components firmly in position. There are two basic types of microscope stand: an upright or an inverted one. The light source is below the condenser lens in the upright microscope and the objectives are above the specimen stage and this is the most commonly used format for viewing specimens. However, in an inverted microscope, the light source and the condenser lens are above the specimen stage and the objective lenses are beneath it. The condenser and light source can often be swung out of the light path and this also allows additional room for manipulating the specimen directly on the stage.

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Specimen stage

A mechanical device that is finely engineered to hold the specimen firmly in place. It enables the specimen to be moved and positioned in fine and smooth increments, both horizontally and transversely, in the X and Y directions (for locating a region of interest). The stage is moved vertically in the Z direction for focusing the specimen unlike in inverted microscopes where the objectives are moved and the stage is fixed. There are usually fine and coarse focusing controls for high and low magnification viewing respectively. The fine focus control can be moved in increments of up to 1 micrometer in the best research microscopes. The specimen stage can either be moved by a hand or a stepper motor (attached to the fine focus control of the microscope) controlled by a computer.

Objective lens

This component is responsible for producing the magnified image and can be the most expensive component of the microscope. Objectives are available in many different varieties and with much information inscribed on each one. This information can include manufacturer, magnification (4X, 10X, 20X, 40X, 60X and 100 X), immersion requirements (air, water or oil), coverslip thickness and more specialized optical properties of the lens. In fact, lens correction for optical artefacts, such as chromatic aberration, may also be included in the lens description. Like the words Fluo st (for fluorite in cases of least correction) and plan or plan apo (for plan apochromat in case of most highly corrected) might also appear on the lens surface. Objective lenses can either be dry (glass/air/coverslip) or immersion lenses (glass/oil or water/coverslip). Generally, most objectives below 40X are air (dry) and those of 40X and above are immersion (oil, glycerol or water) based. An objective designed to operate in oil will be labelled as oil or oel. Other immersion media include glycerol and water and the lens will be marked accordingly. Dipping lenses are specially designed to work without a coverslip and are dipped directly into water or tissue culture medium (like used in case of physiological experiments). There is even a mark for numerical aperture on the lens (a number usually between 0.04 and 1.4). Lenses with a low NA collect less light than those with a high NA. High power objectives with a higher NA have a better resolution than the low power lenses. The objective lens is most prone to be damaged by mishandling although many lenses are coated by a protective coating. Even a scratch in front of lens can lead to serious image degradation.

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Eyepiece It works in combination with the objective lens to further magnify the image and allows it to be detected by eye, or usually to project the image into a digital camera for recording purposes. Eyepieces usually magnify by 10X since an eyepiece of higher magnification merely enlarges the image without any improvement in resolution. There is an upper limit to the useful magnification of the collection of lenses in a microscope.

Phase contrast microscopy

Principle

or examining cells through brightfield microscopy, cells are often killed, sliced into thin sections and stained. Although such procedures are useful for examining the details of cell's internal architecture, however, with an increased emphasis on the study of cell dynamics (especially in the fields of motility, division, and cell death), techniques that allow visualization of intact or still living cells have become increasingly important. This improves contrast (without sectioning and staining) by exploiting differences in the thickness and refractive index of various regions of the cells. Phase contrast microscopy was introduced by the Nobel laureate Frits Zernike. It is a form of interferometry whereby contrast is derived from refractive elements within the sample, which cause phase shifts in light that are subsequently converted into amplitude changes which are registered as light or dark regions against a uniform reference. Cell organelles are made up of different proportions of various molecules like DNA, RNA, protein, lipid, carbohydrate, salts and water. Regions of different composition have different refractive indices. As the rays pass from the light source through the object, their velocity might be affected by the physical property of the specimen. Along its route, the velocity of the rays slows down to varying extent upon encountering different regions of the specimen. Light passing through the thicker parts of the cell is held up relative to the light that passes through thinner parts of the cytoplasm, resulting in a change in phase relative to light waves that have not passed through the specimen. Although such changes in phase are not detected by the human eye, phase contrast microscope overcomes this problem by converting differences in refractive indices into differences in intensity (relative brightness and darkness), which are visible to the eye. Phase contrast microscope accomplishes this result by: 1. separating the direct light that enters the objective lens from the diffracted light emanating from the specimen and 2. by causing these two types of light rays to interfere with one

Career Avenues

another, producing an image with highly contrasting bright and dark areas against a uniformly illuminated background. The relative brightness or darkness of each part of the image reflects the way in which light from that part of the specimen interferes with the direct light. On an average, light passing through the transparent specimens is retarded by approximately ¼ wavelength and the direct, undiffracted light passes through a portion of the phase plate that speeds it up by approximately ¼ wavelength.

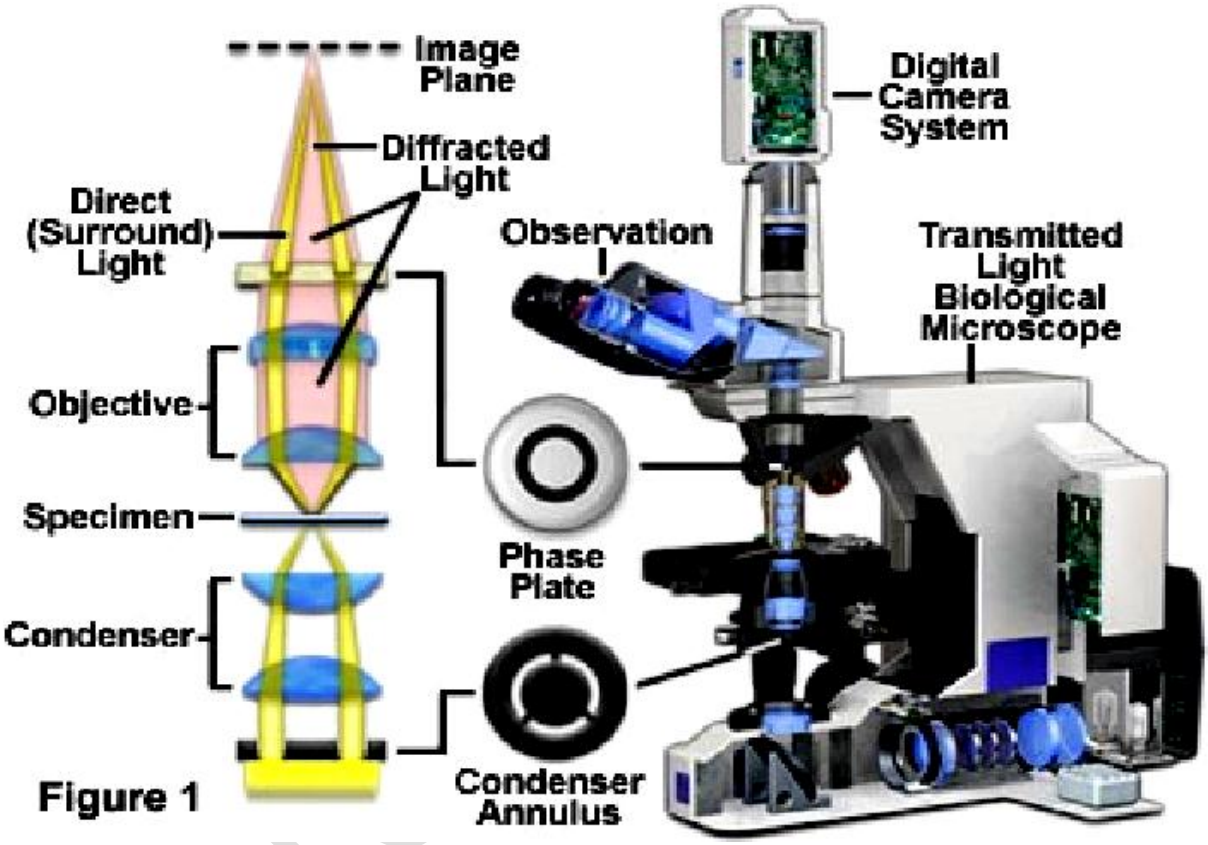


Figure 1

Applications

Phase contrast microscope is most useful for examining intracellular components of living cells at relatively high resolution. It is widely used in microbiology and tissue culture research to detect bacteria, cellular organelles or for testing cell and organelle preparations for lysis. Also, the dynamic motility of mitochondria, mitotic chromosomes, and vacuoles can be followed and filmed using this optics. The greatest benefit derived from its invention has not been in the discovery of new structures, but rather its everyday use in research and teaching laboratories for observing cells in a more revealing way. This microscope is only suitable for observing single

cells or thin cell layers. Additionally, it has optical handicaps that result in loss of resolution, and the image suffers from interfering halos and shading where sharp changes in refractive index occur. Cells and their components appear to be surrounded by a white halo caused by interference fringes. This prevents one from observing distinct edges on the specimen. This restricts the ability to identify and distinguish between organelles within a typical flattened cell. Therefore, many researchers focus on other types of microscopes, like differential interference contrast (DIC) microscope that produces images with a shadow relief.

Fluorescent Microscopy

The absorption and subsequent re-radiation of light by organic and inorganic specimens is typically the result of well-established physical phenomena described as being either **fluorescence** or **phosphorescence**. The emission of light through the fluorescence process is nearly simultaneous with the absorption of the excitation light due to a relatively short time delay between photon absorption and emission, ranging usually less than a microsecond in duration. When emission persists longer after the excitation light has been extinguished, the phenomenon is referred to as phosphorescence.

In microscopy, fluorescence can be used as a label or tag when preparing specific biological probes. Some biological substances, such as chlorophyll and some oils and waxes, have primary fluorescence (auto-fluorescence). But most biological molecules or structures do not fluorescence on their own, so they must be linked with fluorescent molecules fluorochromes) in order to create specific fluorescent probes.

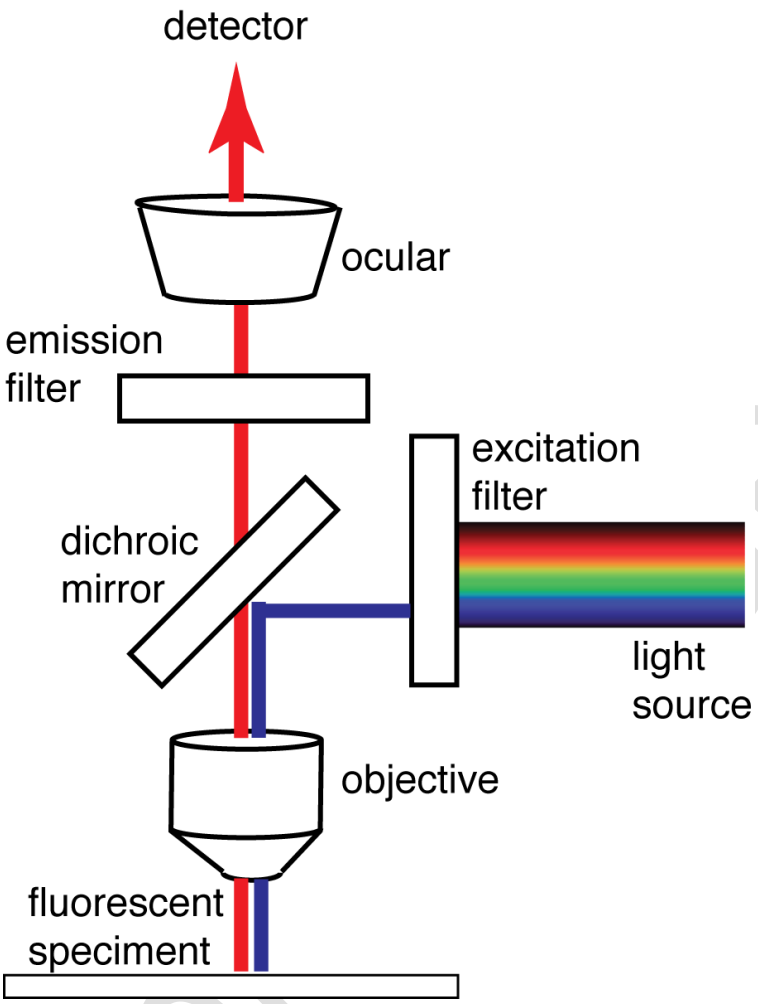
Principle of Fluorescence Microcopy

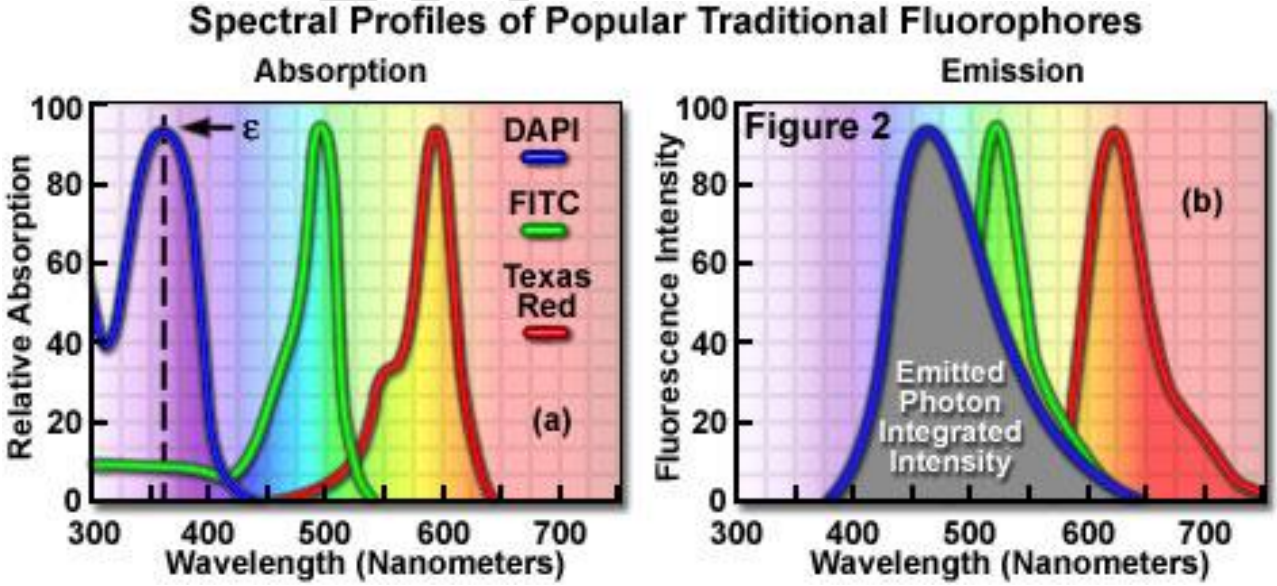
A fluorescence microscope is much the same as a conventional light microscope with added features to enhance its capabilities. The conventional microscope uses visible light (400-700 nanometers) to illuminate and produce a magnified image of a sample. A fluorescence microscope, on the other hand, uses a much higher intensity light source which excites a fluorescent species in a sample of interest. This fluorescent species in turn emits a lower energy light of a longer wavelength that produces the magnified image instead of the original light source. Fluorescent microscopy is often used to image specific features of small specimens such

as microbes. It is also used to visually enhance 3-D features at small scales. This can be accomplished by attaching fluorescent tags to anti-bodies that in turn attach to targeted features, or by staining in a less specific manner. When the reflected light and background fluorescence is filtered in this type of microscopy the targeted parts of a given sample can be imaged. This gives an investigator the ability to visualize desired organelles or unique surface features of a sample of interest. Confocal fluorescent microscopy is most often used to accentuate the 3-D nature of samples. This is achieved by using powerful light sources, such as lasers, that can be focused to a pinpoint. This focusing is done repeatedly throughout one level of a specimen after another. Most often an image reconstruction program pieces the multi level image data together into a 3-D reconstruction of the targeted sample.

In most cases the sample of interest is labeled with a fluorescent substance known as a fluorophore and then illuminated through the lens with the higher energy source. The illumination light is absorbed by the fluorophores (now attached to the sample) and causes them to emit a longer lower energy wavelength light. This fluorescent light can be separated from the surrounding radiation with filters designed for that specific wavelength allowing the viewer to see only that which is fluorescing. The basic task of the fluorescence microscope is to let excitation light radiate the specimen and then sort out the much weaker emitted light from the image. First, the microscope has a filter that only lets through radiation with the specific wavelength that matches your fluorescing material. The radiation collides with the atoms in your specimen and electrons are excited to a higher energy level. When they relax to a lower level, they emit light. To become detectable (visible to the human eye) the fluorescence emitted from the sample is separated from the much brighter excitation light in a second filter. This works because the emitted light is of lower energy and has a longer wavelength than the light that is used for illumination. Most of the fluorescence microscopes used in biology today are epi-fluorescence microscopes, meaning that both the excitation and the observation of the fluorescence occur above the sample. Most use a Xenon or Mercury arc-discharge lamp for the more intense light source.

Career Avenues


detector
ocular
emission filter
excitation filter
dichroic mirror
light source
objective
fluorescent speciment


Spectral Profiles of Popular Traditional Fluorophores
Absorption
Emission
Relative Absorption
Fluorescence Intensity
ε
DAPI
FITC
Texas Red
(a)
(b)
Figure 2
Emitted Photon Integrated Intensity
Wavelength (Nanometers)
Wavelength (Nanometers)

Other popular advanced fluorescence techniques, such as fluorescence resonance energy transfer (**FRET**) and fluorescence recovery after photobleaching (**FRAP**), as well as spectroscopy, are often combined with total internal reflection to achieve additional information. The result is a very powerful tool for the study of individual fluorophores and fluorescently labeled molecules. The advantages resulting from the study of the properties of single molecules are only beginning to be appreciated. Thus, the current range of optical microscopy now extends from the single molecule to the entire animal.

Applications:

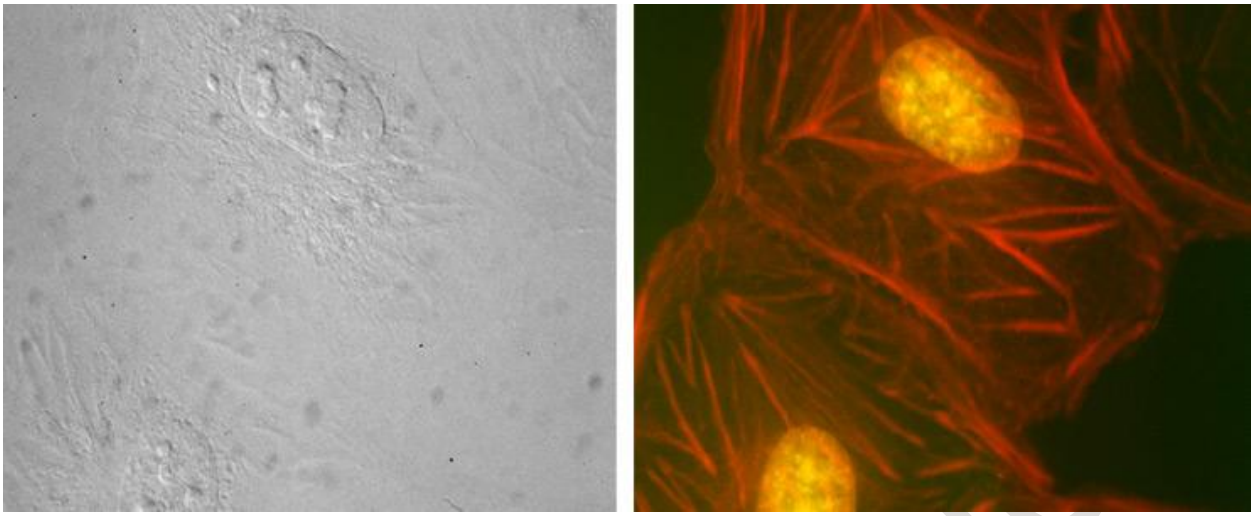
The refinement of epi-fluorescent microscopes and advent of more powerful focused light sources, such as lasers, has led to more technically advanced scopes such as the confocal laser scanning microscopes and total internal reflection fluorescence microscopes (TIRF).

CLSM's are invaluable tools for producing high resolution 3-D images of subsurfaces in specimens such as microbes. Their advantage is that they are able to produce sharp images of thick samples at various depths by taking images point by point and reconstructing them with a computer rather than viewing whole images through an eyepiece.

These microscopes are often used for -

- Imaging structural components of small specimens, such as cells
- Conducting viability studies on cell populations (are they alive or dead?)
- Imaging the genetic material within a cell (DNA and RNA)
- Viewing specific cells within a larger population with techniques such as FISH

Career Avenues



Cells captured using brightfield (left) and fluorescence (right) microscopy. Fluorescent labeling of the nucleus (yellow) and actin (red) makes it possible to see much more detailed cell structure

Confocal Microscopy

Confocal microscopy offers several advantages over conventional optical microscopy, including shallow depth of field, elimination of out-of-focus glare, and the ability to collect serial optical sections from thick specimens. In the biomedical sciences, a major application of confocal microscopy involves imaging either fixed or living cells and tissues that have usually been labeled with one or more fluorescent probes. When these samples are imaged using a conventional light microscope, the fluorescence in the specimen away from the region of interest interferes with resolution of structures in focus, especially for those specimens that are thicker than approx. 2 μm (Fig. 1). The confocal approach provides a slight increase in both lateral and axial resolution, although it is the ability of the instrument to eliminate the "out-of-focus" flare from thick fluorescently labeled specimens that has caused the explosion in its popularity in recent years. Most modern confocal microscopes are now relatively easy to operate and have become integral parts of many multiuser imaging facilities. Because the resolution achieved by the laser scanning confocal microscope (LSCM) is somewhat better than that achieved in a conventional, wide-field light microscope (theoretical maximum resolution of 0.2 μm), but not as great as that in the transmission electron microscope (0.1 nm), it has bridged the gap between these two commonly used techniques. The method of image formation in a confocal microscope is fundamentally different from that in a conventional wide-field microscope in which the entire specimen is bathed in light from a mercury or xenon source, and the image can be viewed

directly by eye. In contrast, the illumination in a confocal microscope is achieved by scanning one or more focused beams of light, usually from a laser, across the specimen Fig. 2. The images produced by scanning the specimen in this way are called optical sections. This refers to the noninvasive method of image collection by the instrument, which uses light rather than physical means to section the specimen. The confocal approach has facilitated the imaging of living specimens, enabled the automated collection of three dimensional (3D) data in the form of Z-series, and improved the images of multi-labeled specimens.

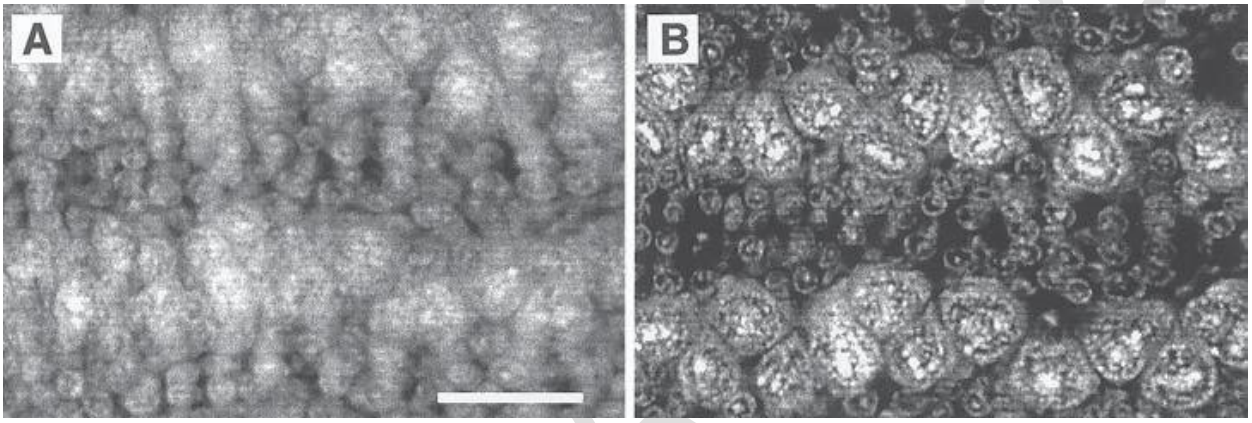
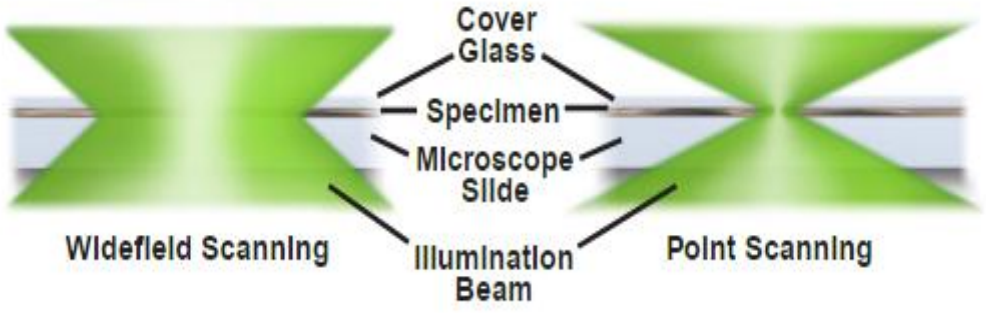


Fig. 1. Conventional epifluorescence image **(A)** compared with a confocal image **(B)** of a similar region of a whole mount of a butterfly pupal wing epithelium stained with propidium iodide. Note the improved resolution of the nuclei in **(B)**, due to the rejection of out-of-focus flares by the LSCM.

Figure 2 - Widefield Versus Point Scanning of Specimens



The basic optics of the optical microscope have remained fundamentally unchanged for decades because the final resolution achieved by the instrument is governed by the wavelength of light, the objective lens, and the properties of the specimen itself. The dyes used to add contrast to specimens, and other technology associated with the methods of optical microscopy, have improved significantly over the past 20 years. The growth and refinement of the confocal approach is a direct result of a renaissance in optical microscopy that has been fueled largely by advancements in modern technology. A number of major technological advances that would have been a benefit to Minsky's confocal design have gradually become available (or more affordable) to biologists and other microscopists. Among these are stable multiwavelength lasers for improved point light sources, improved dichromatic mirrors, sensitive low-noise photodetectors, fast microcomputers with image processing capabilities enhanced by availability of affordable large-capacity memory chips, sophisticated image analysis software packages, and high-resolution video displays and digital image printers.

These technologies were developed independently, and since 1955, have been gradually incorporated into confocal imaging systems. As one example, digital image processing techniques were first applied effectively in the early 1980s by researchers at the Woods Hole Oceanographic Institute. Using what they termed "video-enhanced microscopes" they were able to image cellular structures such as microtubules, which are just beyond the theoretical resolution of the optical microscope. The apparent increase in resolution was enabled by digital enhancement of images that were captured using a low light level silicon intensified target (**SIT**) video camera connected to a digital image processor. The cellular structures were imaged using differential interference contrast (**DIC**) optics, and the images were further enhanced using digital processing methods.

The classification of confocal microscope designs is usually done on the basis of the method by which the specimens are scanned. The two fundamental means of scanning are to scan either the stage or the illumination beam, and there are at least two fundamentally different methods of beam scanning. The original design by Minsky was a stage scanning system that was driven by a primitive tuning fork device, and it was rather slow in building an image. Current stage scanning confocal designs that have evolved from the original concept are primarily used in materials science applications such as the microchip industry. Systems based upon this principle have recently become popular in biomedical applications involving screening of DNA on microchips.

A more practical alternative for most imaging of biological systems is to scan the illumination beam across a stationary specimen. This approach is the basis of many of the systems that have evolved into the research microscopes that are in vogue today. The technical details involved in confocal microscopy are not dealt with in this introduction, but basically two fundamentally different methods of beam scanning are used; multiple-beam scanning and single-beam scanning. Single-beam scanning is currently the most popular, and is the method used in the LSCM. Here the scanning of the beam is most commonly achieved by use of computer-controlled mirrors driven by galvanometers at a rate of one frame per second. To achieve faster scanning, at near video frame rates, some systems use an acousto-optical device or oscillating mirrors. The alternative method uses two beams to scan at near real time, and usually relies on using some form of spinning Nipkow disk. Such systems have been derived from the tandem scanning microscope (**TSM**), with improvements to make them more efficient for collecting images from fluorescently labeled specimens

Principle of Confocal Microscopy

The confocal principle in epi-fluorescence laser scanning microscopy is diagrammatically presented in Figure 2. Coherent light emitted by the laser system (excitation source) passes through a pinhole aperture that is situated in a conjugate plane (confocal) with a scanning point on the specimen and a second pinhole aperture positioned in front of the detector (a photomultiplier tube). As the laser is reflected by a dichromatic mirror and scanned across the specimen in a defined focal plane, secondary fluorescence emitted from points on the specimen (in the same focal plane) pass back through the dichromatic mirror and are focused as a confocal point at the detector pinhole aperture.

Career Avenues

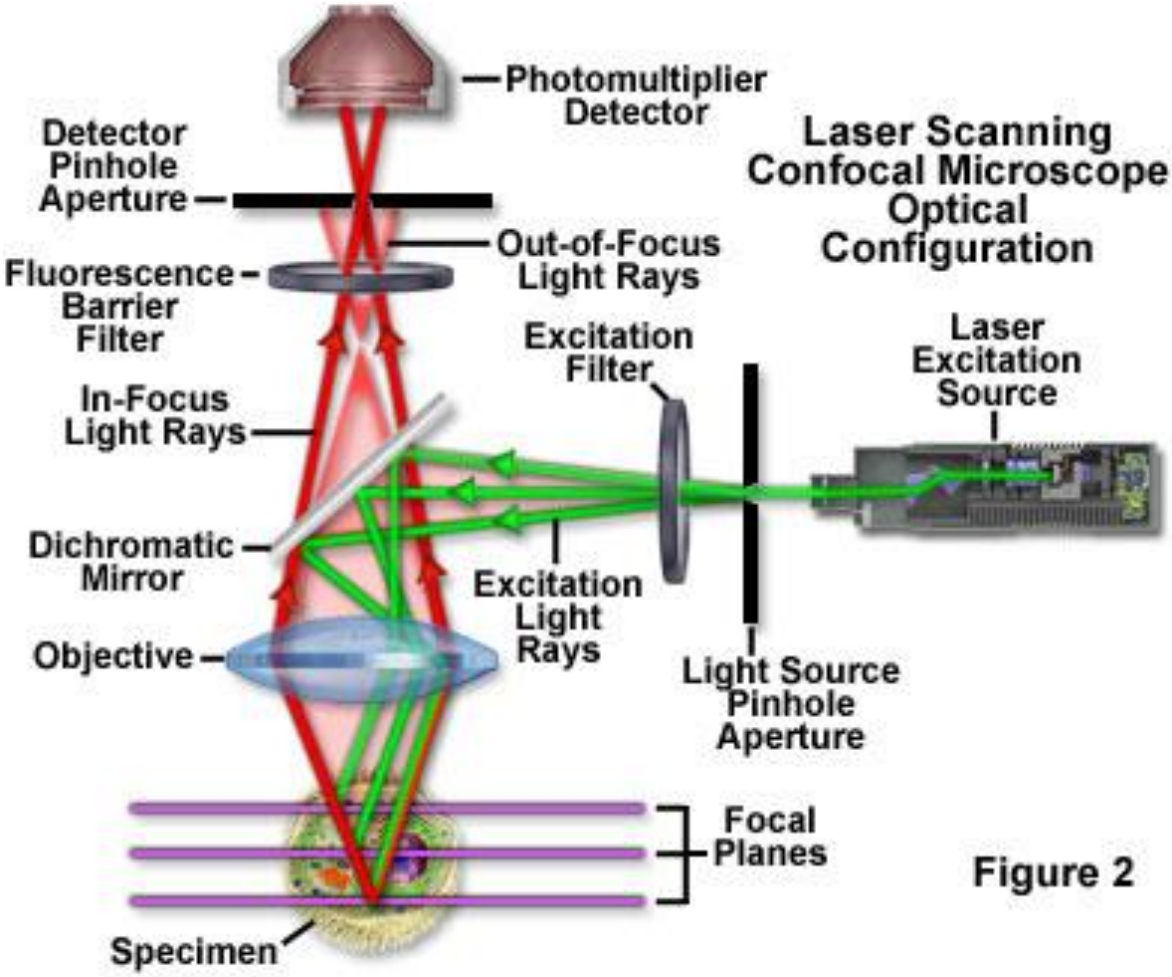


Figure 2

The significant amount of fluorescence emission that occurs at points above and below the objective focal plane is not confocal with the pinhole (termed **Out-of-Focus Light Rays** in Figure 2) and forms extended Airy disks in the aperture plane. Because only a small fraction of the out-of-focus fluorescence emission is delivered through the pinhole aperture, most of this extraneous light is not detected by the photomultiplier and does not contribute to the resulting image. The dichromatic mirror, barrier filter, and excitation filter perform similar functions to identical components in a widefield epi-fluorescence microscope. Refocusing the objective in a confocal microscope shifts the excitation and emission points on a specimen to a new plane that becomes confocal with the pinhole apertures of the light source and detector.

In traditional widefield epi-fluorescence microscopy, the entire specimen is subjected to intense illumination from an incoherent mercury or xenon arc-discharge lamp, and the resulting image of secondary fluorescence emission can be viewed directly in the eyepieces or projected onto the surface of an electronic array detector or traditional film plane. In contrast to this simple concept,

the mechanism of image formation in a confocal microscope is fundamentally different. As discussed above, the confocal fluorescence microscope consists of multiple laser excitation sources, a scan head with optical and electronic components, electronic detectors (usually photomultipliers), and a computer for acquisition, processing, analysis, and display of images.

Advantages and Disadvantages of Confocal Microscopy

The primary advantage of laser scanning confocal microscopy is the ability to serially produce thin (0.5 to 1.5 micrometer) optical sections through fluorescent specimens that have a thickness ranging up to 50 micrometers or more. The image series is collected by coordinating incremental changes in the microscope fine focus mechanism (using a stepper motor) with sequential image acquisition at each step. Image information is restricted to a well-defined plane, rather than being complicated by signals arising from remote locations in the specimen. Contrast and definition are dramatically improved over widefield techniques due to the reduction in background fluorescence and improved signal-to-noise. Furthermore, optical sectioning eliminates artifacts that occur during physical sectioning and fluorescent staining of tissue specimens for traditional forms of microscopy. The non-invasive confocal optical sectioning technique enables the examination of both living and fixed specimens under a variety of conditions with enhanced clarity.

In specimens more complex than a pollen grain, complex interconnected structural elements can be difficult to discern from a large series of optical sections sequentially acquired through the volume of a specimen with a laser scanning confocal microscope. However, once an adequate series of optical sections has been gathered, it can be further processed into a three-dimensional representation of the specimen using volume-rendering computational techniques Fig. 6. This approach is now in common use to help elucidate the numerous interrelationships between structure and function of cells and tissues in biological investigations.

Advances in confocal microscopy have made possible **multi-dimensional** views of living cells and tissues that include image information in the **x**, **y**, and **z** dimensions as a function of time and presented in multiple colors (using two or more fluorophores). After volume processing of individual image stacks, the resulting data can be displayed as three-dimensional multicolor video sequences in real time.

Disadvantages of confocal microscopy are limited primarily to the limited number of excitation wavelengths available with common lasers (referred to as **laser lines**), which occur over very narrow bands and are expensive to produce in the ultraviolet region. In contrast, conventional widefield microscopes use mercury or xenon based arc-discharge lamps to provide a full range of excitation wavelengths in the ultraviolet, visible, and near-infrared spectral regions.

Another downside is the harmful nature of high-intensity laser irradiation to living cells and tissues (an issue that has recently been addressed by multiphoton and Nipkow disk confocal imaging).

Finally, the high cost of purchasing and operating multi-user confocal microscope systems, which can range up to an order of magnitude higher than comparable widefield microscopes, often limits their implementation in smaller laboratories.

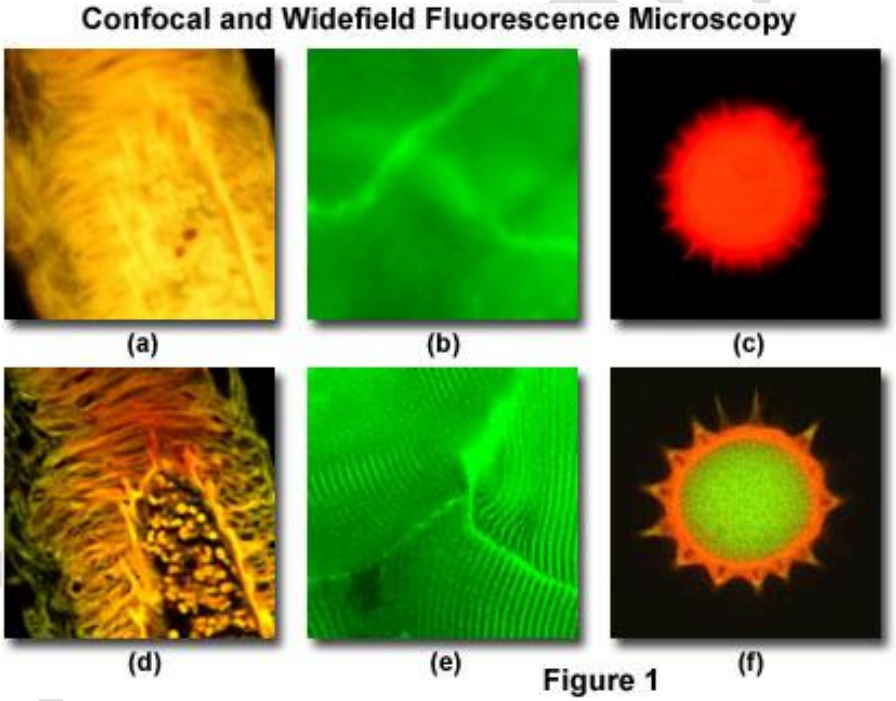


Figure 1

Figures 1 a,b,c are from widefield fluorescence Microscopy while 1 d, e and f are the same specimens imaged using confocal microscopy.

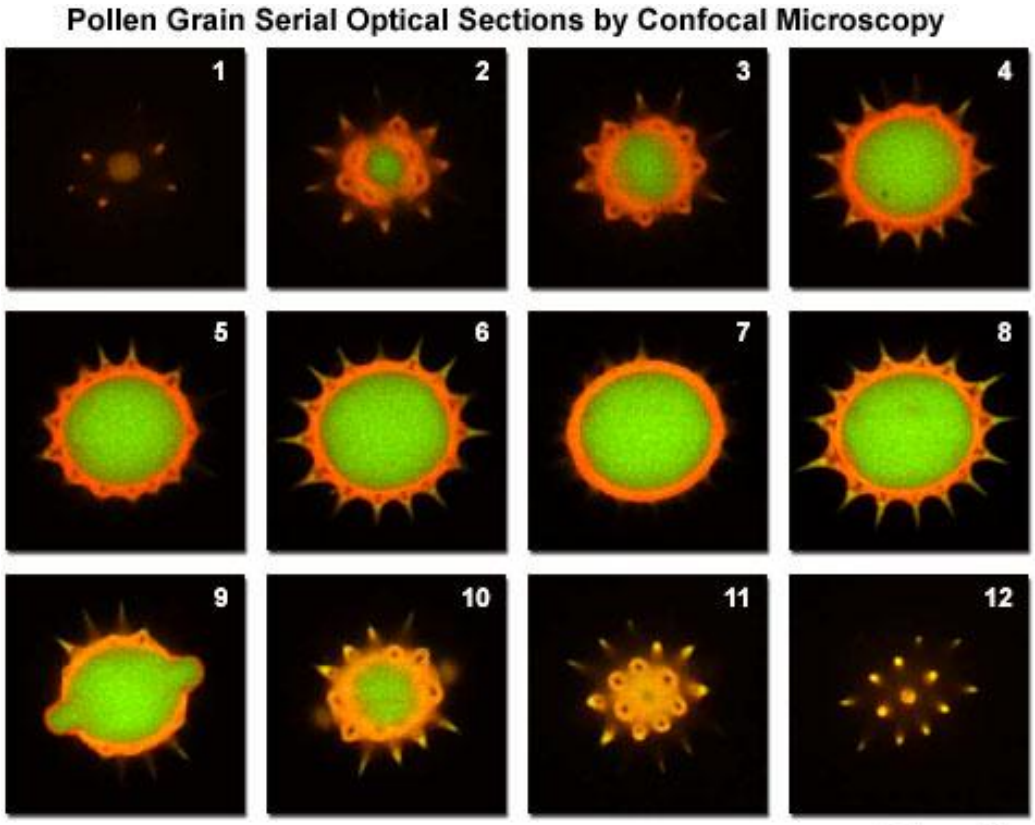


Figure 6

Electron Microscopy

Electron microscopy (EM) is a technique for obtaining high resolution images of biological and non-biological specimens. It is used in biomedical research to investigate the detailed structure of tissues, cells, organelles and macromolecular complexes. The high resolution of EM images results from the use of electrons (which have very short wavelengths) as the source of illuminating radiation. Electron microscopy is used in conjunction with a variety of ancillary techniques (e.g. thin sectioning, immuno-labeling, negative staining) to answer specific questions. EM images provide key information on the structural basis of cell function and of cell disease.

Principle of Electron Microscopy

Resolution in microscopy is limited to about ½ of the wavelength of the illumination source used to image the sample. Because our eyes can only detect photons with wavelengths greater than ~400 nm, the best resolution that can be achieved by light microscopes is about ~200 nm. One way to beat the diffraction limit of light is to use an illumination source with a shorter wavelength than photons – electrons.

Louis de Broglie showed that every particle or matter propagates like a wave. The wavelength of a particle or a matter can be calculated as follows.

$$\lambda = \frac{h}{p}$$

where λ is the wavelength of a particle, h is Planck's constant (6.626 x 10^{-34} J seconds), and p is the momentum of a particle. Since the momentum is the product of the mass and the velocity of a particle,

$$\lambda = \frac{h}{mv}$$

Because the velocity of the electrons is determined by the accelerating voltage, or electron potential where

$$eV = \frac{1}{2}mv^2$$

The velocity of electrons can be calculated by

$$v = \sqrt{\frac{2eV}{m}}$$

Therefore, the wavelength of propagating electrons at a given accelerating voltage can be determined by

$$\lambda = \frac{h}{\sqrt{2meV}}$$

Since the mass of an electron is 9.1 x 10^{-31} kg and e = 1.6 x 10^{-19},

$$\lambda = \frac{6.62*10^{-34}}{\sqrt{2*9.1*10^{-31}*1.6*10^{-19}*V}} = \frac{12.25*10^{-10}}{\sqrt{V}}$$

Thus, the wavelength of electrons is calculated to be 3.88 pm when the microscope is operated at 100 keV, 2.74 pm at 200 keV, and 2.24 pm at 300 keV.

However, because the velocities of electrons in an electron microscope reach about 70% the speed of light with an accelerating voltage of 200 keV, there are relativistic effects on these electrons. These effects include significant length contraction, time dilation, and an increase in mass. By accounting for these changes,

$$\lambda = \frac{12.25 * 10^{-10}}{\sqrt{V}} * \frac{1}{\sqrt{1 + \frac{eV}{2mc^2}}}$$

where c is the speed of light, which is ~3 x 10^8 m/s. Therefore, the wavelength at 100 keV, 200 keV, and 300 keV in electron microscopes is 3.70 pm, 2.51 pm and 1.96 pm, respectively.

The wavelength of electrons is much smaller than that of photons (2.5 pm at 200 keV). Thus the resolution of an electron microscope is theoretically unlimited for imaging cellular structure or proteins. Practically, the resolution is limited to ~0.1 nm due to the objective lens system in electron microscopes. Thus, electron microscopy can resolve subcellular structures that could not be visualized using standard fluorescences microscopy, such as the microvilli of intestinal cells or the internal structure of a bacterium.

Electron microscopes can even resolve molecular structure in a cell, for example the plasma membrane is only about 5nm in diameter. The membrane is composed of a lipid bilayer, each layer is a single molecule thick. These fatty acids are about 20 carbons long with a hydrophilic head group. These individual lipid layers can be distinguished in an electron micrograph.

Types of Electron Microscopy

All electron microscopes use electromagnetic and/or electrostatic lenses to control the path of electrons. Glass lenses, used in light microscopes, have no effect on the electron beam. The basic design of an electromagnetic lens is a solenoid (a coil of wire around the outside of a tube) through which one can pass a current, thereby inducing an electromagnetic field. The electron beam passes through the centre of such solenoids on its way down the column of the electron microscope towards the sample. Electrons are very sensitive to magnetic fields and can therefore be controlled by changing the current through the lenses.

The faster the electrons travel, the shorter their wavelength. The resolving power of a microscope is directly related to the wavelength of the irradiation used to form an image. Reducing wavelength increases resolution. Therefore, the resolution of the microscope is increased if the accelerating voltage of the electron beam is increased. The accelerating voltage

of the beam is quoted in kilovolts (kV). It is now possible to purchase a 1,000kV electron microscope, though this is not commonly found.

Although modern electron microscopes can magnify objects up to about two million times, they are still based upon Ruska's prototype and the correlation between wavelength and resolution. The electron microscope is an integral part of many laboratories such as The John Innes Centre. Researchers can use it to examine biological materials (such as microorganisms and cells), a variety of large molecules, medical biopsy samples, metals and crystalline structures, and the characteristics of various surfaces. Nowadays, electron microscopes have many other uses outside research. They can be used as part of a production line, such as in the fabrication of silicon chips, or within forensics laboratories for looking at samples such as gunshot residues. In the arena of fault diagnosis and quality control, they can be used to look for stress lines in engine parts or simply to check the ratio of air to solids in ice cream!

Transmission Electron Microscope (TEM)

The original form of electron microscopy, Transmission electron microscopy (TEM) involves a high voltage electron beam emitted by a cathode and formed by magnetic lenses. The electron beam that has been partially transmitted through the very thin (and so semitransparent for electrons) specimen carries information about the structure of the specimen. The spatial variation in this information (the "image") is then magnified by a series of magnetic lenses until it is recorded by hitting a fluorescent screen, photographic plate, or light sensitive sensor such as a CCD (charge-coupled device) camera. The image detected by the CCD may be displayed in real time on a monitor or computer.

Transmission electron microscopes produce two-dimensional, black and white images.

Resolution of the TEM is also limited by spherical and chromatic aberration, but a new generation of aberration correctors has been able to overcome or limit these aberrations. Software correction of spherical aberration has allowed the production of images with sufficient resolution to show carbon atoms in diamond separated by only 0.089 nm and atoms in silicon at 0.078 nm at magnifications of 50 million times. The ability to determine the positions of atoms within materials has made the TEM an indispensable tool for nano-technologies research and development in many fields, including heterogeneous catalysis and the development of

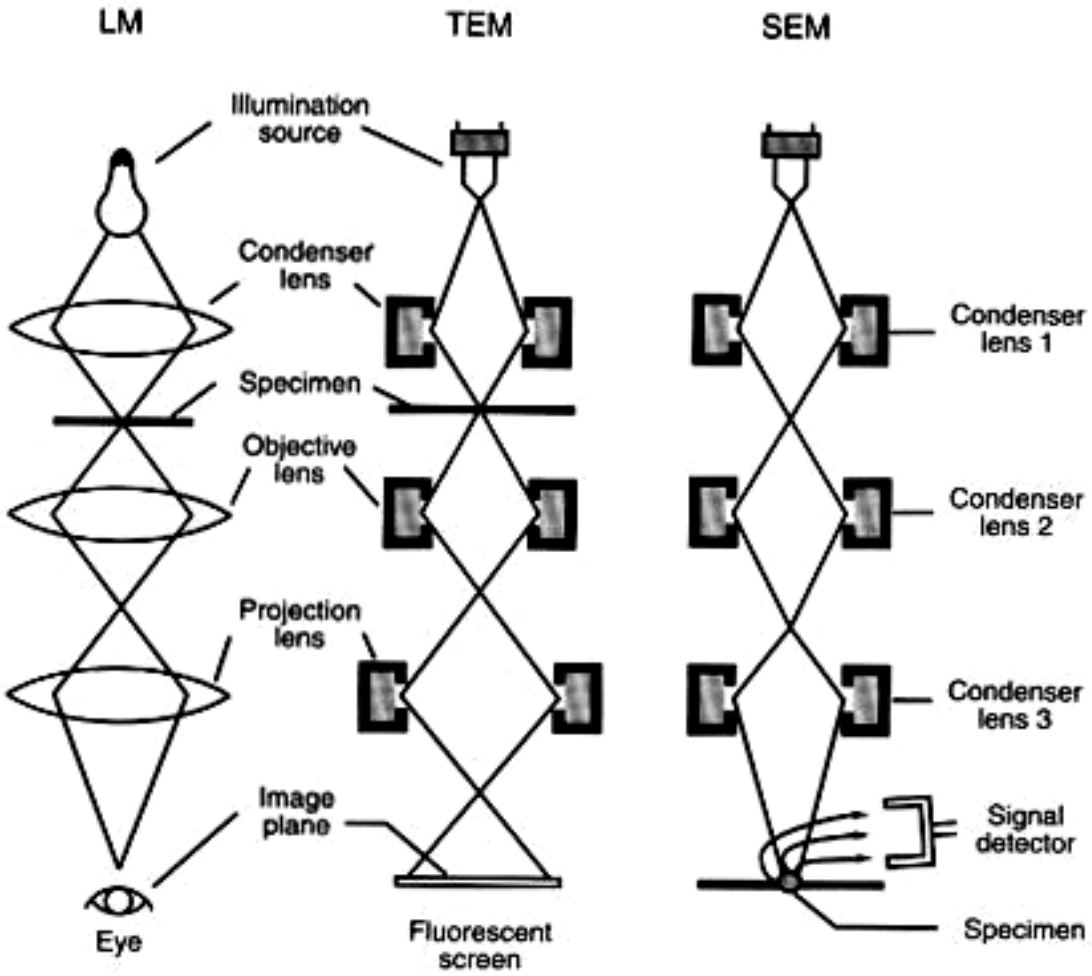
semiconductor devices for electronics and photonics. In the life sciences, it is still mainly the specimen preparation which limits the resolution of what we can see in the electron microscope, rather than the microscope itself.

Scanning Electron Microscope (SEM)

Unlike the TEM, where the electrons in the primary beam are transmitted through the sample, the Scanning Electron Microscope (SEM) produces images by detecting secondary electrons which are emitted from the surface due to excitation by the primary electron beam. In the SEM, the electron beam is scanned across the surface of the sample in a raster pattern, with detectors building up an image by mapping the detected signals with beam position. TEM resolution is about an order of magnitude better than the SEM resolution. TEM can easily resolve details of 0.2nm. Because the SEM image relies on electron interactions at the surface rather than transmission it is able to image bulk samples and has a much greater depth of view, and so can produce images that are a good representation of the 3D structure of the sample. SEM images are therefore considered to provide us with 3D, topographical information about the sample surface but will still always be only in black and white.

In the SEM, much lower accelerating voltages to prevent beam penetration into the sample are used since what we require is generation of the secondary electrons from the true surface structure of a sample. Therefore, it is common to use low KV, in the range 1-5kV for biological samples, even though SEMs are capable of up to 30 kV.

Career Avenues



Sample Preparation

Materials to be viewed in an electron microscope generally require processing to produce a suitable sample. This is mainly because the whole of the inside of an electron microscope is under high vacuum in order to enable the electron beam to travel in straight lines. The technique required varies depending on the specimen, the analysis required and the type of microscope:

Cryofixation - freezing a specimen rapidly, typically to liquid nitrogen temperatures or below, that the water forms ice. This preserves the specimen in a snapshot of its solution state with the minimal of artefacts. An entire field called cryo-electron microscopy has branched from this technique. With the development of cryo-electron microscopy, it is now possible to observe virtually any biological specimen close to its native state.

Fixation - a general term used to describe the process of preserving a sample at a moment in time and to prevent further deterioration so that it appears as close as possible to what it would

be like in the living state, although it is now dead. In chemical fixation for electron microscopy, glutaraldehyde is often used to crosslink protein molecules and osmium tetroxide to preserve lipids.

Dehydration - removing water from the samples. The water is generally replaced with organic solvents such as ethanol or acetone as a stepping stone towards total drying for SEM specimens or infiltration with resin and subsequent embedding for TEM specimens.

Embedding - infiltration of the tissue with wax (for light microscopy) or a resin (for electron microscopy) such as araldite or LR White, which can then be polymerised into a hardened block for subsequent sectioning.

Sectioning - the production of thin slices of the specimen. For light microscopy, the sections can be a few micrometres thick but for electron microscopy they must be very thin so that they are semitransparent to electrons, typically around 90nm. These ultra-thin sections for electron microscopy are cut on an ultramicrotome with a glass or diamond knife. Glass knives can easily be made in the laboratory and are much cheaper than diamond, but they blunt very quickly and therefore need replacing frequently.

Staining - uses heavy metals such as lead and uranium to scatter imaging electrons and thus give contrast between different structures, since many (especially biological) materials are nearly "transparent" to the electron beam. By staining the samples with heavy metals, we add electron density to it which results in there being more interactions between the electrons in the primary beam and those of the sample, which in turn provides us with contrast in the resultant image. In biology, specimens can be stained "en bloc" before embedding and/or later, directly after sectioning, by brief exposure of the sections to solutions of the heavy metal stains.

Freeze-fracture *and* ***freeze-etch*** - a preparation method particularly useful for examining lipid membranes and their incorporated proteins in "face on" view. The fresh tissue or cell suspension is frozen rapidly (cryofixed), then fractured by simply breaking or by using a microtome while maintained at liquid nitrogen temperature. The cold, fractured surface is generally "etched" by increasing the temperature to about -95°C for a few minutes to let some surface ice sublime to reveal microscopic details. For the SEM, the sample is now ready for imaging. For the TEM, it can then be rotary-shadowed with evaporated platinum at low angle (typically about 6°) in a high vacuum evaporator. A second coat of carbon, evaporated perpendicular to the average surface plane is generally performed to improve stability of the replica coating. The specimen is returned

to room temperature and pressure, and then the extremely fragile "shadowed" metal replica of the fracture surface is released from the underlying biological material by careful chemical digestion with acids, hypochlorite solution or SDS detergent. The floating replica is thoroughly washed from residual chemicals, carefully picked up on an EM grid, dried then viewed in the TEM.

Sputter Coating - an ultra-thin coating of electrically-conducting material, deposited by low vacuum coating of the sample. This is done to prevent charging of the specimen which would occur because of the accumulation of static electric fields due to the electron irradiation required during imaging. It also increases the amount of secondary electrons that can be detected from the surface of the sample in the SEM and therefore increases the signal to noise ratio. Such coatings include gold, gold/palladium, platinum, chromium etc.

Disadvantages of Electron Microscopy

Electron microscopes are very expensive to buy and maintain. They are dynamic rather than static in their operation: requiring extremely stable high voltage supplies, extremely stable currents to each electromagnetic coil/lens, continuously-pumped high/ultra-high vacuum systems and a cooling water supply circulation through the lenses and pumps. As they are very sensitive to vibration and external magnetic fields, microscopes aimed at achieving high resolutions must be housed in buildings with special services.

A significant amount of training is required in order to operate an electron microscope successfully and electron microscopy is considered a specialised skill.

The samples have to be viewed in a vacuum, as the molecules that make up air would scatter the electrons. This means that the samples need to be specially prepared by sometimes lengthy and difficult techniques to withstand the environment inside an electron microscope. Recent advances have allowed some hydrated samples to be imaged using an environmental scanning electron microscope, but the applications for this type of imaging are still limited.

Career Avenues

Centrifugation

Principle

The earth's gravitational force is sufficient to separate many types of particles over time. A tube of anticoagulated whole blood left standing on a bench top will eventually separate into plasma, red blood cell and white blood cell fractions. However, the length of time required precludes this manner of separation for most applications. In practice, centrifugal force is necessary to separate most particles. In addition, the potential degradation of biological compounds during prolonged storage means faster separation techniques are needed. The rate of separation in a suspension of particles by way of gravitational force mainly depends on the particle size and density. Particles of higher density or larger size typically travel at a faster rate and at some point will be separated from particles less dense or smaller. This sedimentation of particles, including cells, can be explained by the Stokes equation, which describes the movement of a sphere in a gravitational field.

From the Stokes equation five important behaviors of particles can be explained:

1. The rate of particle sedimentation is proportional to the particle size.
2. The sedimentation rate is proportional to the difference in density between the particle and the medium.
3. The sedimentation rate is zero when the particle density is the same as the medium density.
4. The sedimentation rate decreases as the medium viscosity increases.
5. The sedimentation rate increases as the gravitational force increases.

Most particles are so small that gravitational force is insufficient to overcome the random molecular forces of particles to influence separation. **Centrifugation**, the name given to separation applications which involve spinning around an axis to produce a centrifugal force, is a way to increase the magnitude of the gravitational field. The particles in suspension experience a radial centrifugal force moving them away from the axis of rotation.2 The radial force generated by the spinning rotor is expressed relative to the earth's gravitational force and therefore is known as the relative centrifugal force (RCF) or the "g force." The g force acting on particles is exponential to the speed of rotation (defined as revolutions per minute; rpm). Doubling the speed of rotation increases the centrifugal force by a factor of four. The centrifugal force also increases

with the distance from the axis of rotation. These two parameters are of considerable significance when selecting the appropriate centrifuge.

An object moving in a circle at a steady angular velocity will experience a centrifugal force (F) directed outwards. This is the basis of centrifugation. Angular velocity in radians (ω) and the radius of rotation (r) in centimeters collectively determine the magnitude of the force (F).

$$F=\omega^2 r \qquad \text{.....1}$$

(F) might be expressed in terms of earth's gravitational force if it is divided by 980. The resultant is referred to as the relative centrifugal force (RCF). RCF is more frequently referred to as the 'number times g'.

$$RCF = \omega^2 r / 980 \qquad \text{.....2}$$

For everyday use the relationship will be better if expressed in terms of 'revolutions per minute' (rpm) the common way in which the operating speed of a centrifuge rotor is expressed. Radius (r), Angular velocity in radians (ω) and rpm share the following relationship-

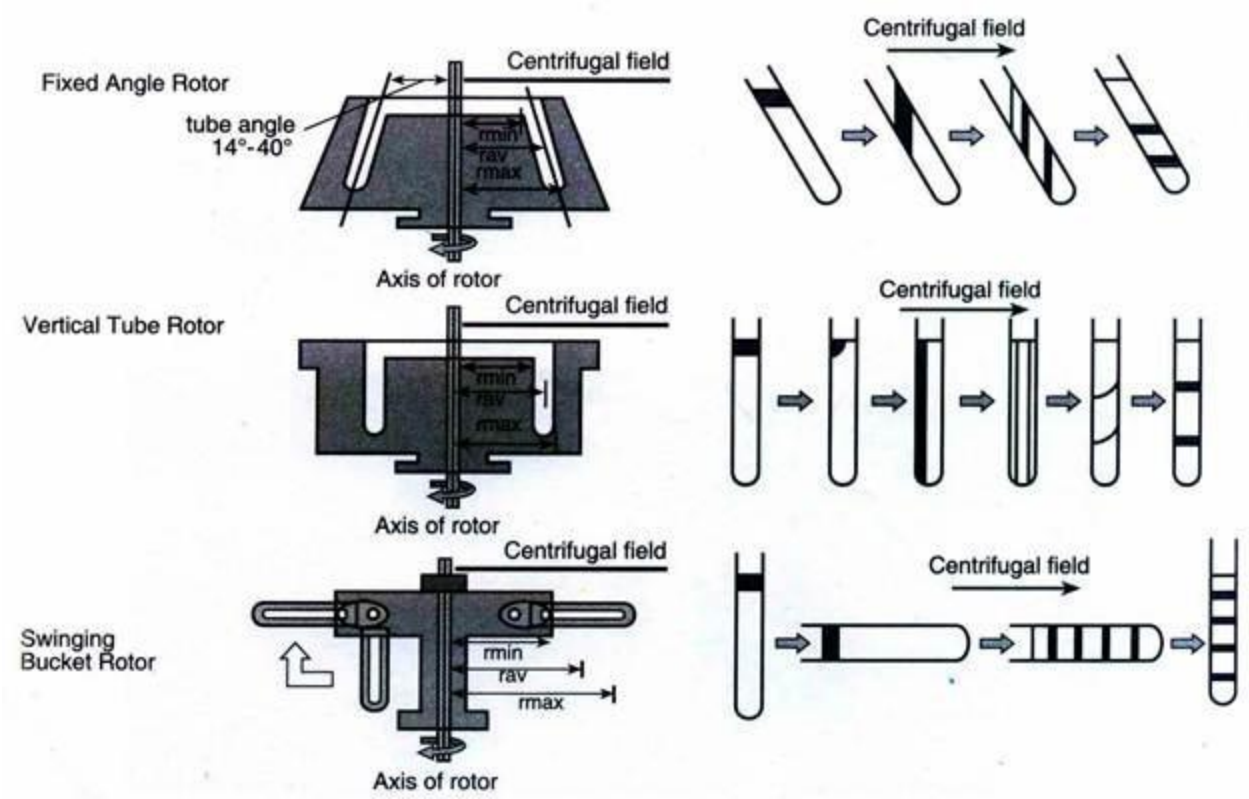
$$\omega = 2\pi (rpm) / 60 \qquad \text{.....3}$$

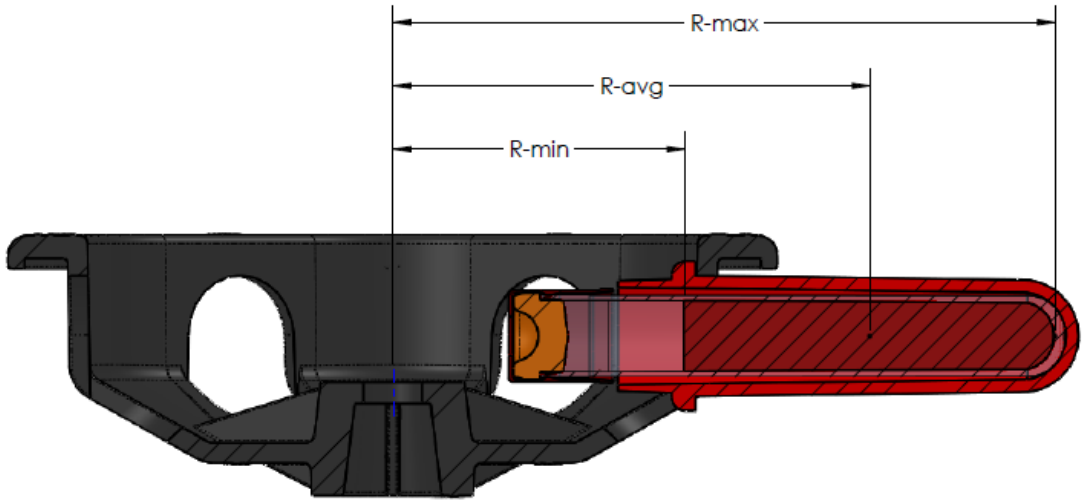
$$\omega = \pi (rpm) / 30 \qquad \text{.…3a}$$

Substituting for ω it can be written as

$$RCF = \frac{\pi^2 (rpm)^2 r}{(30)^2 / 980} \qquad \text{..…4}$$

$$RCF = (1.119 \times 10^{-5}) (rpm)^2 r \qquad \text{..…5}$$





Figures depicting the various types of centrifuge rotors, radius from the axis of rotation in a centrifuge and the pattern in which the particles move in the tubes when centrifugal force is applied.

Apart from RCF, the rate of sedimentation of a given particle would also depend upon its own characteristics such as its density, and its radius, the characteristics of the medium in which the particle is suspended; its density, and viscosity will also tend to affect the rate of sedimentation of the given particle. Incorporating all of these factors the sedimentation rate (V) of a particle in a centrifuge can be given by this equation

$$v = \frac{dr}{dt} = \frac{2}{9} \frac{r_p^2 (\rho_p - \rho_m) \omega^2 r}{\eta} \quad \text{.....6}$$

Where V= Sedimentation rate

r = radius (distance from the center of rotation to the particle)

ρ_p = Density of the particle

ρ_m = Density of the medium

ω = Angular velocity in radians

η = Viscosity coefficient of the medium

The above expression involves in the numerator the square of the particle radius. It thus automatically becomes clear that the most influencing factor determining the sedimentation rate of the particle is the size of the particle itself. Therefore, particles with slight differences in their size will have vastly different sedimentation velocities. The above mathematical expression says that all other things being equal:

1. The larger the particle. the faster it will move (the size is the single most important factor)
2. The denser a particle is the faster it will move.
3. The denser a solution is the slower will be the movement of the particle.
4. The higher the viscosity of the medium, the slower will be the particle movement

An expression for sedimentation time can be obtained by integrating equation ..(6)

$$t = \frac{9}{2} \times \frac{\eta}{\omega^2 r_p^2 \left(\rho_p - \rho_m\right)} \ln \frac{r_b}{r_t}$$

Where t = Sedimentation time

r_t = radial distance from the centre of rotation to the liquid meniscus, and

r_b = radial distance from the centre of rotation to the bottom of the tube

Parameters	Low speed	High speed	Ultracentrifuge
Speed ranges (r.p.m. × 10^3)	2-6	18-22	35-120
Maximum RCF (× 10^3)	8	60	700
Pelleting Applications			
Bacteria	–	Yes	Yes*
Animal and Plant cells	Yes	Yes	Yes*
Nuclei	Yes	Yes	Yes*
Precipitates	Some	Most	Yes*
Membrane organelles	Some	Yes	Yes
Membrane factions	Some	Some	Yes

Classes of centrifuges and their applications

Types of Centrifugation

Broadly there are two types of centrifugation methods depending on the purpose for which centrifugation is being done

1. Preparative centrifugation
2. Analytical centrifugation

Preparative Centrifugation

Preparative centrifugation is concerned with the actual isolation of biological material for subsequent biochemical investigations. Preparative centrifugation methods can be divided into two main techniques depending upon the medium of suspension in which the separation is carried out. Separations carried out in a suspending medium which is homogenous are known as **differential centrifugation** while those carried out in a suspending medium having density gradients are known as **density gradient centrifugations**. Density gradient centrifugation can further be divided into rate-zonal and isopycnic centrifugation.

Differential Centrifugation

The simplest form of separation by centrifugation is differential centrifugation, sometimes called differential pelleting (see Figure below). Particles of different densities or sizes in a suspension will sediment at different rates, with the larger and denser particles sedimenting faster. These sedimentation rates can be increased by using centrifugal force. A suspension of cells subjected to a series of increasing centrifugal force cycles will yield a series of pellets containing cells of decreasing sedimentation rate.

Differential pelleting is commonly used for harvesting cells or producing crude subcellular fractions from tissue homogenate. For example, a rat liver homogenate containing nuclei, mitochondria, lysosomes, and membrane vesicles that is centrifuged at low speed for a short time will pellet mainly the larger and more dense nuclei. Subsequent centrifugation at a higher centrifugal force will pellet particles of the next lower order of size (e.g., mitochondria) and so on. It is unusual to use more than four differential centrifugation cycles for a normal tissue homogenate. Due to the heterogeneity in biological particles, differential centrifugation suffers from contamination and poor recoveries. Contamination by different particle types can be addressed by resuspension and repeating the centrifugation steps (i.e., washing the pellet).

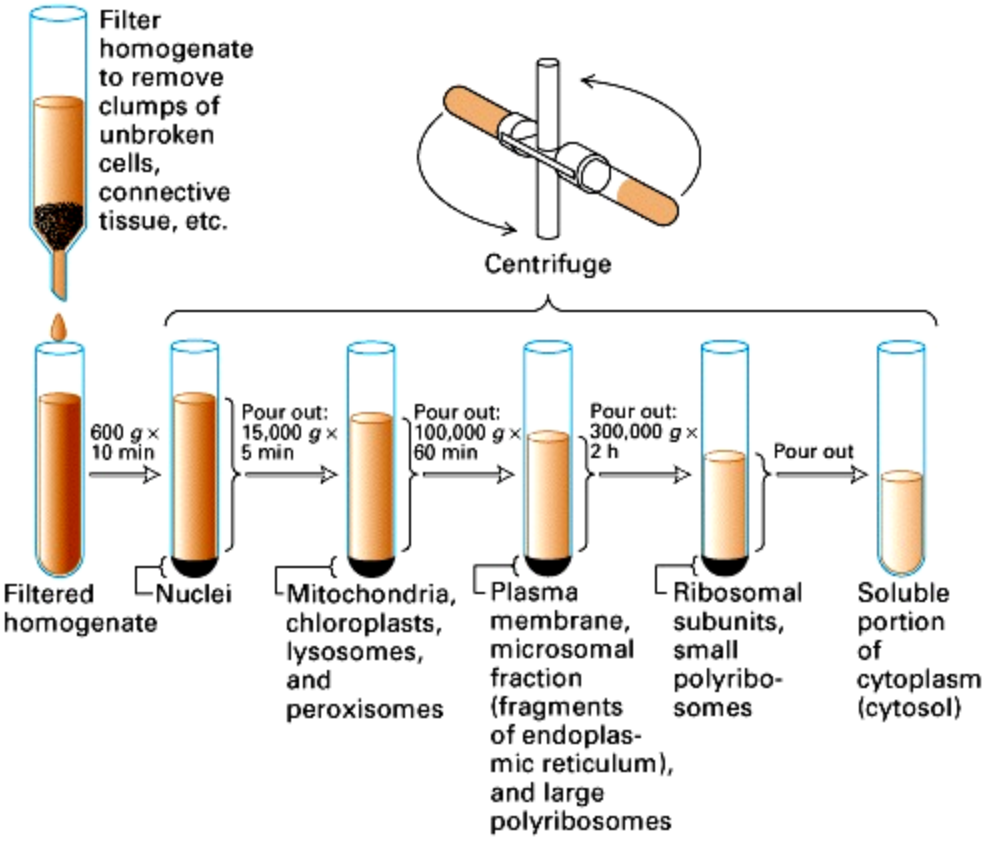


Figure: Particles of different densities or size will sediment at different rates with the largest and most dense particles sedimenting the fastest followed by less dense and smaller particles.

Density gradient Centrifugation

As opposed to differential centrifugation where a homogenous medium is used for separation density gradient centrifugation employs medium which has gradients. The separation under the centrifugal field is therefore dependent upon the buoyant densities of the particles. The gradients apart from exerting their separating effect eliminate mixing of separated components due to convection and mechanical vibrations. This method gives a much better separation than differential centrifugation. Density gradient centrifugation has two variations;

1. Rate-zonal centrifugation and
2. Isopycnic centrifugation.

Rate-Zonal Centrifugation

In rate-zonal centrifugation the problem of cross-contamination of particles of different sedimentation rates may be avoided by layering the sample as a narrow zone on top of a density gradient (see Figures below). In this way the faster sedimenting particles are not contaminated by

the slower particles as occurs in differential centrifugation. However, the narrow load zone limits the volume of sample (typically 10%) that can be accommodated on the density gradient. The gradient stabilizes the bands and provides a medium of increasing density and viscosity. The speed at which particles sediment depends primarily on their size and mass instead of density. As the particles in the band move down through the density medium, zones containing particles of similar size form as the faster sedimenting particles move ahead of the slower ones. Because the density of the particles is greater than the density of the gradient, all the particles will eventually form a pellet if centrifuged long enough.

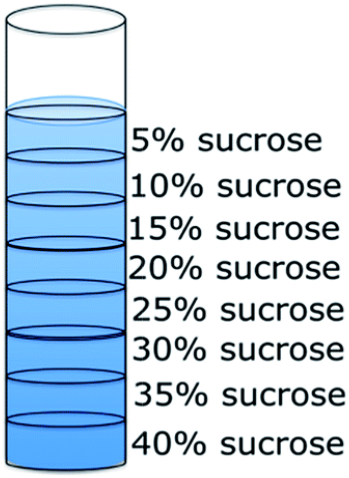


Figure depicting how a density gradient is created by carefully layering solution (in this case sucrose solution) of different densities.

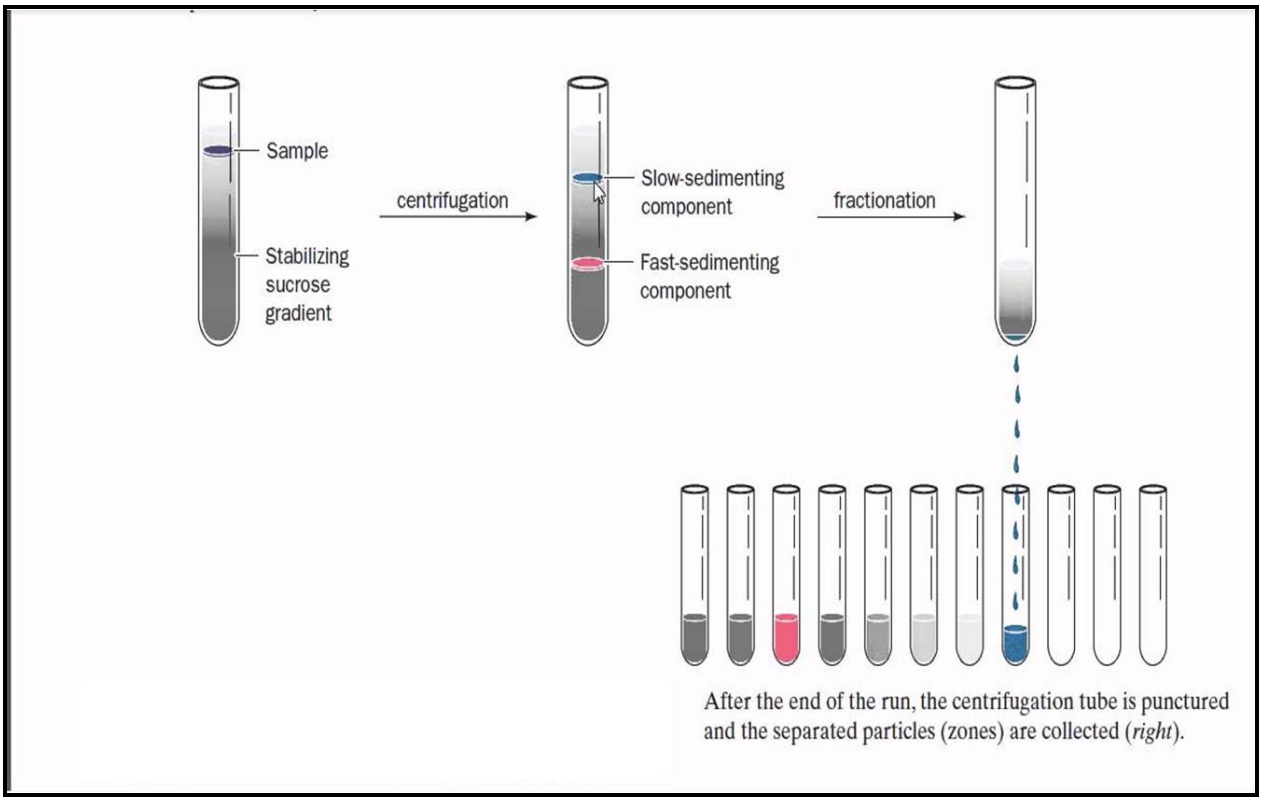


Figure depicting rate-zonal centrifugation. The sample is layered on a sucrose gradient (left). During Centrifugation (middle), each particle sediments at a rate that depends largely on the mass.

Isopycnic Centrifugation

In isopycnic separation, also called buoyant or equilibrium separation, particles are separated solely on the basis of their density. Particle size only affects the rate at which particles move until their density is the same as the surrounding gradient medium. The density of the gradient medium must be greater than the density of the particles to be separated. By this method, the particles will never sediment to the bottom of the tube, no matter how long the centrifugation time (see Figure below). Upon centrifugation, particles of a specific density sediment until they reach the point where their density is the same as the gradient media (i.e., the equilibrium position). The gradient is then said to be isopycnic and the particles are separated according to their buoyancy. Since the density of biological particles is sensitive to the osmotic pressure of the gradient, isopycnic separation may vary significantly depending on the gradient medium used. Although a continuous gradient may be more suited for analytical purposes, preparative techniques commonly use a discontinuous gradient in which the particles band at the interface

between the density gradient layers. This makes harvesting certain biological particles (e.g., lymphocytes) easier.

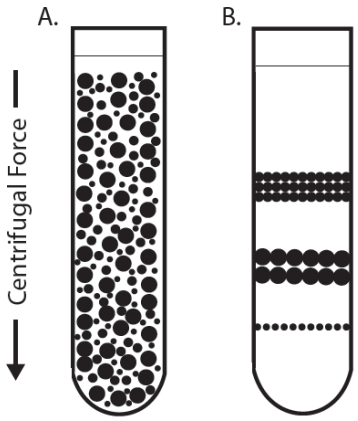


Figure depicting isopycnic centrifugation : it starts with a uniform mixture of sample and density gradient (3A) under centrifugal force, particles move until their density is the same as the surrounding medium (3B).

Differential Centrifugation	Rate zonal	Isopycnic centrifugation:
Separation is attained principally based on the size of the particles.	particles differs in their size, however not in density .	particles differ in their density, but not in size .
This separation is mostly used in simple pelleting and in locating partially-pure research of subcellular organelles and macromolecules.	Position of sample is related to the time of sedimentation (time dependent)	Here size is only affecting the rate of migration (time-independent)
Larger and more colossal components will sediment at lower speeds, while smaller components necessitate higher centrifugal force.	Its a Flat gradient.	It is a Steep grandient.
	Maximum density of gradient does not exceed that of the most dense particle that is the pellet.	Separation is based on the buoyant densities of the particles, so no pellet.

Analytical centrifugation

Analytical ultracentrifugation (AUC) is the study of the behavior of macromolecules in solution under the influence of a strong gravitational force. Most macromolecules have a different density from the solvent surrounding them and so will sink (or float) in a strong enough field. Observations of dynamic behaviour ("sedimentation velocity") or of systems in equilibrium

("sedimentation equilibrium") can provide information about size, shape, density and conformational changes in proteins and other macromolecules. A wide variety of materials have been investigated by AUC, ranging in size from peptides to viruses, in origin from inorganic particles through synthetic polymers to living cells.

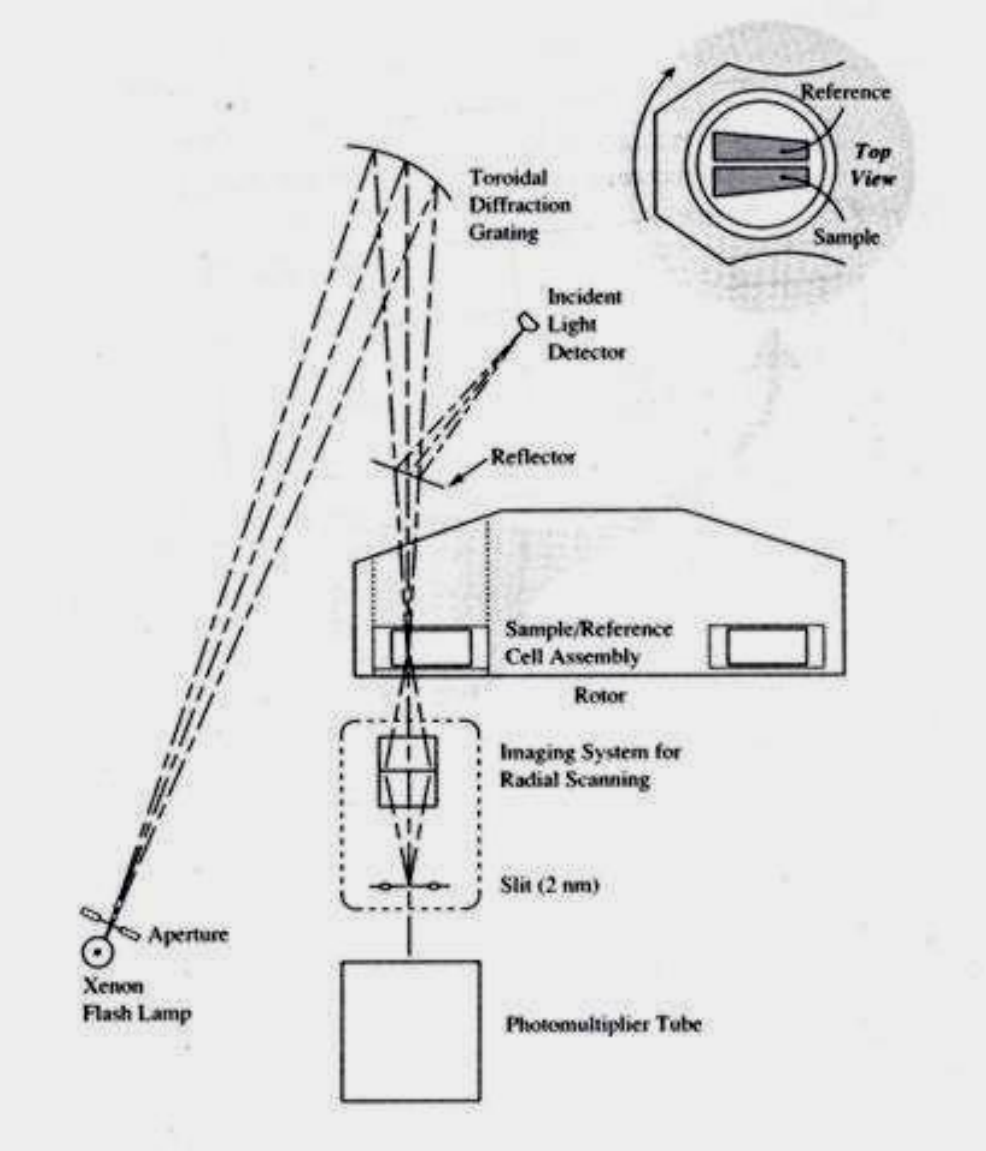
An analytical ultracentrifuge spins a rotor at an accurately controlled speed and temperature. The concentration distribution of the sample is determined at known times using absorbance measurements. The concentration, c, is determined for solutes obeying the Beer-Lambert law:

$$A = \varepsilon \times c \times l$$

where the absorbance, A, of the sample is measured at a given wavelength, ε, knowing the fixed position in the cell whose path length is l,

This figure displayed below is a schematic diagram of the Beckman Optima XL-A absorbance system. A high intensity xenon flask lamp allows the use of wavelengths between 190 and 800nm. The lamp is fired briefly as a selected sector passes the detector.

Career



Applications

The main use of sedimentation velocity experiments is to examine whether a protein sample is homogeneous in solution or a mixture of forms (e.g. monomer/dimer, or aggregates). It may be possible to tell whether its shape changes significantly or if association with another component in the solution is taking place. Sedimentation equilibrium is particularly useful for determination of molecular weights or subunit stoichiometry (monomer, dimer, trimer etc) in solution. Other methods such as size exclusion chomatography or electrophoresis are susceptible to errors caused by interactions with the matrix, and mass spectrometry, though very accurate, does not normally preserve subunit interactions. AUC has also been used to analyse nucleic acids on

density gradients - Meselson and Stahl's original experiments using 15N labelled DNA to show semiconservative replication were done by AUC.

Sedimentation Coefficient

The velocity of a particle per unit centrifugal field is referred to as its sedimentation coefficient. It is denoted by the symbol's'. The mathematical expression of (s) is therefore

$$s = \frac{v}{\omega^2 r}$$

Where V = velocity of a particle per unit centrifugal field

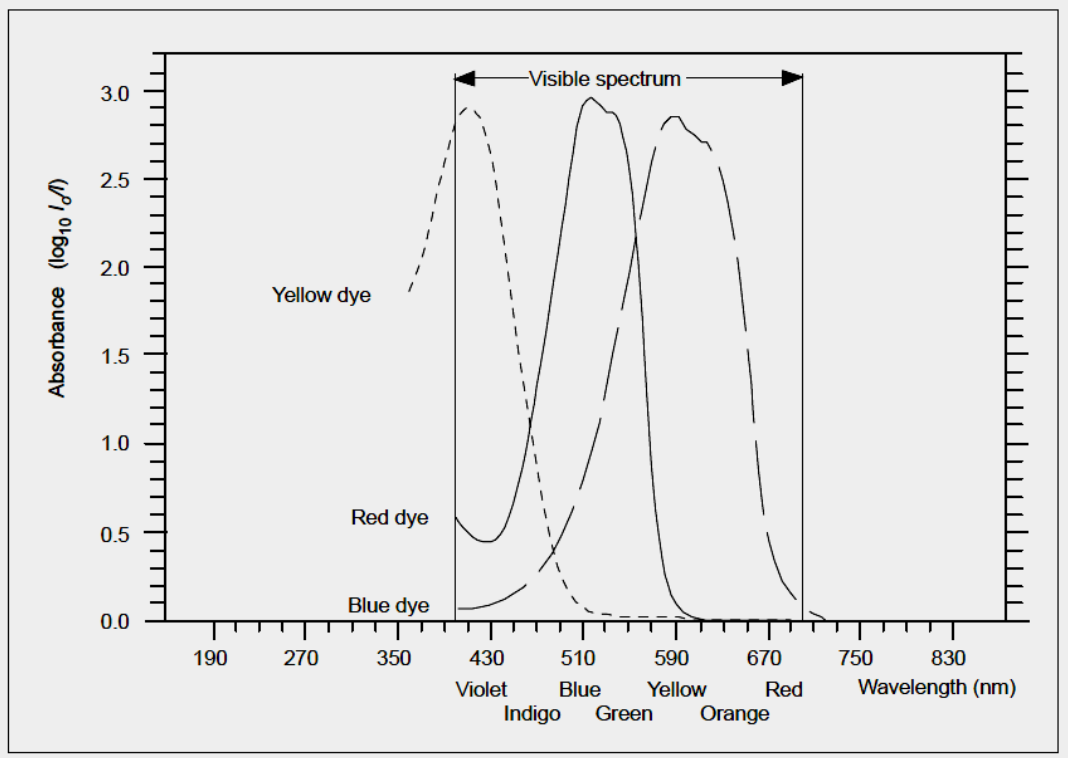
Sedimentation coefficient is affected by molecular weight, molecular shape and binding of small solute molecules to large macromolecules

Spectroscopy

Spectroscopy is a scientific measurement technique. It measures light that is emitted, absorbed, or scattered by materials and can be used to study, identify and quantify those materials. Spectroscopy is the study of the interaction of electromagnetic radiation in all its forms with matter. The interaction might give rise to electronic excitations, (e.g. UV), molecular vibrations (e.g. IR) or nuclear spin orientations (e.g. NMR). When a beam of white light strikes a triangular prism it is separated into its various components (ROYGBIV). This is known as a spectrum. The optical system which allows production and viewing of the spectrum is called a spectroscope. There are many other forms of light which are not visible to the human eye and spectroscopy is extended to cover all these.

UV-Visible spectrophotometry

Visible light absorption is known to all of us, because this is what causes objects to be coloured. For example, a blue dye appears blue because the light at the red end of the spectrum is absorbed, leaving the blue light to be transmitted. See figure below



Absorption of light by dyes

Visible light lies in the wavelength range 4.0 – 7.0 x 10^{-7} m (Fig below). To keep the numbers more manageable it is usually quoted in nanometres (10^{-9} m) so that the range becomes 400–700 nm. When light is absorbed by a material, valence (outer) electrons are promoted from their normal (ground) states to higher energy (excited) states. The energy of visible light depends on its frequency, and is approximately equivalent to 170 kJ mol^{-1} (mole of photons) for red light and 300 kJ mol^{-1} for blue light. The promotion of electrons to different energy levels is not restricted to electromagnetic radiation in the visible part of the spectrum; it can also occur in the ultraviolet region. To encompass the majority of electron transitions the spectrum between 190 and 900 nm is usually considered.

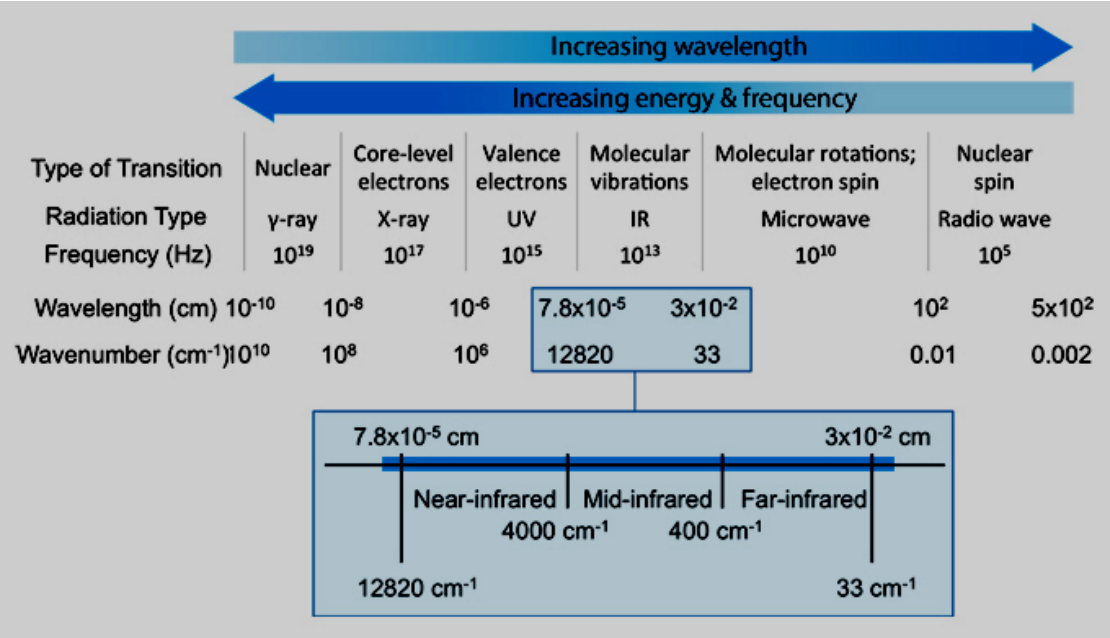


Table 1 Frequency, wavelength and energy of radiation in the part of the spectrum used for ultraviolet/visible spectroscopy

Frequency (ν) (Hz)	Wavelength (λ) (m)	(nm)	Energy (kJ mol^{-1})	
3.33×10^{14}	9.0×10^{-7}	900	137.5	(infrared)
4.29×10^{14}	7.0×10^{-7}	700	171.2	(red light)
7.50×10^{14}	4.0×10^{-7}	400	299.3	(blue light)
1.58×10^{15}	1.9×10^{-7}	190	630.5	(ultraviolet)

Frequency, wavelength and energy are interrelated:

$$c = \nu\lambda$$

where c = velocity of light (3.00×10^{8} ms^{-1})
ν = frequency in Hz
λ = wavelength in m

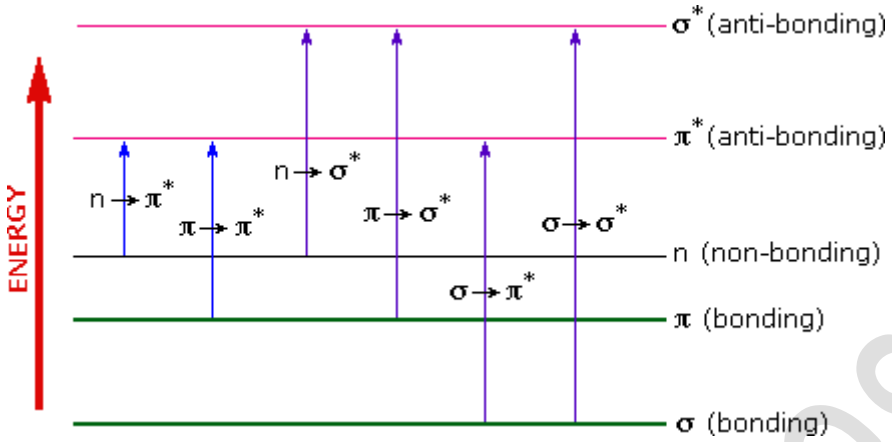
and

$E = h\nu$ or $E = h\nu L$ for one mole of photons

where E = energy of one mole of radiation
h = Planck's constant (6.63×10^{-34} Js)
L = Avogadro constant (6.02×10^{23} mol^{-1})

The origin of the absorptions

Valence electrons can generally be found in one of three types of electron orbital: 1 single, or σ , bonding orbitals; 2 double or triple bonds (π bonding orbitals); and 3 non-bonding orbitals (lone pair electrons). Sigma bonding orbitals tend to be lower in energy than π bonding orbitals, which in turn are lower in energy than non-bonding orbitals. When electromagnetic radiation of the correct frequency is absorbed, a transition occurs from one of these orbitals to an empty orbital, usually an antibonding orbital, σ * or π *. The exact energy differences between the orbitals depends on the atoms present and the nature of the bonding system. Most of the transitions from bonding orbitals are of too high a frequency (too short a wavelength) to measure easily, so most of the absorptions observed involve only π→π*, n→σ* and n→π* transitions. A common exception to this is the d→d transition of d-block element complexes, but in most cases a knowledge of the precise origin of transitions is not required.



Molecular (sub-)structures responsible for interaction with electromagnetic radiation are called chromophores. In proteins, there are three types of chromophores relevant for UV/Vis spectroscopy:

• peptide bonds (amide bond);

• certain amino acid side chains (mainly tryptophan and tyrosine); and

• certain prosthetic groups and coenzymes (e.g. porphyrine groups such as in haem).

Principle of UV-Visible spectroscopy

The chance for a photon to be absorbed by matter is given by an extinction coefficient which itself is dependent on the wavelength l of the photon. If light with the intensity I_0 passes through a sample with appropriate transparency and the path length (thickness) d, the intensity I drops along the pathway in an exponential manner. The characteristic absorption parameter for the sample is the extinction coefficient a, yielding the correlation $I= I_0 e^{-\alpha d.}$ The ratio $T= I/I_0$ is called transmission.

Beer's law tells us that absorption is proportional to the number of absorbing molecules – *ie* to the concentration of absorbing molecules (this is only true for dilute solutions) – and **Lambert's law** tells us that the fraction of radiation absorbed is independent of the intensity of the radiation. Combining these two laws, we can derive the **Beer-Lambert Law**:

$$\log_{10} \frac{I_o}{I} = \varepsilon l c$$

where I_o = the intensity of the incident radiation

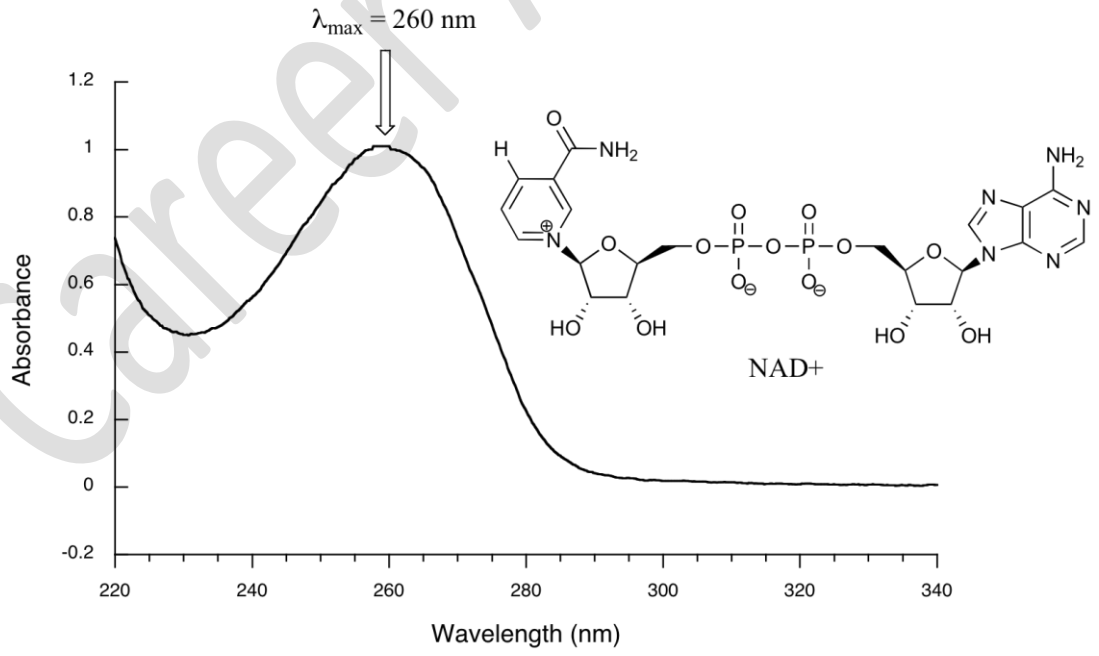
I = the intensity of the transmitted radiation

ε = a constant for each absorbing material, known as the molar absorption coefficient (called the molar extinction coefficient in older texts) and having the units mol^{-1} dm^3 cm^{-1}, but by convention the units are not quoted

l = the path length of the absorbing solution in cm

C = the concentration of the absorbing species in mol dm^{-3}

The value of $\log_{10}$ (I_o/I) is known as the absorbance of the solution (OD or optical density), and can be read directly from the spectrum, often as 'absorbance units'. A useful constant is the molar absorption coefficient, ε, because it is independent of concentration and path length, whereas absorption depends upon both. The other useful piece of information is the wavelength at which maximum absorption occurs. This is given the symbol λ_{max} (Fig below). These two pieces of information alone are frequently sufficient to identify a substance, although identification is not the most common use of this technique. Conversely, if the values of ε and λ_{max} are known, the concentration of its solution can be calculated – this is the more common application. The values of both ε and λ_{max} are strongly influenced by the nature of the solvent, and for organic compounds, by the degree of substitution and conjugation.



Career Avenues

The absorbance spectrum of an important biological molecule called nicotinamide adenine dinucleotide, abbreviated NAD^+. This compound absorbs light in the UV range due to the presence of conjugated pi-bonding systems.

Stating Beer's law a bit differently, we can say that optical density is proportional to the sample concentration if the path-length is constant. It follows then, that if the optical density of a standard solution is known, then an unknown sample concentration can be calculated from its optical density. The formula used for such calculations is

$$\text{concentration of the unknown} = \frac{\text{O.D. of the unknown} \times \text{conc. of a standard}}{\text{O.D. of the standard}}$$

The standard chosen should have an optical density nearer to the O.D. of the unknown.

Limitations of the Beer-Lambert law

The linearity of the Beer-Lambert law is limited by chemical and instrumental factors. Causes of nonlinearity include:

1. Deviations in absorptivity coefficients at high concentrations (>0.01M) due to electrostatic interactions between molecules in close proximity scattering of light 2. Due to particulates in the sample
3. Fluoresecence or phosphorescence of the sample
4. Changes in refractive index at high analyte concentration
5. Shifts in chemical equilibria as a function of concentration
6. Non-monochromatic radiation, deviations can be minimized by using a relatively flat part of the absorption spectrum such as the maximum of an absorption band
7. Stray light, sample instability and temperature of the sample.

Factors affecting UV/Vis absorption

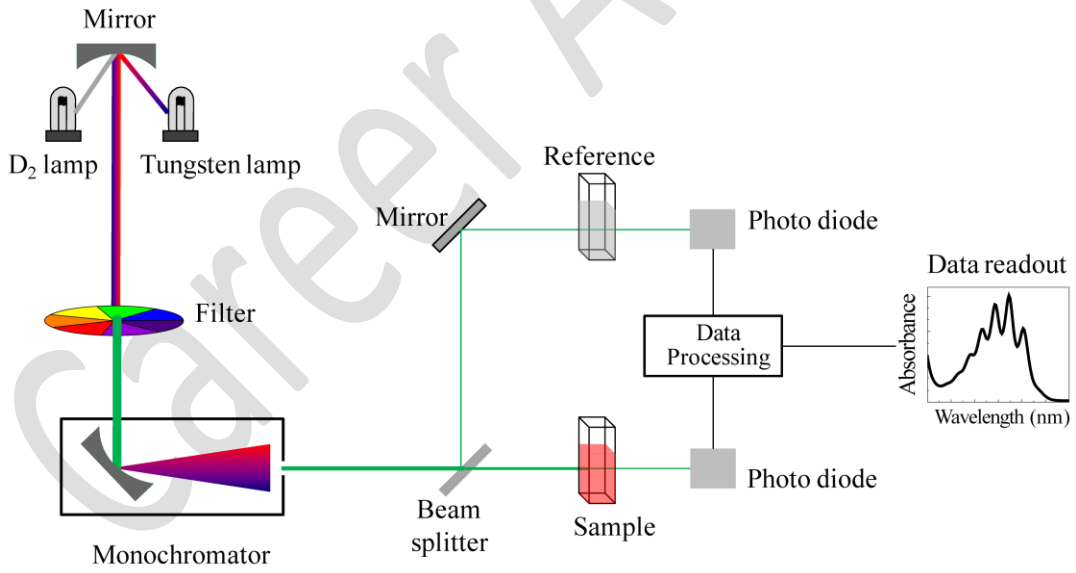
Biochemical samples are usually buffered aqueous solutions, which has two major advantages. Firstly, proteins and peptides are comfortable in water as a solvent, which is also the 'native' solvent. Secondly, in the wavelength interval of UV/Vis (700–200 nm) the water spectrum does not show any absorption bands and thus acts as a silent component of the sample. The absorption spectrum of a chromophore is only partly determined by its chemical structure. The environment also affects the observed spectrum, which mainly can be described by three parameters:

• protonation/deprotonation (pH, RedOx);
• solvent polarity (dielectric constant of the solvent); and
• orientation effects.

Vice versa, the immediate environment of chromophores can be probed by assessing their absorption, which makes chromophores ideal reporter molecules for environmental factors. Four effects, two each for wavelength and absorption changes, have to be considered:
• a wavelength shift to higher values is called red shift or bathochromic effect;
• similarly, a shift to lower wavelengths is called blue shift or hypsochromic effect; • an increase in absorption is called hyperchromicity ('more colour'),
• while a decrease in absorption is called hypochromicity ('less colour').

Spectrophometer - Instrumentation

UV/Vis spectrophotometers are usually dual-beam spectrometers where the first channel contains the sample and the second channel holds the control/reference (buffer) for correction.



Alternatively, one can record the control spectrum first and use this as internal reference for the sample spectrum. The latter approach has become very popular as many spectrometers in the laboratories are computer-controlled, and baseline correction can be carried out using the software by simply subtracting the control from the sample spectrum.

The light source is a tungsten filament bulb for the visible part of the spectrum, and a deuterium bulb for the UV region. Since the emitted light consists of many different wavelengths, a monochromator, consisting of either a prism or a rotating metal grid of high precision called grating, is placed between the light source and the sample.

Wavelength selection can also be achieved by using coloured filters as monochromators that absorb all but a certain limited range of wavelengths. This limited range is called the bandwidth of the filter. Filter-based wavelength selection is used in colorimetry, a method with moderate accuracy, but best suited for specific colorimetric assays where only certain wavelengths are of interest. If wavelengths are selected by prisms or gratings, the technique is called spectrophotometry.

Since borosilicate glass and normal plastics absorb UV light, such cuvettes can only be used for applications in the visible range of the spectrum (up to 350 nm). For UV measurements, quartz cuvettes need to be used. However, disposable plastic cuvettes have been developed that allow for measurements over the entire range of the UV/Vis spectrum.

Most detectors depend on the photoelectric effect, where incident light (photons) liberates electrons from a metal or other material surface. Some sort of external circuitry collects these electrons and measures their number as current. The current is then proportional to the light intensity and therefore a measure of it.

Important requirements for a detector include

(i) high sensitivity to allow the detection of low levels of radiant energy.

(ii) short response time.

(iii)long term stability. and

(iv)(iv) an electronic signal which is easily amplified for typical readout apparatus.

Change in thermal energy, however, is the basis of detection for infrared radiation. Consequently detection devices used for this region are different than those operating in the ultraviolet and visible regions. Ultraviolet and visible radiation detectors are basically of three types; Photocells, Phototubes and Photomultiplier tubes.

Peaks in UV spectra tend to be quite broad, often spanning well over 20 nm at half-maximal height. Typically, there are two things that we look for and record from a UV-Vis spectrum. The

first is λ_{max}, which is the wavelength at maximal light absorbance. As you can see, NAD^+ has λ_{max}=260nm. We also want to record how much light is absorbed at λ_{max}. Here we use a unitless number called **absorbance**, abbreviated 'A'. This contains the same information as the 'percent transmittance' number used in IR spectroscopy, just expressed in slightly different terms. To calculate absorbance at a given wavelength, the computer in the spectrophotometer simply takes the intensity of light at that wavelength *before* it passes through the sample (I_0), divides this value by the intensity of the same wavelength *after* it passes through the sample (I), then takes the $\log_{10}$ of that number:

Applications

UV-vis spectroscopy has many different applications in organic and biological chemistry. One of the most basic of these applications is the use of the Beer - Lambert Law to determine the concentration of a chromophore.

Biochemists and molecular biologists often determine the concentration of a DNA sample by assuming an average value of $\varepsilon = 0.020\ ng^{-1} \times mL$ for double-stranded DNA at its λ_{max} of 260 nm. UV spectroscopy is also very useful in the study of proteins. Proteins absorb light in the UV range due to the presence of the aromatic amino acids tryptophan, phenylalanine, and tyrosine, all of which are chromophores.

Biochemists frequently use UV spectroscopy to study conformational changes in proteins - how they change shape in response to different conditions. When a protein undergoes a conformational shift (partial unfolding, for example), the resulting change in the environment around an aromatic amino acid chromophore can cause its UV spectrum to be altered

UV spectroscopy is also used for the identification of an unknown compound- The spectrum of unknown compound is compared with the spectrum of a reference compound and if both the spectrums coincide then it confirms the identification of the unknown substance.

Career Avenues

Circular Dichroism or CD Spectroscopy

Circular Dichroism, an absorption spectroscopy, uses circularly polarized light to investigate structural aspects of optically active chiral media. It is mostly used to study biological molecules, their structure, and interactions with metals and other molecules.

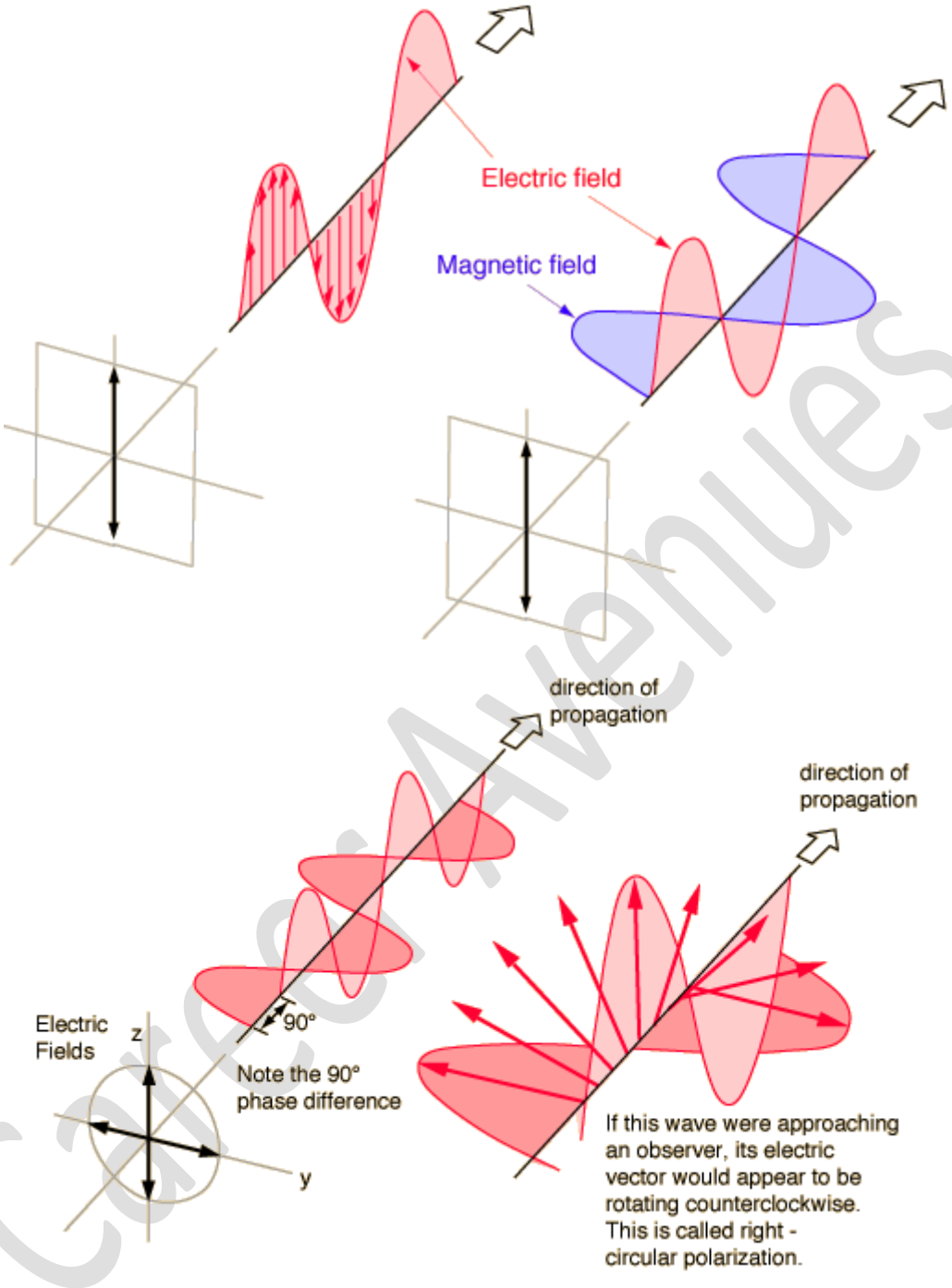
Circular Dichroism (CD) is an absorption spectroscopy method based on the differential absorption of left and right circularly polarized light. Optically active chiral molecules will preferentially absorb one direction of the circularly polarized light. The difference in absorption of the left and right circularly polarized light can be measured and quantified. UV CD is used to determine aspects of protein secondary structure. Vibrational CD, IR CD, is used to study the structure of small organic molecules, proteins and DNA. UV/Vis CD investigates charge transfer transitions in metal-protein complexes.

Linearly and circularly polarised light

Electromagnetic radiation consists of oscillating electric and magnetic fields perpendicular to each other and the direction of propagation. Most light sources emit waves where these fields oscillate in all directions perpendicular to the propagation vector. Linear polarized light occurs when the electric field vector oscillates in only one plane. In circularly polarized light, the electric field vector rotates around the propagation axis maintaining a constant magnitude. When looked at down the axis of propagation the vector appears to trace a circle over the period of one wave frequency (one full rotation occurs in the distance equal to the wavelength). In linear polarized light the direction of the vector stays constant and the magnitude oscillates. In circularly polarized light the magnitude stays constant while the direction oscillates. As the radiation propagates the electric field vector traces out a helix.

The magnetic field vector is out of phase with the electric field vector by a quarter turn. When traced together the vectors form a double helix. Light can be circularly polarized in two directions: left and right. If the vector rotates counterclockwise when the observer looks down the axis of propagation, the light is left circularly polarized (LCP). If it rotates clockwise, it is right circularly polarized (RCP). If LCP and RCP of the same amplitude, they are superimposed on one another and the resulting wave will be linearly polarized.

Career Avenues



Physical principles of CD

Inherently asymmetric chromophores (uncommon) or symmetric chromophores in asymmetric environments will interact differently with right- and left-circularly polarized light resulting in two related phenomena. Circularly-polarized light rays will travel through an optically active medium with different velocities due to the different indices of refraction for right- and left-

circularly polarized light called optical rotation or circular birefringence. The variation of optical rotation as a function of wavelength is called optical rotary dispersion (ORD).

In addition to changing the plane of polarization, an optically active sample also shows unusual absorption behavior. Left-and right-handed polarized components of the incident light are absorbed differently by the sample, which yields a difference in the absorption coefficient $\Delta\varepsilon = \varepsilon_{left} - \varepsilon_{right}$. This latter difference is called circular dichroism (CD). The difference in absorption coefficients $\Delta\varepsilon$ (i.e. CD) is measured in units of cm^2g^{-1}, and is the observed quantity in CD experiments. Historically, results from CD experiments are reported as ellipticity Θ_λ. Normalisation of Θ_λ similar to the ORD yields the molar ellipticity:

It is common practice to display graphs of CD spectra with the molar ellipticity in units of 1° cm^2 $dmol^{-1}$ = $10^o cm^2$ mol^{-1} on the ordinate axis.

Three important conclusions can be drawn:

– ORD and CD are the manifestation of the same underlying phenomenon;

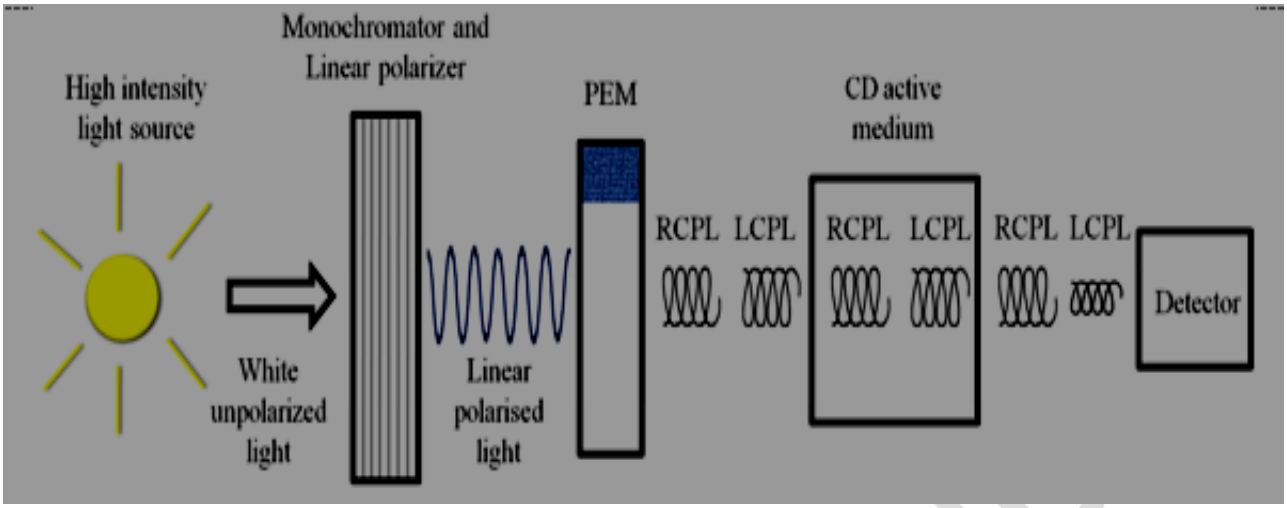
– if an optically active molecule has a positive CD, then its enantiomer will have a negative CD of exactly the same magnitude; and

– the phenomenon of CD can only be observed at wavelengths where the optically active molecule has an absorption band.

Instrumentation

Light is linearly polarized and passed through a monochromator. The single wavelength light is then passed through a modulating device, usually a photoelastic modulator (PEM), which transforms the linear light to circular polarized light. The incident light on the sample switches between LCP and RCP light. As the incident light swtches direction of polarization the absorption changes and the differention molar absorptivity can be calculated.

Career Avenues



Applications

The most widely used application of CD spectroscopy is identifying structural aspects of proteins and DNA. It has been shown that CD spectra between 260 and approximately 180 nm can be analyzed for the different secondary structural types of proteins: alpha helix, parallel and antiparallel beta sheet, turn, and others.

CD spectroscopy can also be used to monitor changes of secondary structure within a sample over time. Frequently, CD instruments are equipped with temperature control units and the sample can be heated in a controlled fashion. As the protein undergoes its transition from the folded to the unfolded state, the CD at a certain wavelength (usually 222 nm) is monitored and plotted against the temperature, thus yielding a thermal denaturation curve which can be used for stability analysis.

Infra-red Spectroscopy or IR Spectroscopy

An invaluable tool in organic structure determination and verification involves the class of electromagnetic (EM) radiation with frequencies between 4000 and 400 cm-1 (wavenumbers). The category of EM radiation is termed infrared (IR) radiation, and its application to organic chemistry known as IR spectroscopy. Radiation in this region can be utilized in organic structure determination by making use of the fact that it is absorbed by interatomic bonds in organic compounds. Chemical bonds in different environments will absorb varying intensities and at varying frequencies. Thus IR spectroscopy involves collecting absorption information and analyzing it in the form of a spectrum -- The frequencies at which there are absorptions of IR

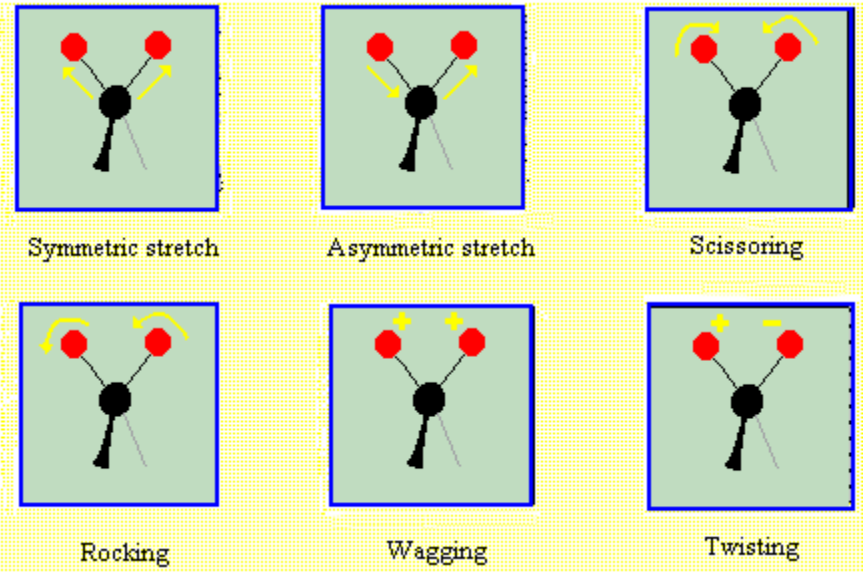
radiation ("peaks" or "signals") can be correlated directly to bonds within the compound in question.

Principle

IR radiation does not have enough energy to induce electronic transitions as seen with UV. Absorption of IR is restricted to compounds with small energy differences in the possible vibrational and rotational states. For a molecule to absorb IR, the vibrations or rotations within a molecule must cause a net change in the dipole moment of the molecule. The alternating electrical field of the radiation (remember that electromagnetic radiation consists of an oscillating electrical field and an oscillating magnetic field, perpendicular to each other) interacts with fluctuations in the dipole moment of the molecule. If the frequency of the radiation matches the vibrational frequency of the molecule then radiation will be absorbed, causing a change in the amplitude of molecular vibration.

Using the molecule HBr as an example, the charge distribution between hydrogen and bromine is not evenly distributed since bromine is more electronegative than hydrogen and has a higher electron density. HBr thus has a large dipole moment and is thus polar. The dipole moment is determined by the magnitude of the charge difference and the distance between the two centers of charge. As the molecule vibrates, there is a fluctuation in its dipole moment; this causes a field that interacts with the electric field associated with radiation. If there is a match in frequency of the radiation and the natural vibration of the molecule, absorption occurs and this alters the amplitude of the molecular vibration. This also occurs when the rotation of asymmetric molecules around their centers results in a dipole moment change, which permits interaction with the radiation field. For the purpose of this discussion, the bonds between atoms can be considered as flexible springs, illustrating the constant vibrational motion within a molecule (Fig. below). Bond vibrations can thus be either stretching or bending (deformation) actions.

Career Avenues



Various vibrational modes of the molecules upon IR absorption

A molecule consisting of (N) number of atoms has a total of 3N degrees of freedom, corresponding to the Cartesian coordinates of each atom in the molecule. In a non-linear molecule, 3 of these degrees of freedom are rotational, 3 are translational and the remainder is fundamental vibrations. In a linear molecule, there are 3 translational degrees of freedom and 2 are rotational. This is because in a linear molecule, all of the atoms lie on a single straight line and hence rotation about the bond axis is not possible.

Mathematically the normal modes for a linear and non linear can be expressed as

Linear Molecules: (3N - 5) degrees of freedom

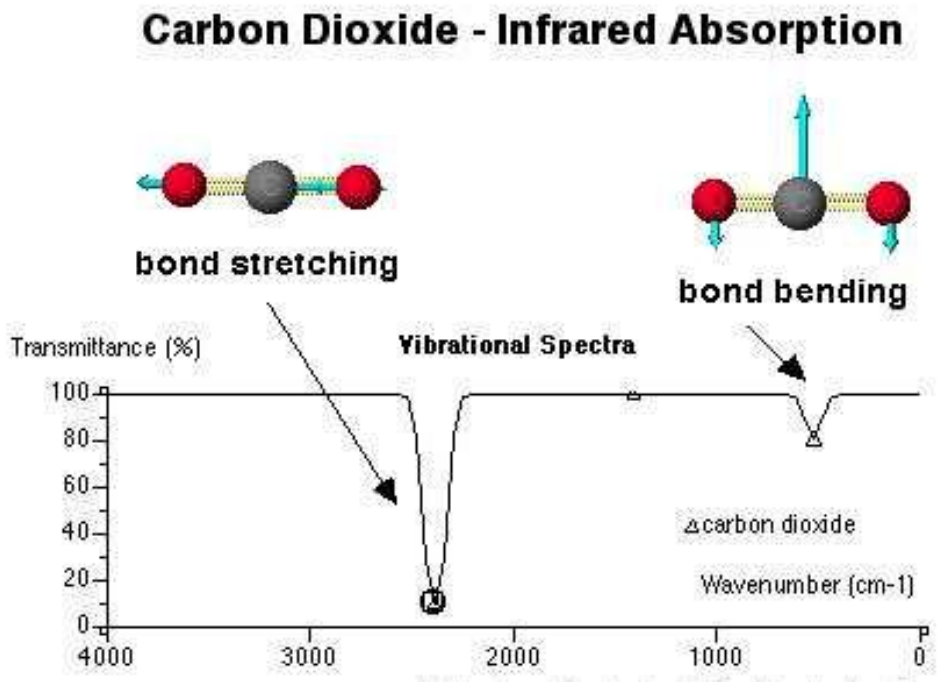
Non-Linear molecules: (3N - 6) degrees of freedom

An infrared spectrum arises from the fact that a molecule absorbs incident light of a certain wavelength which will then be 'missing' from the transmitted light. The recorded spectrum will show an absorption band. The criterion for a band to appear in the infrared spectrum is that the transition to the excited state is accompanied by a change in dipole moment, i.e. a change in charge displacement.

Over time organic chemists have recorded and catalogued the types and locations of IR absorptions produced by a wide variety of chemical bonds in various chemical environments. These data can be quickly referenced through tables of IR absorption ranges and compared to the

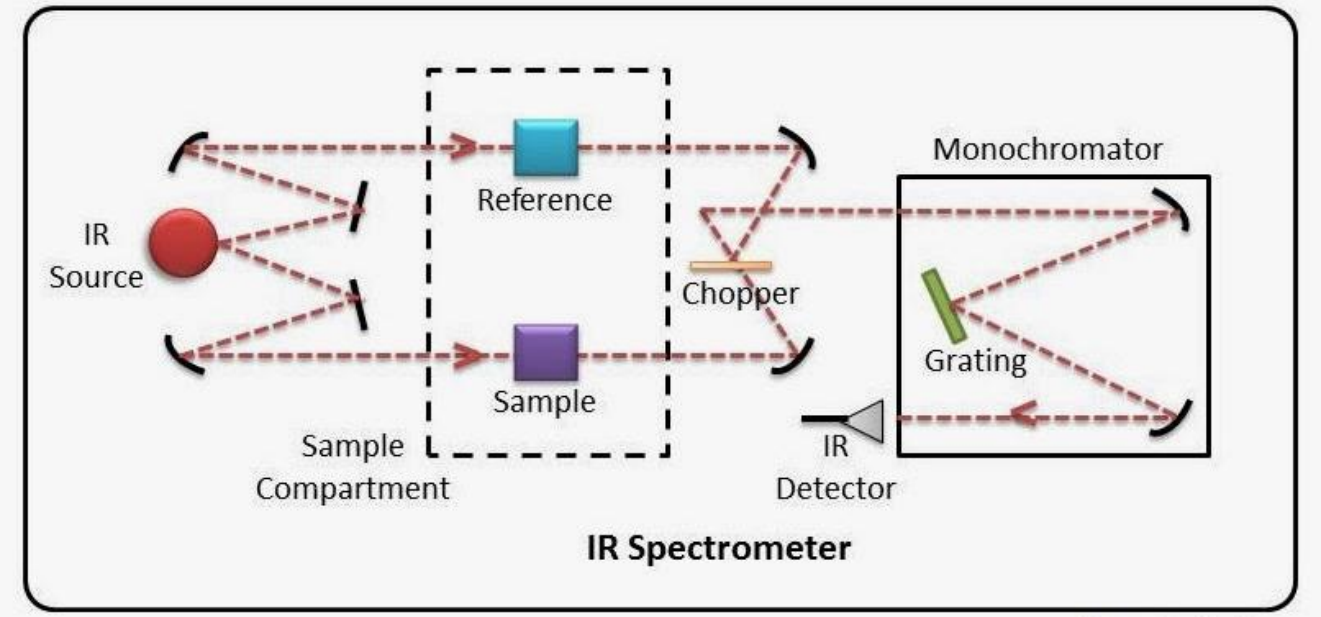
spectrum under consideration. As a general rule, the most important factors determining where a chemical bond will absorb are the bond order and the types of atoms joined by the bond. Conjugation and nearby atoms shift the frequency to a lesser degree. Therefore the same or similar functional groups in different molecules will typically absorb within the same, specific frequency ranges. Consequently tables of IR absorptions are arranged by functional group. The fundamental frequencies observed are characteristic of the functional groups concerned, hence the term fingerprint.

CO_2 is a linear molecule and thus has the formula (3N-5). It has 4 modes of vibration (3(3)-5). CO_2 has 2 stretching modes, symmetric and asymmetric. The CO_2 symmetric stretch is not IR active because there is no change in dipole moment because the net dipole moments are in opposite directions and as a result, they cancel each other. In the asymmetric stretch, O atom moves away from the C atom and generates a net change in dipole moments and hence absorbs IR radiation at 2350 cm^{-1}. The other IR absorption occurs at 666 cm^{-1}. CO_2 has a total of four of stretching and bending modes but only two are seen. Two of its bands are degenerate and one of the vibration modes is symmetric hence it does not cause a dipole moment change because the polar directions cancel each other. The IR spectrum for CO_2 can be seen below in the figure



Instrumentation

The most common source for infrared light is white-glowing zircon oxide or the so-called globar made of silicium carbide with a glowing temperature of 1500 K. The beam of infrared light passes a monochromator and splits into two separate beams: one runs through the sample, the other through a reference made of the substance the sample is prepared in. After passing through a splitter alternating between both beams, they are reflected into the detector. The reference is used to compensate for fluctuations in the source, as well as to cancel possible effects of the solvent. Samples of solids are either prepared in thick suspensions (mulls) such as nujol, and held as layers between NaCl planes or pressed into KBr disks. Non-covalent materials must be used for sample containment and in the optics, as these materials are transparent to infrared. All materials need to be free of water, because of the strong absorption of the O–H vibration.



Applications

IR spectroscopy is a great method for identification of compounds, especially for identification of functional groups. Therefore, we can use group frequencies for structural analysis. Group frequencies are vibrations that are associated with certain functional groups. It is possible to identify a functional group of a molecule by comparing its vibrational frequency on an IR spectrum to an IR stored data bank.

Raman Spectroscopy

A Raman spectrum arises from the analysis of scattered light. The largest part of an incident light beam passes through the sample (transmission). A small part is scattered isotropically, i.e. uniformly in all directions (Rayleigh scatter), and possesses the same wavelength as the incident beam. The Raman spectrum arises from the fact that a very small proportion of light scattered by the sample will have a different frequency than the incident light. As different vibrational states are excited, energy portions will be missing, thus giving rise to peaks at lower frequencies than the incident light (Stokes lines). Notably, higher frequencies are also observed (anti- Stokes lines); these arise from excited molecules returning to ground state. The emitted energy is dumped onto the incident light which results in scattered light of higher energy than the incident light. The criterion for a band to appear in the infrared spectrum is that the transition to the excited state is accompanied by a change in dipole moment, i.e. a change in charge displacement. Conversely, the criterion for a peak to appear in the Raman spectrum is a change in polarisability of the molecule during the transition.

The assignment of peaks in Raman spectra usually requires consideration of peak position, intensity and form, as well as depolarisation. This allows identification of the type of symmetry of individual vibrations, but not the determination of structural elements of a molecule. The depolarisation is calculated as the ratio of two intensities with perpendicular and parallel polarisation with respect to the incident beam. The use of lasers as light source for Raman spectroscopy easily facilitates the use of linearly polarised light. Practically, the Raman spectrum is measured twice. In the second measurement, the polarisation plane of the incident beam is rotated by 90π.

For Raman spectroscopy, aqueous solutions are frequently used, since water possesses a rather featureless weak Raman spectrum. The Raman effect can principally be observed with bright, monochromatic light of any wavelength; however, light between the visible region of the spectrum is normally used due to few unwanted absorption effects. The ideal light source for Raman spectrometers is therefore a laser. Because the Raman effect is observed in light scattered off the sample, typical spectrometers use a 90-degree configuration.

Career Avenues

Differences between IR and Raman Methods

Sr.No.	Raman	IR
01	It is due to the scattering of light by the vibrating molecules.	It is the result of absorption of light by vibrating molecules.
02	The vibration is Raman active if it causes a change in polarizability.	Vibration is IR active if there is change in dipole moment.
03	The molecule need not possess a permanent dipole moment	The vibration concerned should have a change in dipole moment due to that vibration.
04	Water can be used as a solvent	Water cannot be used due to its intense absorption of IR.
05	Sample preparation is not very elaborate, it can be in any state.	Sample preparation is elaborate Gaseous samples can rarely be used.
06	Gives an indication of covalent character in the molecule.	Gives an indication of ionic character in the molecule.
07	Cost of instrumentation is very high	Comparatively inexpensive.

FTIR Spectroscopy

The dispersive infrared spectrometer emerged in the 1940's. This design helped to spread the use of infrared spectroscopy as a common analytical technique for organic compound characterization in laboratories. Fourier transform infrared (FT-IR) spectrometers were developed for commercial use in the 1960's, but at that time tended to be used for advanced research only. This was due to the cost of the instrument components and the large computers required to run them. Gradually, technology advancements in computers and instruments have reduced the cost and enhanced the capabilities of an FT-IR. Today, an FT-IR instrument is the standard for organic compound identification work in modern analytical laboratories.

An FT-IR instrument uses a system called an interferometer to collect a spectrum. The interferometer consists of a source, beamsplitter, two mirrors, a laser and a detector. The energy

goes from the source to the beamsplitter which splits the beam into two parts. One part is transmitted to a moving mirror; one part is reflected to a fixed mirror. The moving mirror moves back and forth at a constant velocity. This velocity is timed according to the very precise laser wavelength in the system which also acts as an internal wavelength calibration. The two beams are reflected from the mirrors and recombined at the beamsplitter. The beam from the moving mirror has traveled a different distance than the beam from the fixed mirror. When the beams are combined an interference pattern is created, since some of the wavelengths recombine constructively and some destructively. This interference pattern is called an interferogram. This interferogram then goes from the beamsplitter to the sample, where some energy is absorbed and some is transmitted. The transmitted portion reaches the detector. The detector reads information about every wavelength in the infrared range simultaneously.

To obtain the infrared spectrum, the detector signal is sent to the computer, and an algorithm called a Fourier transform is performed on the interferogram to convert it into a single beam spectrum. A reference or "background" single beam is also collected without a sample and the sample single beam is ratio-ed to the background single beam to produce a transmittance or "%T" spectrum. This transmittance spectrum can be converted to absorbance by taking the negative $\log_{10}$ of the data points.

The x-axis of the FT-IR spectrum is displayed in "wave numbers", or cm^{-1}. This unit is a product of the Fourier transform algorithm operating on the interferogram and is the reciprocal of the actual wavelength of light measured in centimeters at a point in the infrared spectrum.

Applications

FT–IR is increasingly used for analysis of peptides and proteins. The peptide bond gives rise to nine characteristic bands, named amide A, B, I, II, III, . . ., VII. The amide I (1600–1700 cm^{-1}) and amide II (1500–1600 cm^{-1}) bands are the major contributors to the protein infrared spectrum. Both bands are directly related to the backbone conformation and have thus been used for assessment of the secondary structure of peptides and proteins. The interpretation of spectra of molecules with a large number of atoms usually involves deconvolution of individual bands and

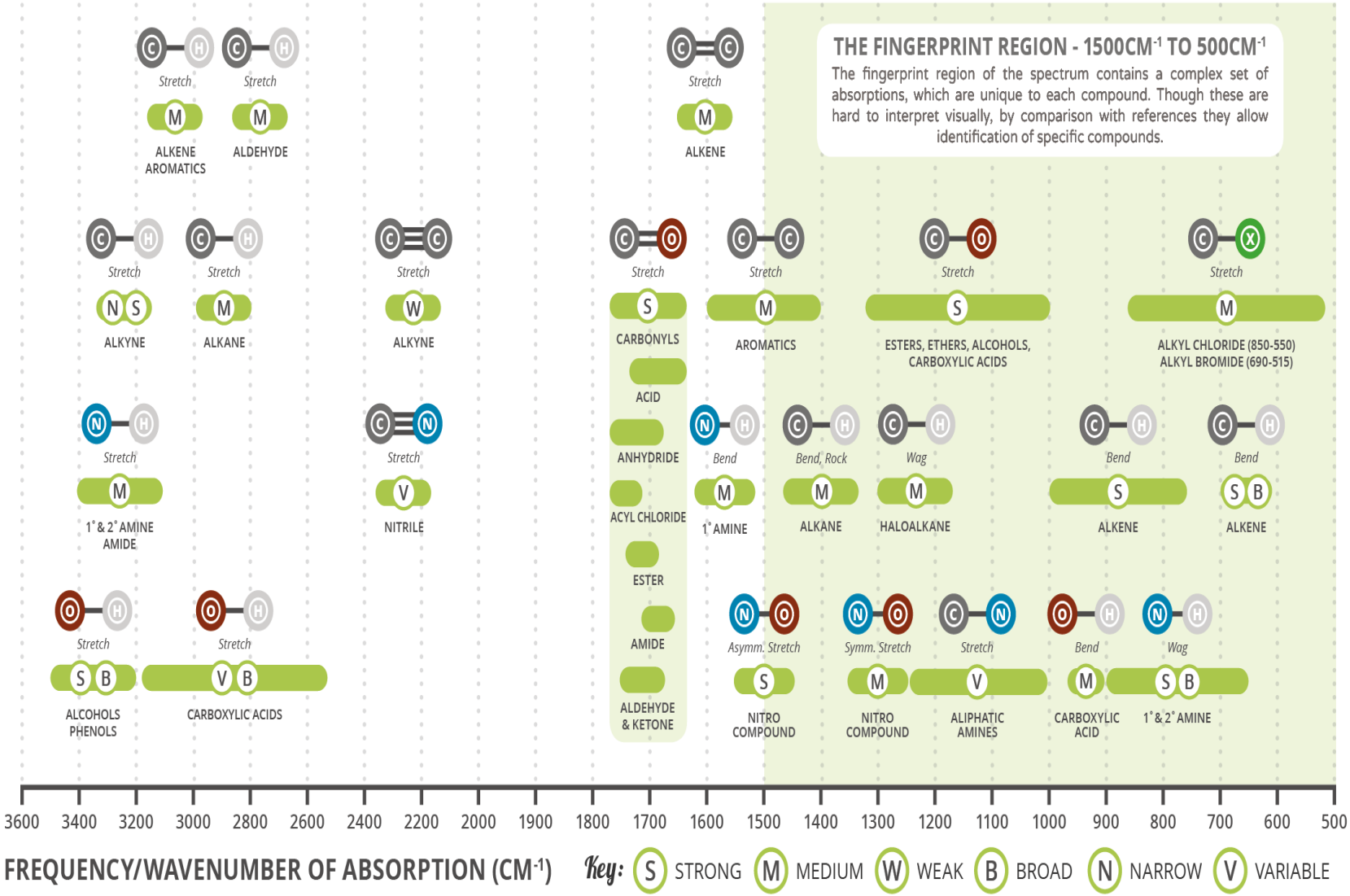
second derivative spectra. Time-resolved FT–IR (trFT–IR) enables the observation of protein reactions at the sub-millisecond timescale.

PROPERTIES	DISPERSIVE IR	FTIR
1.Components	Many moving parts result in mechanical slippage.	Only mirror moves during an experiment
2.Calibration	Calibration against reference spectra required to measure frequency	Use of laser provides high frequency precision (to 0.01 cm-1 (Connes' advantage)
1.Stray light	Stray light within instrument causes spurious readings	Stray light does not affect detector, since all signals are modulated.
2.Resolution	In order to improve resolution, only small amount of IR beam is allowed to pass through the slits.	Much larger beam aperture used; higher energy throughput (Throughput or Jacquinot's advantage).
3.No. of Frequency	Only narrow-frequency radiation falls on the detector at any one time.	All frequencies of radiation fall on detector simultaneously; improved S/N obtained quickly (Multiplex or Fellgett's advantage)
4.Scan speed	Slow scan speeds make dispersive instruments too slow for monitoring systems undergoing rapid change (e.g. GC effluents)	Rapid scan speeds permit monitoring samples undergoing rapid change.
5.Source effects	Sample subject to thermal effects from the source due to length of scan time.	Short scan times, hence sample is not subject to thermal effects.
6. IR from sample itself	Any emission of IR radiation by sample will fall on detector due to the conventional positioning of the sample before the monochromator.	Any emission of IR radiation by the sample will not be detected.
7.Advantage of Beam optics	Double beam optics permit continuous real-time background substraction.	Single beam optics; background spectrum collected separately in time from sample spectrum. Can result in error if background spectra not collected frequently.

Career

ANALYTICAL CHEMISTRY - INFRARED SPECTROSCOPY

Commonly referred to as IR spectroscopy, this technique allows chemists to identify characteristic groups of atoms (functional groups) present in molecules.



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Mass Spectroscopy

Mass spectrometry is an analytical tool used for measuring the molecular mass of a sample. For large samples such as biomolecules, molecular masses can be measured to within an accuracy of 0.01% of the total molecular mass of the sample i.e. within a 4 Daltons (Da) or atomic mass units (amu) error for a sample of 40,000 Da. This is sufficient to allow minor mass changes to be detected, e.g. the substitution of one amino acid for another, or a post-translational modification. For small organic molecules the molecular mass can be measured to within an accuracy of 5 ppm or less, which is often sufficient to confirm the molecular formula of a compound, and is also a standard requirement for publication in a chemical journal. Structural information can be generated using certain types of mass spectrometers, usually those with multiple analysers which are known as tandem mass spectrometers. This is achieved by fragmenting the sample inside the instrument and analysing the products generated. This procedure is useful for the structural elucidation of organic compounds and for peptide or oligonucleotide sequencing.

Principle

A mass spectrometer generates multiple ions from the sample under investigation, it then separates them according to their specific mass-to-charge ratio (m/z), and then records the relative abundance of each ion type. The first step in the mass spectrometric analysis of compounds is the production of gas phase ions of the compound, basically by electron ionization. This molecular ion undergoes fragmentation. Each primary product ion derived from the molecular ion, in turn, undergoes fragmentation, and so on. The ions are separated in the mass spectrometer according to their mass-to-charge ratio, and are detected in proportion to their abundance. A mass spectrum of the molecule is thus produced. It displays the result in the form of a plot of ion abundance versus mass-to-charge ratio. Ions provide information concerning the nature and the structure of their precursor molecule. In the spectrum of a pure compound, the molecular ion, if present, appears at the highest value of m/z (followed by ions containing heavier isotopes) and gives the molecular mass of the compound.

Instrumentation

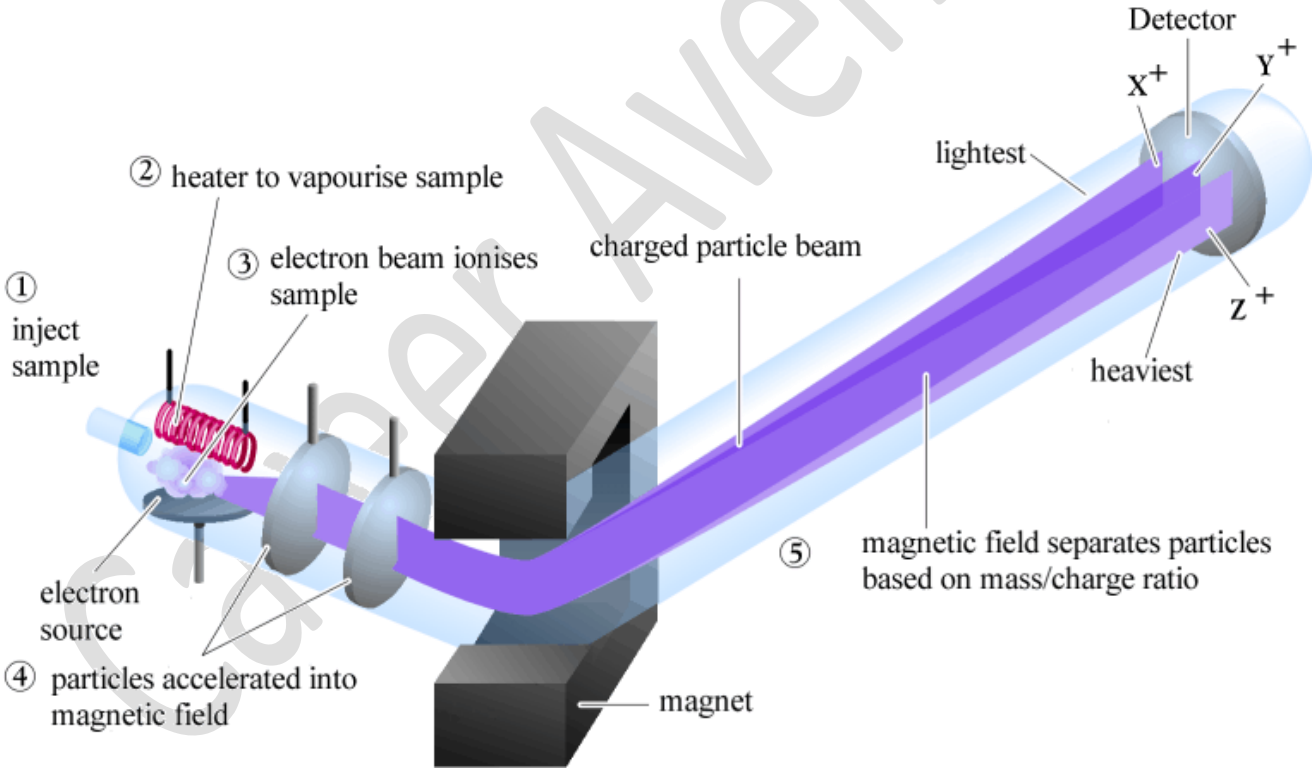
All mass spectrometers are basically similar (Fig. 9.1). They consist of the following:

• A high vacuum system (10^6 torr or 1 mtorr): These include turbomolecular pumps, diffusion pumps and rotary vane pumps.

• A sample inlet: This comprises a sample or target plate; a high-performance liquid chromatography (HPLC), gas chromatography (GC) or capillary electrophoresis system; solids probe; electron impact or direct chemical ionisation chamber.

• An ion source (to convert molecules into gas-phase ions): This can be MALDI; ESI; fast atom bombardment (FAB); electron impact or direct chemical ionisation. • A mass filter/analyser: This can be: TOF; quadrupole; ion trap; magnetic sector or ion cyclotron Fourier transform (the last is also actually a detector).

• A detector: This can be a conversion dynode, electron multiplier, microchannel plate or array detector.



The separation of ions according to their m/z values is as follows. They are accelerated to a constant velocity in a vacuum by a series of negatively charged electrostatic plates and are then deflected from their original trajectory by a magnetic field (Figure above). For ions carrying the same charge the amount of deflection is dependent on the mass of the ion, so that the ions of

smallest mass undergo the greatest degree of deflection. By varying either the magnetic field or the accelerating voltage the ion trajectories may be varied so that fragment ions of a given m/z value impinge on the collector slit of the detector. The number of ions of a given mass impinging on the detector is a measure of the abundance of that particular ion. Any material that can be ionised and whose ions can exist in the gas phase can be investigated by MS

Common applications and fields that use mass spectrometry

Field of Study	Applications
Proteomics	– Determine protein structure, function, folding and interactions – Identify a protein from the mass of its peptide fragments – Detect specific post-translational modifications throughout complex biological mixtures – Quantitate (relative or absolute) proteins in a given sample – Monitor enzyme reactions, chemical modifications and protein digestion
Drug Discovery	– Determine structures of drugs and metabolites – Screen for metabolites in biological systems
Clinical Testing	– Perform forensic analyses such as confirmation of drug abuse – Detect disease biomarkers (e.g., newborns screened for metabolic diseases)
Genomics	– Sequence oligonucleotides
Environment	– Test water quality or food contamination
Geology	– Measure petroleum composition – Perform carbon dating

NMR or Nuclear Magnetic Resonance spectroscopy

NMR or nuclear magnetic resonance spectroscopy is a technique used to determine a compound's unique structure. It identifies the carbon-hydrogen framework of an organic

compound. Using this method and other instrumental methods including infrared and mass spectrometry, scientists are able to determine the entire structure of a molecule.

Principle

The atomic nucleus is charged particle which keeps spinning because of which it generates a magnetic field. Such a nucleus is then supposed to possess an angular momentum represented by a spin quantum number, I. which is assigned half integral values, O 1/2, 1, 3/2, ... 9/2 depending on the particular nucleus. Without an external applied magnetic field, the nuclear spins are random and spin in random directions. But, when an external magnetic field is present, the nuclei align themselves either with or against the field of the external magnet (Figure below). Nuclei possessing odd numbers of both neutrons and protons have an integral spin number of I = 1. Thus, the charge on them is distributed non-symmetrically (e.g .. ^{2}H. ^{14}N). Nuclei possessing even numbers of both protons and neutrons (^{l2}C. ^{16}0. ^{32}S. etc.) are not measured in NMR experiments because they do not possess an angular momentum (I = 0) and do not exhibit magnetic properties.

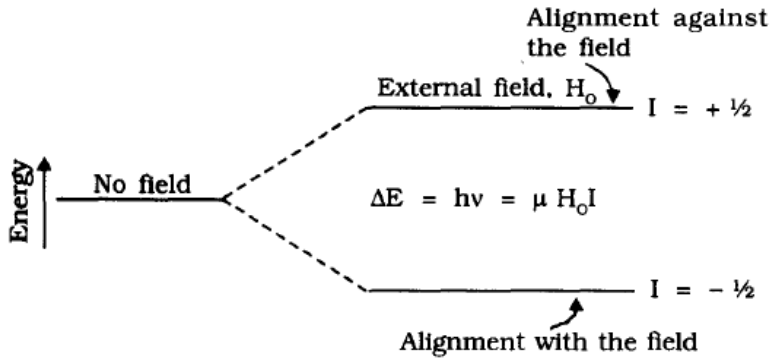
Magnetic nuclei assume discrete orientations with corresponding energy levels under the influence of external magnetic fields. The value of I determines the number of quantized energy

$$I, I - 1, I - 2, \ldots \ldots - I$$

Any given nucleus can have 2I + 1 possible levels or orientations. Thus for ^{1}H, ^{15}N, ^{19}F and ^{31}P, with I = $\frac{1}{2}$ only two orientations are available. These levels are described as (i) aligned with the applied field (lower energy), and (ii0 antiparallel to the field (higher energy). These two levels will have the energy of $-\mu H_o$ and $+\mu H_o$ for low and high energy respectively, where H_o is the intensity of the applied magnetic field and μ is the nuclear magnetic moment. The difference in energy ΔE is then equal to $2\mu H_o$.

The general relationship is as follows:

$$\Delta E = \mu H_o / I$$



In order to change from low energy state to high energy state, such nuclei must absorb the appropriate quantum of energy. In a magnetic field of several hundred millitesla (several thousand gauss) such nuclei absorb radiation in the radiowave region of the electromagnetic spectrum. This phenomenon is known as nuclear resonance or nuclear magnetic resonance. The above equation can, thus, also be written as

$$\Delta E = h\nu = 2\mu H_o \text{ (for cases where } I = \frac{1}{2})$$

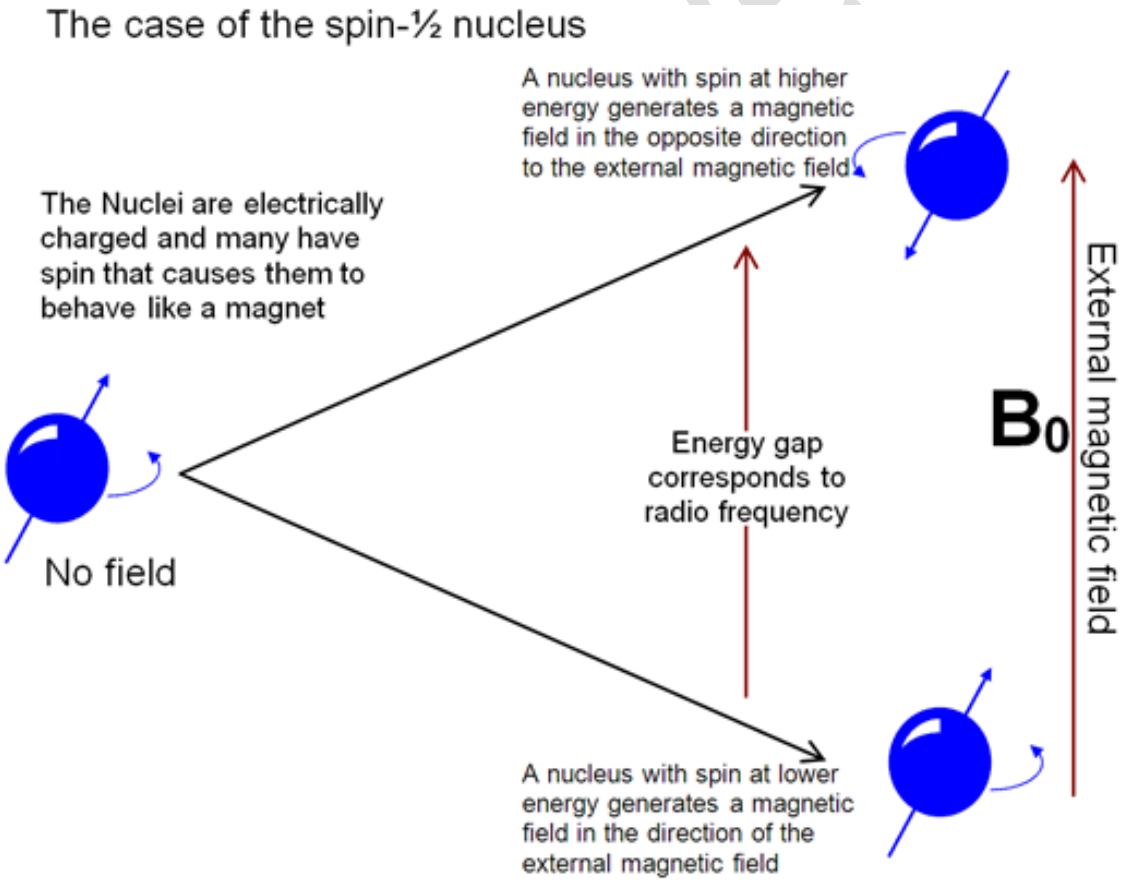
If we combine the two equations, we get

$$2\pi\nu = \frac{2\pi\mu H_o}{hI} = \gamma H_o$$

where 2π serves to convert linear frequency to angular units of frequency. γ, which is obviously equal to $2\pi\mu/hI$ is known as the *gyromagnetic ratio* and is a characteristic property of the nucleus. Another mathematical expression for γ can be derived from the above relationships:

$$\text{Gyromagnetic ratio} = \gamma = \frac{2\pi\nu}{H_o}$$

This relationship is a fundamental equation for n.m.r. The relationship obviously describes the ratio between the frequency and the field. The ratio is sometimes termed "resonance condition".



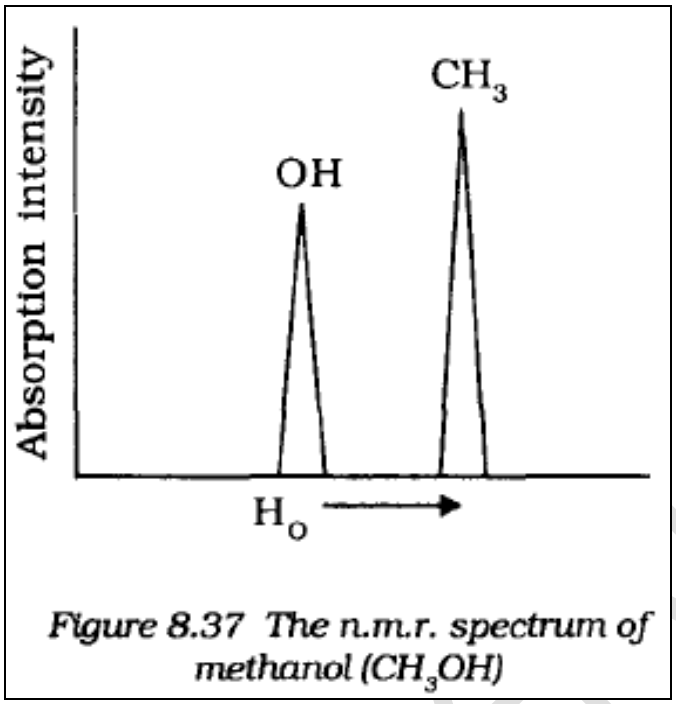


Figure 8.37 The n.m.r. spectrum of methanol (CH_3OH)

Obviously all nuclei do not absorb at the same applied field but the absorption depends upon the magnetic field which a particular nucleus feels. Clearly, the effective field strength is different for different nuclei as one nucleus will have slightly different environment from any other nucleus. Thus, at a given radio frequency, different nuclei will require slightly different magnetic field strength to give rise to the same effective magnetic strength which causes absorption. It is, therefore, a normal practice to measure the absorbance of a sample at a monochromatic radiowave frequency while varying the magnetic field intensity. n.m.r. spectra are thus a plot of energy absorbed against the magnetic field strength applied. Such a plot is shown in Figure 8.37. The number signals obtained on such a plot are indicative of the different sets of equivalent nuclei or protons. For example, in Figure 8.37 there are two signals which are indicative of two sets of protons existing in different environment, viz., CH_3, and – OH. The area under the peak in such plots is directly proportional to the number of nuclei or protons which are responsible for the resonance. For an example one can refer to Figure 8.37 again; the peak corresponding to CH_3 is much higher than the peak corresponding to – OH.

Applications

Traditionally, NMR spectroscopy is the main method of structure determination for organic compounds.

The structures of proteins up to a mass of about 50 kDa can be determined with biomolecular NMR spectroscopy.

The basic principles of NMR can be applied to imaging of live samples. Because the proton is one of the more sensitive nuclides and is present in all biological systems abundantly, ^{1}H resonance is used almost exclusively in the clinical environment. The most important compound in biological samples in this context is water. It is distributed differently in different tissues, but constitutes about 55% of body mass in the average human. Magnetic resonance imaging (MRI) can be applied to large volumes in whole living organisms and has a central role in routine clinical imaging of large-volume soft tissues.

Career Avenues

Microarray

Molecular Biology research evolves through the development of the technologies used for carrying them out. It is not possible to research on a large number of genes using traditional methods. DNA Microarray is one such technology which enables the researchers to investigate and address issues which were once thought to be non traceable. One can analyze the expression of many genes in a single reaction quickly and in an efficient manner. DNA Microarray technology has empowered the scientific community to understand the fundamental aspects underlining the growth and development of life as well as to explore the genetic causes of anomalies occurring in the functioning of the human body.

A typical microarray experiment involves the hybridization of an mRNA molecule to the DNA template from which it is originated. Many DNA samples are used to construct an array. The amount of mRNA bound to each site on the array indicates the expression level of the various genes. This number may run in thousands. All the data is collected and a profile is generated for gene expression in the cell.

Microarray provides a basis to genotype thousands of different loci at a time, which is useful for association and linkage studies to isolate chromosomal regions related to a particular disease. This array also can be used to locate chromosomal aberrations related to cancer, such as segments of allelic imbalance, which can be identified by loss of heterozygosity. By comparative genomic hybridization techniques on genomic DNA, amplified or deleted regions in the chromosomes can be identified, such as in the case of oral cancer.

Technique

Gene microarray technology rests on the ability to deposit many (tens of thousands) different DNA sequences on a small surface, usually a glass slide (often referred to as a "chip"). The different DNA fragments are arranged in rows and columns such that the identity of each fragment is known through its location on the array. Two types of microarrays are gene expression microarray and tissue microarray (TMA). Techniques like Northern blot and reverse transcriptase-polymerase chain reaction (RT-PCR) allow testing for only a few genes per

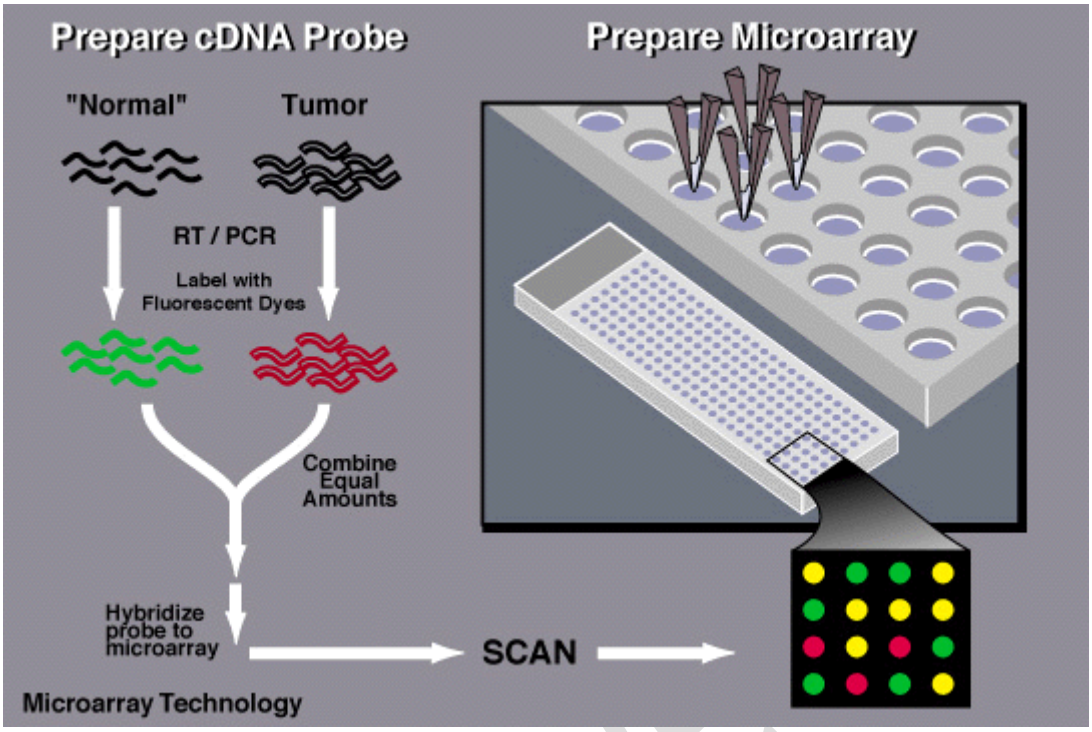
experiment. But microarray or "global expression profiling" not only looks at orders of magnitude more genes than was possible previously, but also has the advantage that the genes examined are not influenced by preselection of genes.

Microarray Principle

mRNA is an intermediary molecule which carries the genetic information from the cell nucleus to the cytoplasm for protein synthesis. Whenever some genes are expressed or are in their active state, many copies of mRNA corresponding to the particular genes are produced by a process called transcription. These mRNAs synthesize the corresponding protein by translation. So, indirectly by assessing the various mRNAs, we can assess the genetic information or the gene expression. This helps in the understanding of various processes behind every altered genetic expression. Thus, mRNA acts as a surrogate marker. Since mRNA is degraded easily, it is necessary to convert it into a more stable cDNA form. Labeling of cDNA is done by fluorochrome dyes Cy3 (green) and Cy5 (red). The principle behind microarrays is that complementary sequences will bind to each other.

The unknown DNA molecules are cut into fragments by restriction endonucleases; fluorescent markers are attached to these DNA fragments. These are then allowed to react with probes of the DNA chip. Then the target DNA fragments along with complementary sequences bind to the DNA probes. The remaining DNA fragments are washed away. The target DNA pieces can be identified by their fluorescence emission by passing a laser beam. A computer is used to record the pattern of fluorescence emission and DNA identification. This technique of employing DNA chips is very rapid, besides being sensitive and specific for the identification of several DNA fragments simultaneously.

Career Avenues



How to set up for a Microarray experiment.

There are three parts to the experiments:

I. Preparing a DNA chip (or buying one)

II. Carrying out the reaction (isolate mRNA, process the mRNA and hybridize to DNA chip)

III. Collecting and analyzing the results

I. Preparing a DNA chip

- You must know the sequence (or part of the sequence) of all the genes you want to study.
- You must synthesize a piece of each gene as a short single strand DNA (oligonucleotide) about 25 bases long. This 25-base sequence will be sufficient to hybridize specifically to a complementary sequence of RNA or DNA.
- You must fix the short DNA sequences for each gene on a tiny spot on a glass slide (usually the DNA fragments are synthesized directly on the slide or can be linked to the slide after synthesis). You can fit 10,000 different genes (10,000 tiny spots) on a single slide (DNA chip). Each spot represent one single gene and has billions of copies of the same 25-base sequence.

- **II. Carrying out the reaction**
- You first need to choose what cells or tissue you want to study and grow them under specific conditions.
- For example if you want to find out:
- How gene expression changes when cells are irradiated with UV light or exposed to a toxic chemical?
- What happens to the activity of various genes in yeast cells when the cells are shifted from 25 ºC to 37 ºC?
- What genes are only expressed in fish embryos and not in adult fish?
- What genes have increased expression in cancer cells compared to normal cells?
- You isolate total mRNA from the cells you are studying (both normal and treated or cancerous cells)

- You want to be able to count how many copies of each mRNA there is in the cell in order to determine the activity of each gene. So you need to label the mRNA (so that you can detect it and count how much of each mRNA sequence there is). The easiest way to do that is to reverse transcribe the mRNA into cDNA and introduce modified fluorescent bases into the DNA. In order to distinguish later between the mRNAs coming from the normal cells and those coming from the treated or diseased cells you label the two cDNAs populations with different fluorescent colors (such as green and red)

- You mix the two populations of fluorescently labeled cDNAs and hybridize them to the same DNA chip in a special chamber. You then wash away all the unbound cDNA.

III. Collecting and analyzing the results

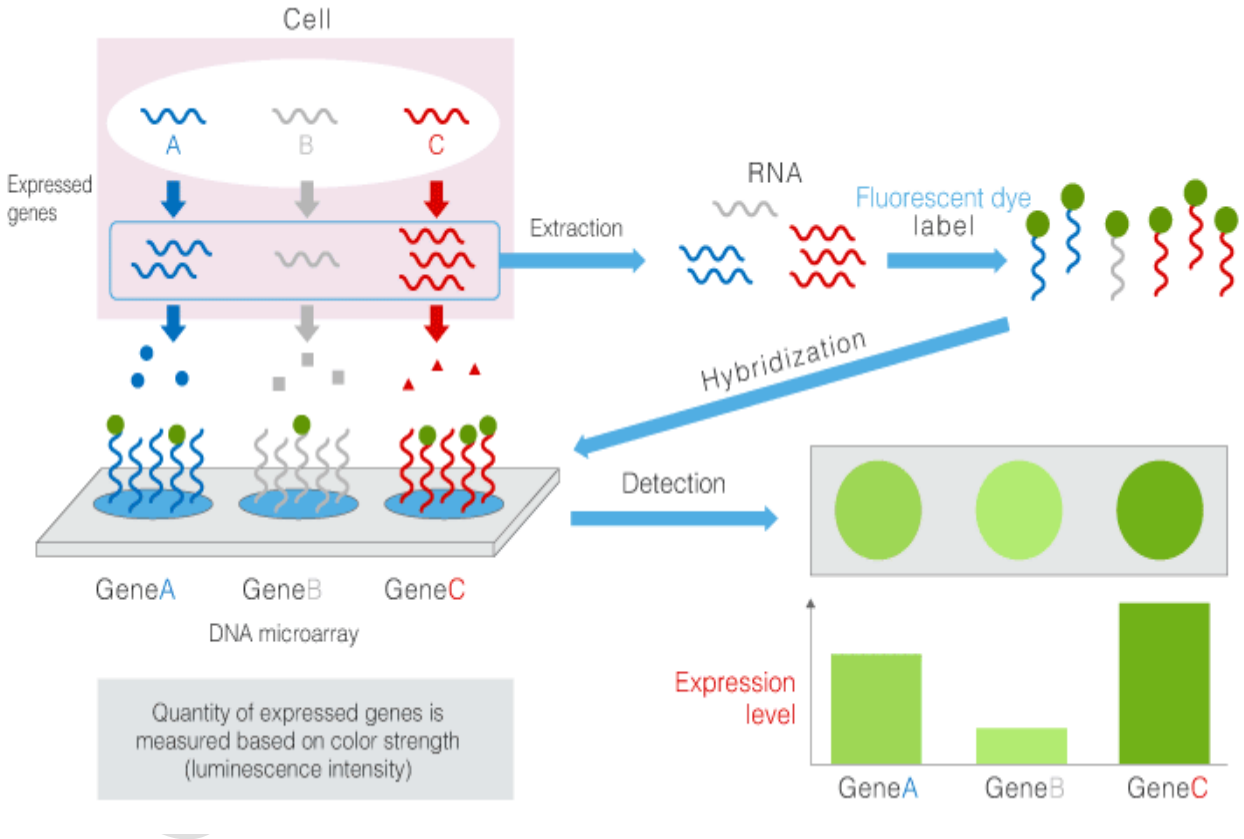
- Using a scanner hooked to a computer you measure how much of each labeled cDNA (green and red) is bound to each spot on the slide. The more label on a spot the more active the gene is. BUT remember! You are comparing two populations of cells. So what counts is not how much total cDNA binds to each spot but how much of each color bind to the same spot. The color intensities are subtracted from each other.

Career Avenues

• By measuring the difference of intensity between the two colors for each spot you can determine for example if a gene is more active or less active in a cancer cell compared to a normal cell.

• You generate a grid that uses a color code to show the change in activity for each gene

• You try to figure out if there is a pattern in gene expression that is clearly visible, and reproducible.



DNA microarray Technology flow chart

Advantages of the DNA microarray technology;

• Can follow the activity of MANY genes at the same time.

• Can get a lot of results fast

• Can COMPARE the activity of many genes in diseased and healthy cells

• Can categorize diseases into subgroups

Drawback:

• Too much data all at once. Can take quite a while to analyze all the results.

• The results may be too complex to interpret

• The results are not always reproducible

• The results are not always quantitative enough

• The technology is still too expensive

Career Avenues

Chromatography

Biomolecules are purified using chromatography techniques that separate them according to differences in their specific properties. Proteins can be selectively precipitated by the addition of certain salts. A wide range of chromatographic procedures makes use of differences in size, binding affinities, charge and other properties. These include ion-exchange chromatography, size-exclusion, affinity, and high-performance liquid chromatography.

Property	Technique
Charge	Ion exchange chromatography (IEX), chromatofocusing (CF)
Size	Gel filtration(GF), also called size exclusion
Hydrophobicity	Hydrophobic interactionchromatography (HIC), Reversed phase chromatography (RPC)
Biorecognition (ligand specificity)	Affinity chromatography (AC)

COLUMN CHROMATOGRAPHY

The most powerful methods for fractionating proteins make use of **column chromatography,** which takes advantage of differences in protein charge, size, binding affinity, and other properties. A porous solid material with appropriate chemical properties (the stationary phase) is held in a column, and a buffered solution (the mobile phase) percolates through it. The protein-containing solution, layered on the top of the column, percolates through the solid matrix as an ever-expanding band within the larger mobile phase. Individual proteins migrate faster or more slowly through the column depending on their properties. The standard elements of a chromatographic column include a solid, porous material supported inside a column, generally made of plastic or glass, as shown in Figure 1. The solid material (matrix) makes up the stationary phase through which flows a solution, the mobile phase. The solution that passes out of the column at the bottom (the effluent) is constantly replaced by solution supplied from a reservoir at the top. The protein solution to be separated is layered on top of the column and allowed to percolate into the solid matrix. Additional solution is added on top. The protein solution forms a band within the mobile phase that is initially the depth of the protein solution applied to the column. As proteins migrate through the column, they are retarded to different degrees by their different interactions with the matrix material. The overall protein band thus widens as it moves through the column. Individual types of proteins (such as A, B, and C) gradually separate from each other, forming bands within the broader protein band. Separation improves (resolution increases) as the length of the column increases. However, each individual protein band also broadens with time due to diffusional spreading, a process that decreases

resolution. In this example, protein A is well separated from B and C, but diffusional spreading prevents complete separation of B and C under these conditions.

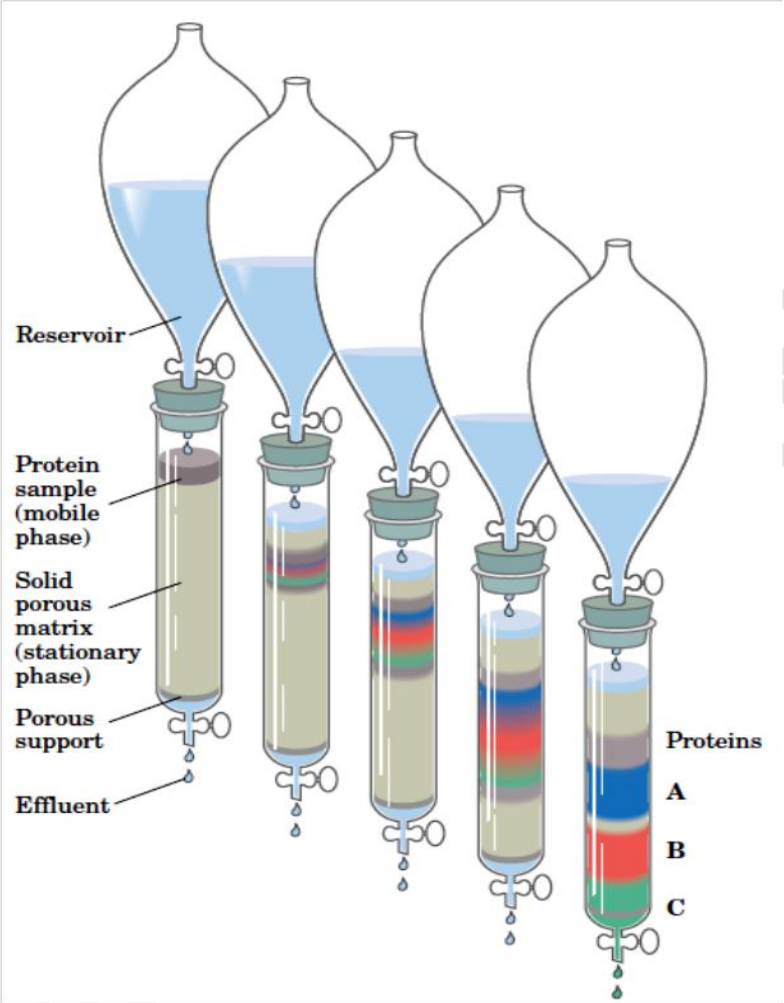


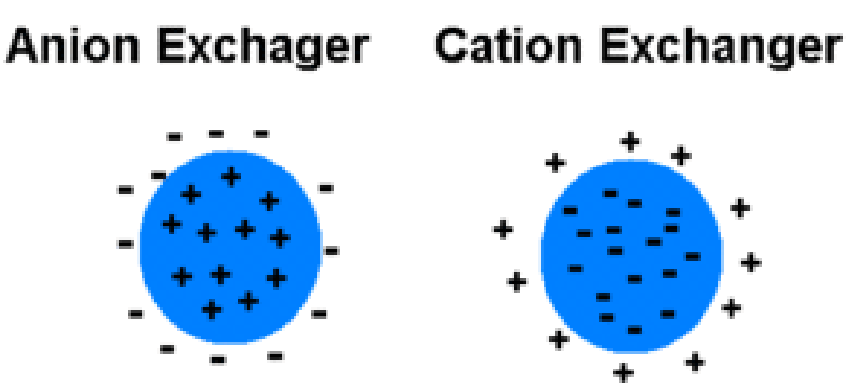
Figure 1 – Column Chromatography

ION EXCHANGE CHROMATOGRAPHY (IEX)

IEX is based on the reversible interaction between a charged protein and an oppositely charged chromatography medium. Biomolecules with even small differences in net surface charge can be separated, and very high resolution is obtained by choosing the optimal ion exchanger and separation conditions.

The net surface charge of a protein varies according to the surrounding pH. Typically, when the pH is above its iso electric point (pI), a protein will bind to a positively charged anion exchanger. Below its pI, a protein will bind to a negatively charged cation exchanger. Anion and cation exchangers are classified as strong or weak, depending on how much the ionization state of the functional groups vary with pH. A strong ion exchanger has the same

charge density on its surface over a broad pH range, whereas the charge density of a weak ion exchanger changes with pH. The selectivity and the capacity of a weak ion exchanger are different at different pH values. We recommend to start with a strong ion exchanger (e.g., Q, S, or SP), and then to try a weak ion exchanger (e.g., DEAE or CM) if the selectivity is unsatisfactory.

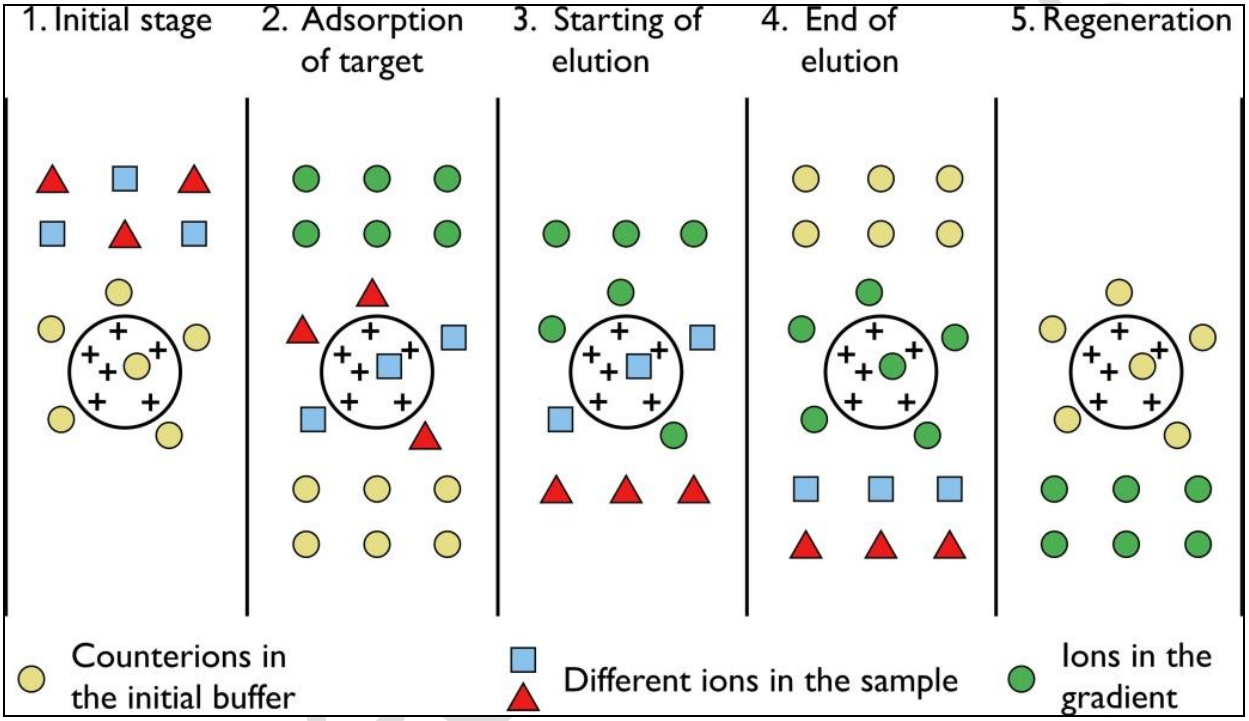


Resin Type	Cation Exchanger	Anion Exchanger
Net charge of molecule of interest	+	−
Charge of resin	−	+
Running conditions	0.5–1.5 pH units below the pI of the molecule of interest	0.5–1.5 pH units above the pI of the molecule of interest

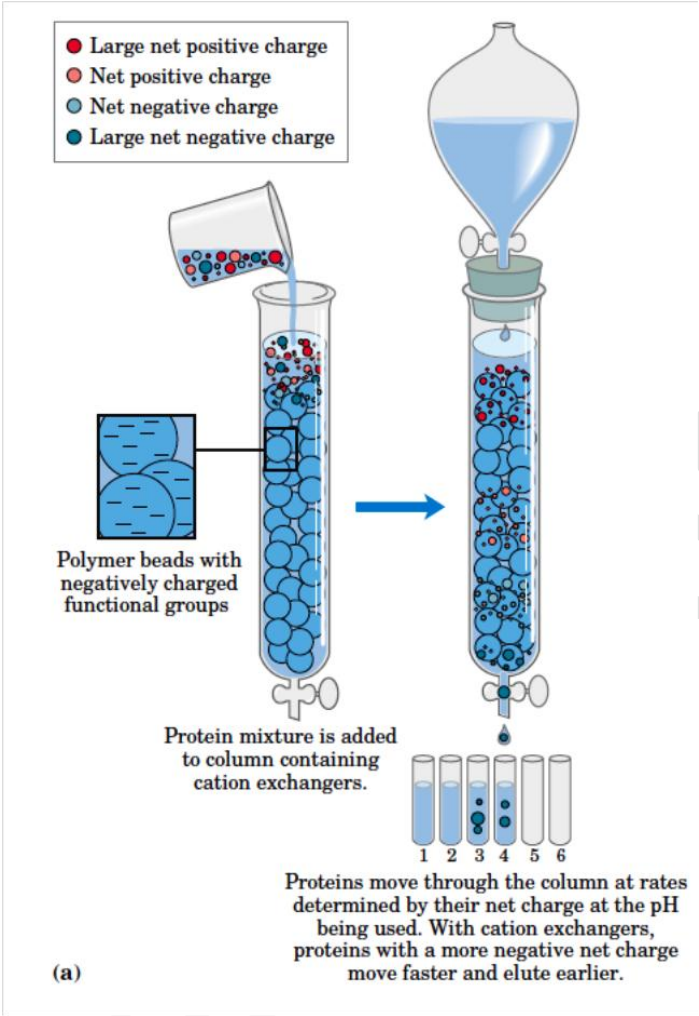
CATION-EXCHANGE CHROMATOGRAPHY

The solid matrix has negatively charged groups. In the mobile phase, proteins with a net positive charge migrate through the matrix more slowly than those with a net negative charge, because the migration of the former is retarded more by interaction with the stationary phase. The two

types of protein can separate into two distinct bands. The expansion of the protein band in the mobile phase (the protein solution)is caused both by separation of proteins with different properties and by diffusional spreading. As the length of the column increases, the resolution of two types of protein with different net charges generally improves. However, the rate at which the protein solution can flow through the column usually decreases with column length. And as the length of time spent on the column increases, the resolution can decline as a result of diffusional spreading within each protein band.



Stages in ion-exchange chromatography



Ion-exchange chromatography

Ion-exchange chromatography exploits differences in the sign and magnitude of the net electric charges of proteins at a given pH. The column matrix is a synthetic polymer containing bound charged groups; those with bound anionic groups are called **cation exchangers,** and those with bound cationic groups are called **anion exchangers.** Ion-exchange chromatography on a cation exchanger is shown here. The affinity of each protein for the charged groups on the column is affected by the pH (which determines the ionization state of the molecule) and the concentration of competing free salt ions in the surrounding solution. Separation can be optimized by gradually changing the pH and/or salt concentration of the mobile phase so as to create a pH or salt gradient.

In IEX, proteins bind as they are loaded onto the column at low ionic strength. The conditions are then altered so that bound substances are eluted differentially. Elution is usually performed by increasing salt concentration or changing pH in a gradient, or stepwise. The most common salt is NaCl, but other salts can also be used.

Type	Functional groups	Functional group name	Matrices
Weakly acidic (cation exchanger)	$-COO^-$	Carboxy	Agarose
	$-CH_2COO^-$	Carboxymethyl	Cellulose
			Dextran
			Polyacrylate
Strongly acidic (cation exchanger)	$-SO_3^-$	Sulpho	Cellulose
	$-CH_2SO_3^-$	Sulphomethyl	Dextran
	$-CH_2CH_2CH_2SO_3^-$	Sulphopropyl	Polystyrene
			Polyacrylate
Weakly basic (anion exchanger)	$-CH_2CH_2N^+H_3$	Aminoethyl	Agarose
	$-CH_2CH_2N^+H$ $\vert$ $(CH_2CH_3)_2$	Diethylaminoethyl	Cellulose
			Dextran
			Polystyrene
			Polyacrylate
Strongly basic (anion exchanger)	$-CH_2N^+(CH_3)_3$	Trimethylaminomethyl	Cellulose
	$-CH_2CH_2N^+(CH_2CH_3)_3$	Triethylaminoethyl	Dextran
	$-CH_2N^+(CH_3)_2$ $\vert$ CH_2CH_2OH	Dimethyl-2-hydroxyethyl-aminomethyl	Polystyrene
	$-CH_2CH_2N^+(CH_2CH_3)_2$ $\vert$ $CH_2CH(OH)CH_3$	Diethyl-2-hydroxypropyl-aminoethyl	

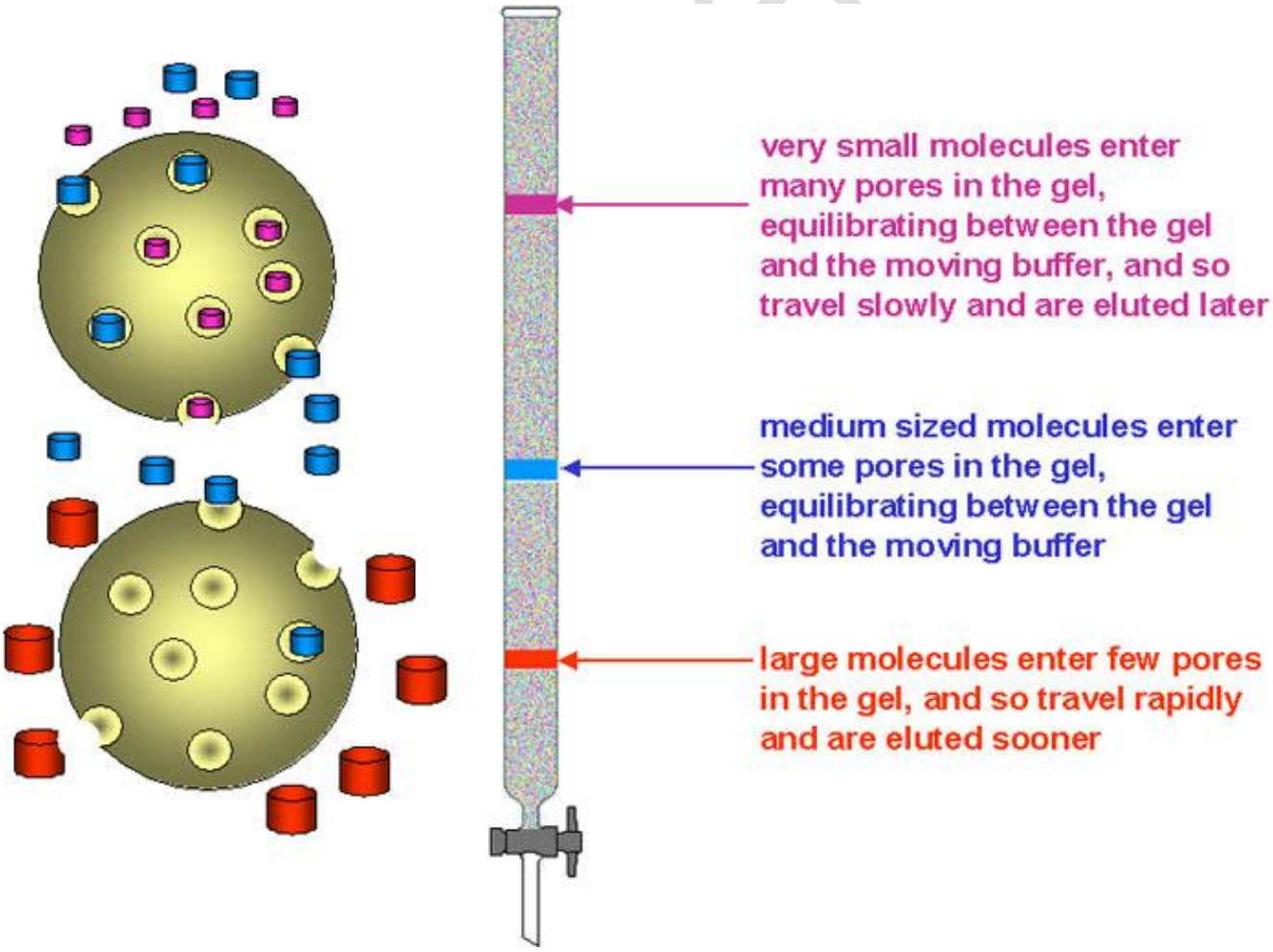
Examples of ion-exchange matrices

SIZE-EXCLUSION CHROMATOGRAPHY (SEC)

Also known as Gel Filtration (GF) is the simplest and mildest of all chromatography techniques and separates molecules based on differences in size.

Size exclusion chromatography separates proteins according to size. In this method, large proteins emerge from the column sooner than small ones—a somewhat counterintuitive result. The solid phase consists of beads with engineered pores or cavities of a particular size. Large proteins cannot enter the cavities, and so take a short (and rapid) path through the column, around the beads. Small proteins enter the cavities, and migrate through the column more slowly as a result.

The column matrix is a cross-linked polymer with pores of selected size, as shown in Figure 1 (b).Larger proteins migrate faster than smaller ones, because they are too large to enter the pores in the beads and hence take a more direct route through the column. The smaller proteins enter the pores and are slowed by their more labyrinthine path through the column.



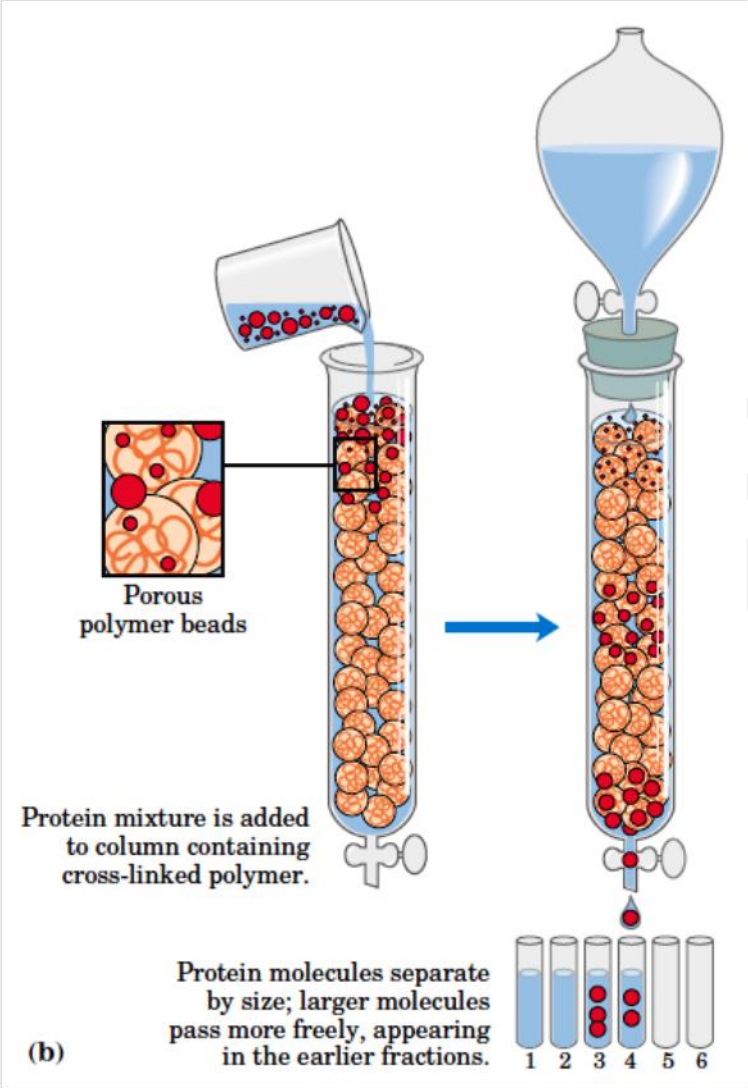


Figure 1 (b) – Size-exclusion chromatography

AFFINITY CHROMATOGRAPHY

This is based on the binding affinity of a protein. The beads in the column have a covalently attached chemical group. A protein with affinity for this particular chemical group will bind to the beads in the column, and its migration will be retarded as a result. As shown in Figure 1(c), Affinity chromatography separates proteins by their binding specificities. The proteins retained on the column are those that bind specifically to a ligand cross-linked to the beads. (In biochemistry, the term "ligand" is used to refer to a group or molecule that binds to a macromolecule such as a protein).After proteins that do not bind to the ligand are washed through the column, the bound protein of particular interest is eluted (washed out of the column) by a solution containing free ligand.

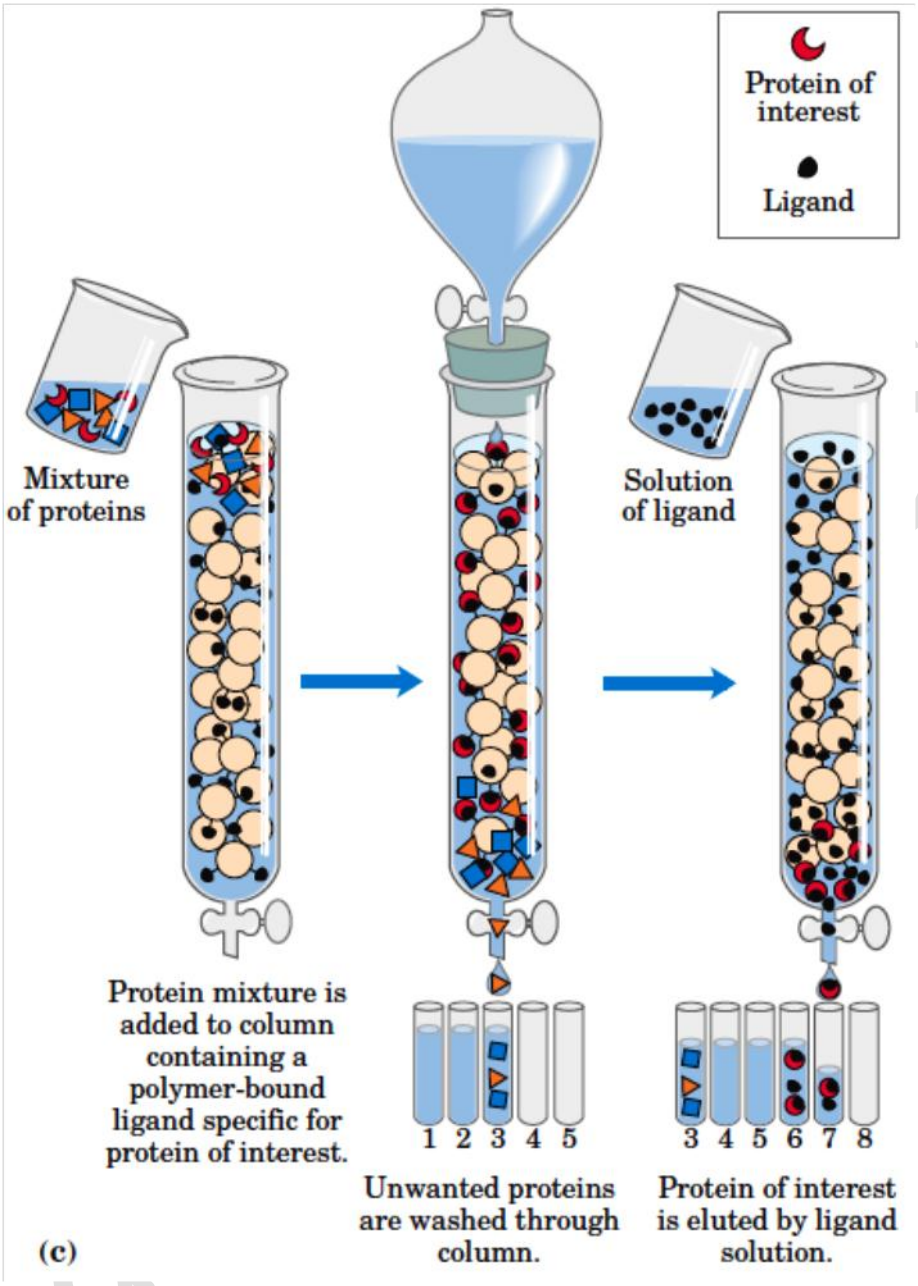
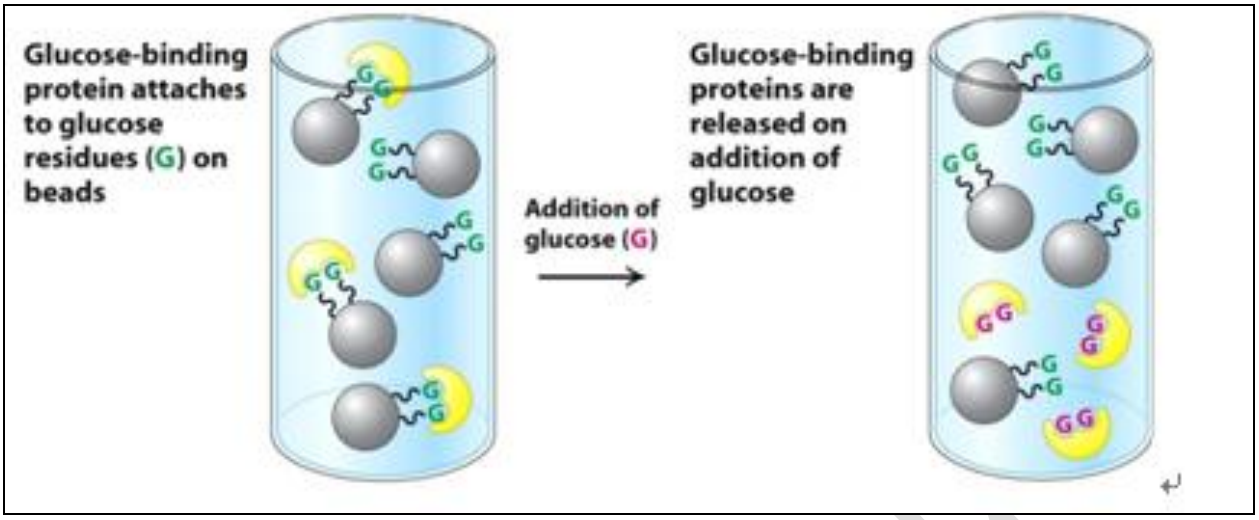
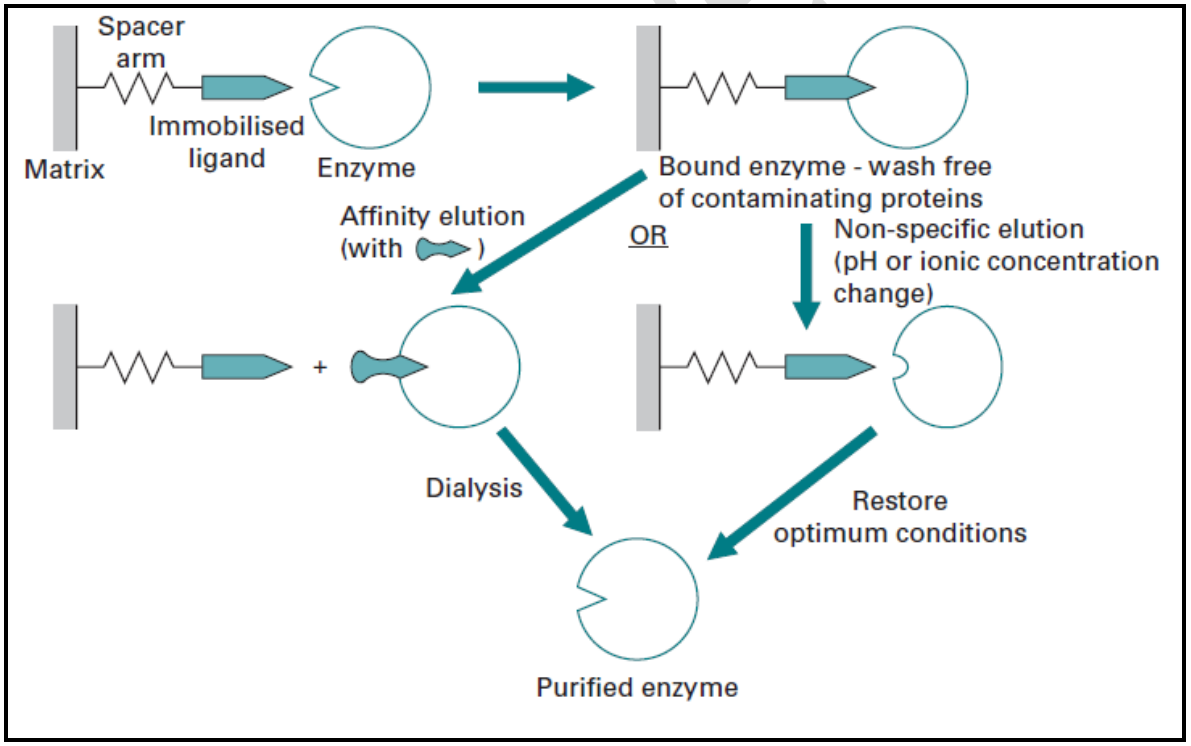


Figure 1 (c) – Affinity Chromatography





Schematic representation of affinity chromatography

Ligand	Affinity
Nucleotides	
5′-AMP	NAD^+-dependent dehydrogenases, some kinases
2′5′-ADP	$NADP^+$-dependent dehydrogenases
Calmodulin	Calmodulin-binding enzymes
Avidin	Biotin-containing enzymes
Fatty acids	Fatty-acid-binding proteins
Heparin	Lipoproteins, lipases, coagulation factors, DNA polymerases, steroid receptor proteins, growth factors, serine protease inhibitors
Proteins A and G	Immunoglobulins
Concanavalin A	Glycoproteins containing α-D-mannopyranosyl and α-D-glucopyranosyl residues
Soybean lectin	Glycoproteins containing *N*-acetyl-α-(or β)-D-galactopyranosyl residues
Phenylboronate	Glycoproteins
Poly(A)	RNA containing poly(U) sequences, some RNA-specific proteins
Lysine	rRNA
Cibacron Blue F3G-A	Nucleotide-requiring enzymes, coagulation factors

Examples of group-specific ligands commonly used in affinity chromatography

HIGH PERFORMANCE LIQUID CHROMATOGRAPHY

High performance liquid chromatography is now one of the most powerful tools in analytical chemistry. It has the ability to separate, identify, and quantitate the compounds that are present in any sample that can be dissolved in a liquid. Today, compounds in trace concentrations as low as *parts per trillion* [ppt] may easily be identified. HPLC can be, and has been, applied to just about any sample, such as pharmaceuticals, food, nutraceuticals, cosmetics, environmental matrices, forensic samples, and industrial chemicals.

HPLC makes use of high-pressure pumps that speed the movement of the protein molecules down the column, as well as higher-quality chromatographic materials that can withstand the crushing force of the pressurized flow. By reducing the transit time on the column, HPLC can limit diffusional spreading of protein bands and thus greatly improve resolution. The approach to purification of a protein that has not previously been isolated is guided both by established precedents and by common sense. In most cases, several different methods must be used sequentially to purify a protein completely. The choice of method is somewhat empirical, and

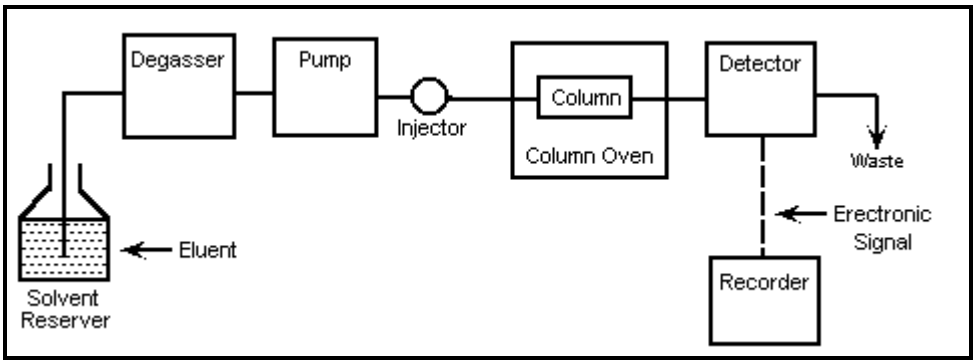
many protocols may be tried before the most effective one is found. Trial and error can often be minimized by basing the procedure on purification techniques developed for similar proteins.

HPLC stands for High Performance Liquid Chromatography. Before HPLC was available, LC analysis was carried by gravitational flow of the eluent (the solvent used for LC analysis) thus required several hours for the analysis to be completed. Even the improvements added in later time were able to shorten the analysis time slightly. Those classical/initial LC systems are called "low pressure chromatography" or "column chromatography".

In 1970s in the US, Jim Waters founded Waters Corporation and started to sell HPLC instruments. This promoted the use of HPLC in practical analysis areas. The LC systems that Waters Corporation developed used high-pressure pump that generates rapid-flow of eluent, and thus resulted in dramatic improvement in the analysis time. Compared to the "low pressure chromatography" the newer types were called "high pressure liquid chromatography". Therefore it was used to be thought that HPLC stands for High Pressure Liquid Chromatography, however nowadays it is a common agreement that HPLC stands for High Performance Liquid Chromatography. Another big change from Tswett's date was the data acquisition methods. Instead of observing the changes of layers by eyes, detector system was coupled to the LC and out-put was recorded on paper chart. If we were to demonstrate Tswett's analysis result on a chart (chromatogram), it will be like figure 3. Initially, HPLC system was referred to Waters Corporation's system. Still now, Waters Corporation is the HPLC pioneer, but there are several other companies that manufacture and sell HPLC systems. Technically speaking, the word LC represents all the Liquid Chromatography, including low pressure LC, however most LC systems used these days are HPLC thus often the word LC is used as comparable as HPLC.

Components of HPLC

Typical HPLC system consists of followings (Figure below). Details of each are explained below.



Components of HPLC system

1. **Pump**

   In the earlier state of HPLC development, the pump was the most important part of the system. The development of HPLC can be said that it was a development of pump system. Pump is positioned in the most upper stream of the LC system and generates a flow of eluent from the solvent reservoir to the system. In the earlier stage of LC development, to be able to generate the high pressure was the one of the most important system requirements. However, nowadays, the high pressure generation is a "standard" requirement and what is more concerned nowadays is to be able to provide a consistent pressure at any condition, to provide a controllable and reproducible flow rate. Since a change in the flow rate can influence the analysis largely.
   Most pumps used in current LC systems generate the flow by back-and-forth motion of a motor-driven piston (reciprocating pumps). Because of this piston motion, it produces "pulses". There have been large system improvements to reduce this pulsation and the recent pumps create much less pulse compared to the older ones. However, recent analysis requires very high sensitivity to quantify a small amount of analytes, and thus even a minor change in the flow rate can influence the analysis. Therefore, the pumps required for the high sensitivity analysis needs to be highly precise.

2. **Injector**

   An injector is placed next to the pump. The simplest method is to use a syringe, and the sample is introduced to the flow of eluent. Since the precision of LC measurement is largely affected by the reproducibility of sample injection, the design of injector is an important factor. The most widely used injection method is based on sampling loops. The use of autosampler (auto-injector) system is also widely used that allows repeated injections in a set scheduled-timing.

3. **Column**

   The separation is performed inside the column; therefore, it can be said that the column is the heart of an LC system. The theory of chromatography column has not changed since Tswett's time; however there has been continuous improvement in column development. The recent columns are often prepared in stainless steel housing, instead of glass columns used in Tswett's experiment. The packing material generally used is silica or polymer gels compared to calcium carbonate used by Tswett.
   The eluent used for LC varies from acidic to basic solvents. Most column housing is made of stainless steel, since stainless is tolerant towards a large variety of solvents. However, for the analysis of some analytes such as biomolecules and ionic compounds, contact with metal is not desired, thus a polyether ether ketone (PEEK) column housing is used instead.

Career Avenues

4. Detector

Separation of analytes is performed inside the column, whereas a detector is used to observe the obtained separation. The composition of the eluent is consistent when no analyte is present. While the presence of analyte changes the composition of the eluent. What detector does is to measure these differences. This difference is monitored as a form of electronic signal. There are different types of detectors available. Different detector types are explained in Lesson 6.

5. Recorder

The change in eluent detected by a detector is in the form of electronic signal, and thus it is still not visible to our eyes. In older days, pen (paper)-chart recorder was popularly used. Nowadays, computer based data processor (integrator) is more common. There are various types of data processors; examples include a simple system consisting of in-built printer and word processor, and a personal computer type consisting of display monitor, keyboard, and printer. Also there are software that are specifically designed for LC system. It provides not only data acquisition, but features like peak-fitting, base line correction, automatic concentration calculation, molecular weight determination, etc…

The components introduced so far are the basics of LC system. Below are some optional equipment used with the basic LC system.

6. Degasser

The eluent used for LC analysis may contain gases such as oxygen that are non-visible to our eyes. When gas is present in the eluent, this is detected as a noise and causes unstable baseline. Generally used method includes sparging (bubbling of inert gas), use of aspirator, distillation system, and/or heating and stirring. However, the method is not convenient and also when the solvent is left for a certain time period (e.g., during the long analysis), gas will dissolve back gradually. Degasser uses special polymer membrane tubing to remove gases. The numerous very small pores on the surface of the polymer tube allow the air to go through while preventing any liquid to go through the pore. By placing this tubing under low pressure container, it created pressure differences inside and outside the tubing (higher inside the tubing). This difference let the dissolved gas to move through the pores and remove the gas. Compared to classical batch type degassing, the degasser can be used on-line, it is more convenient and efficient. Many of new HPLC unit system contain a degasser.

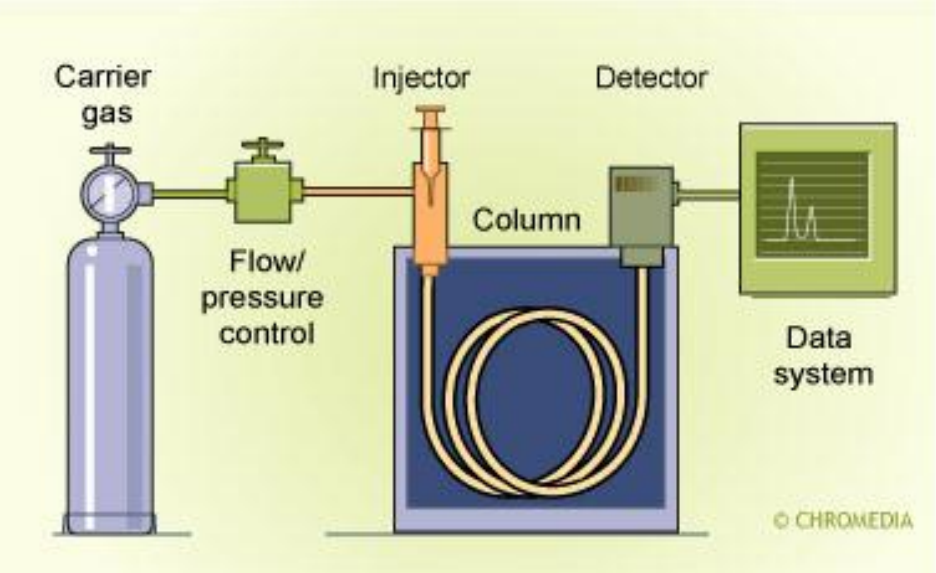
Career Avenues

S.NO.	Conventional Chromatography	HPLC
1.	Plastic or glass columns used	Stainless steel columns used
2.	Generally length of columns10-100 cms	Generally length of columns 10-30 cms
3.	Low resolution than HPLC	Higher resolution than other conventional chromatography
4.	Low pressure sampling	High pressure sampling
5.	Slower process than HPLC	Faster process than other conventional chromatography

Differences between conventional column chromatography and HPLC

Gas Chromatography

The principles of gas chromatography (GC) are similar to those of HPLC but the apparatus is significantly different. It exploits differences in the partition coefficients between a stationary liquid phase and a mobile gas phase of the volatilised analytes as they are carried through the column by the mobile gas phase. Its use is therefore confined to analytes that are volatile but thermally stable. The partition coefficients are inversely proportional to the volatility of the analytes so that the most volatile elute first. The temperature of the column is raised to 50-300^0 C to facilitate analyte volatilisation. The stationary phase consists of a high-boiling-point liquid material such as silicone grease or wax that is either coated onto the internal wall of the column or supported on an inert granular solid and packed into the column. There is an optimum flow rate of the mobile gas phase for maximum column efficiency. Very high resolutions are obtained hence the technique is very useful for the analysis of complex mixtures. Gas chromatography is widely used for the qualitative and quantitative analysis of a large number of low-polarity compounds because it has high sensitivity, reproducibility and speed of resolution. Analytically, it is a very powerful technique when coupled to mass spectrometry.



1. **Choice of gas** : The choice of a practical carrier gas is simple: nitrogen or helium. Air may be used as a carrier gas under certain conditions with portable or on-site chromatographs but this is uncommon with laboratory-scale instruments. The choice of nitrogen or helium is made, in part, on the principles of separation and, in part, on economics
2. **Sample injection port**
   For optimum column efficiency, the sample should not be too large, and should be introduced onto the column as a "plug" of vapour - slow injection of large samples causes band broadening and loss of resolution. The most common injection method is where a microsyringe is used to inject sample through a rubber septum into a flash vapouriser port at the head of the column. The temperature of the sample port is usually about 50°C higher than the boiling point of the least volatile component of the sample. For packed columns, sample size ranges from tenths of a microliter up to 20 microliters. Capillary columns, on the other hand, need much less sample, typically around 10^{-3} mL. The injector can be used in one of two modes; split or splitless. The injector contains a heated chamber containing a glass liner into which the sample is injected through the septum. The carrier gas enters the chamber and can leave by three routes (when the injector is in split mode). The sample vapourises to form a mixture of carrier gas, vapourised solvent and vapourised solutes. A proportion of this mixture passes onto the column, but most exits through the split outlet. The septum purge outlet prevents septum bleed components from entering the column.

3. Columns

There are two general types of column, *packed* and *capillary* (also known as *open tubular*). Packed columns contain a finely divided, inert, solid support material (commonly based on *diatomaceous earth*) coated with liquid stationary phase. Most packed columns are 1.5 - 10m in length and have an internal diameter of 2 - 4mm.

Capillary columns have an internal diameter of a few tenths of a millimeter. They can be one of two types; *wall-coated open tubular* (WCOT) or *support-coated open tubular* (SCOT). Wall-coated columns consist of a capillary tube whose walls are coated with liquid stationary phase. In support-coated columns, the inner wall of the capillary is lined with a thin layer of support material such as diatomaceous earth, onto which the stationary phase has been adsorbed. SCOT columns are generally less efficient than WCOT columns. Both types of capillary column are more efficient than packed columns.

4. Detectors

There are many detectors which can be used in gas chromatography. Different detectors will give different types of selectivity. A *non-selective* detector responds to all compounds except the carrier gas, a *selective detector* responds to a range of compounds with a common physical or chemical property and a *specific detector* responds to a single chemical compound. Detectors can also be grouped into *concentration dependant detectors* and *mass flow dependant detectors*. The signal from a concentration dependant detector is related to the concentration of solute in the detector, and does not usually destroy the sample Dilution of with make-up gas will lower the detectors response. Mass flow dependant detectors usually destroy the sample, and the signal is related to the rate at which solute molecules enter the detector. The response of a mass flow dependant detector is unaffected by make-up gas. Have a look at this tabular summary of common GC detectors:

Detector	Support gases	Selectivity	Detectability
Flame ionization (FID)	Hydrogen and air	Most organic cpds.	100 pg
Thermal conductivity (TCD)	Reference	Universal	1 ng
Electron capture (ECD)	Make-up	Halides, nitrates, nitriles, peroxides, anhydrides, organometallics	50 fg
Nitrogen-phosphorus	Hydrogen and air	Nitrogen, phosphorus	10 pg
Flame photometric (FPD)	Hydrogen and air possibly oxygen	Sulphur, phosphorus, tin, boron, arsenic, germanium, selenium, chromium	100 pg
Photo-ionization (PID)	Make-up	Aliphatics, aromatics, ketones, esters, aldehydes, amines, heterocyclics, organosulphurs, some organometallics	2 pg

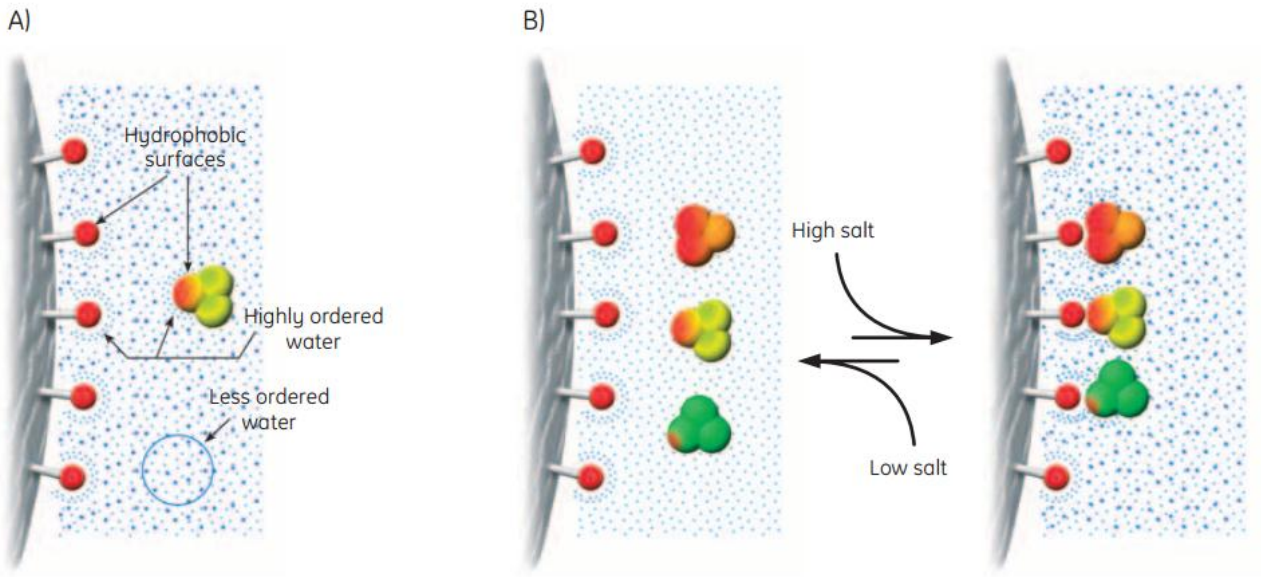
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Hydrophobic interaction chromatography

Hydrophobic interaction chromatography (HIC) was developed to purify proteins by exploiting their surface hydrophobicity which is a measure of their dislike of binding molecules of water. Groups of hydrophilic amino acid residues are scattered over the surface of proteins in a way that gives characteristic properties to each protein. In aqueous solution, these hydrophilic regions on the protein are covered with an ordered layer of water molecules that effectively mask the hydrophobic groups of the proteins the majority of which are in the interior of the folded molecule. These hydrophobic groups can, however, be exposed by the addition of salt ions, which preferentially take up the ordered water molecules. The exposed hydrophobic regions can then interact with each other by weak van der Waals' forces causing protein– protein aggregation. In HIC, the presence of hydrophobic groups such as butyl, octyl and phenyl attached to a matrix facilitates protein–matrix interaction rather than facilitating protein–protein interaction. Commercial materials include Phenyl Sepharose and Phenyl SPW, both for low-pressure HIC, and Poly PROPYL Aspartamide, Bio-Gel TSK Phenyl and Spherogel TSK Phenyl for HPLC HIC.

Since HIC requires the presence of salting-out compounds such as ammonium sulphate to facilitate the exposure of the hydrophobic regions on the protein molecule, it is commonly used immediately after fractionation of protein mixtures with ammonium sulphate. To maximise the process, it is advantageous to adjust the pH of the protein sample to that of its isoelectric point. Once the proteins have been adsorbed to the stationary phase, selective elution can be achieved in a number of ways, including the use of an eluent of gradually decreasing ionic strength or of increasing pH (this increases the hydrophilicity of the protein) (Fig. 11.8) or by selective displacement by a displacer that has a stronger affinity for the stationary phase than has the protein. Examples include non-ionic detergents such as Tween 20 and Triton X-100, aliphatic alcohols such as 1-butanol and ethylene glycol, and aliphatic amines such as 1-aminobutane. HIC has many similarities with reversed-phase HPLC (RPC) but has two advantages over it. The first is that it uses aqueous elution conditions that minimise protein denaturation whereas RPC requires non-polar solvents for elution. The second is that it has a higher capacity.

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A) Highly ordered water shells surround the hydrophobic surfaces of ligands and proteins. Hydrophobic substances are forced to merge to minimize the total area of such shells (maximize entropy). Salts enhance the hydrophobic interaction. B) The equilibrium of the hydrophobic interaction is controlled predominantly by the salt concentration.

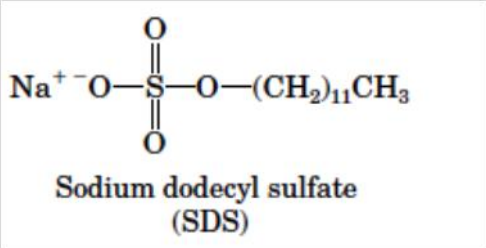
PROTEINS CAN BE SEPARATED AND CHARACTERIZED BY ELECTROPHORESIS

Another important technique for the separation of proteins is based on the migration of charged proteins in an electric field, a process called **electrophoresis.** These procedures are not generally used to purify proteins in large amounts, because simpler alternatives are usually available and electrophoretic methods often adversely affect the structure and thus the function of proteins. Electrophoresis is, however, especially useful as an analytical method. Its advantage is that proteins can be visualized as well as separated, permitting are searcher to estimate quickly the number of different proteins in a mixture or the degree of purity of a particular protein preparation. Also, electrophoresis allows determination of crucial properties of a protein such as its iso electric point and approximate molecular weight.

Electrophoresis of proteins is generally carried out in gels made up of the cross-linked polymer polyacrylamide. The polyacrylamide gel acts as a molecular sieve, slowing the migration of proteins approximately in proportion to their charge-to-mass ratio. Migration may also be affected by protein shape. In electrophores is, the force moving the macromolecule is the electrical potential, *E*. The electrophoretic mobility of the molecule, μ, is the ratio of the velocity of the particle molecule, *V*, to the electrical potential. Electrophoretic mobility is also equal to the net charge of the molecule, *Z*, divided by the frictional coefficient, *f*, which reflects in part a protein's shape. Thus:

$$\mu = \frac{V}{E} = \frac{Z}{f}$$

The migration of a protein in a gel during electrophoresisis therefore a function of its size and its shape. An electrophoretic method commonly employed for estimation of purity and molecular weight makes use of the detergent **sodium dodecyl sulfate (SDS).**



Sodium dodecyl sulfate (SDS)

SDS binds to most proteins in amounts roughly proportional to the molecular weight of the protein, about one molecule of SDS for every two amino acid residues. The bound SDS contributes a large net negative charge, rendering the intrinsic charge of the protein insignificant and conferring on each protein a similar charge-to-mass ratio. In addition, the native conformation of a protein is altered when SDS is bound, and most proteins assume a similar shape. Electrophoresis in the presence of SDS therefore separates proteins almost exclusively on the basis of mass (molecular weight), with smaller polypeptides migrating more rapidly.

After electrophoresis, the proteins are visualized by adding a dye such as Coomassie blue, which binds to proteins but not to the gel itself. Thus, a researcher can monitor the progress of a protein purification procedure as the number of protein bands visible on the gel decreases after each new fractionation step. When compared with the positions to which proteins of known molecular weight migrate in the gel, the position of an unidentified protein can provide an excellent measure of its molecular weight. If the protein has two or more different subunits, the subunits will generally be separated by the SDS treatment and a separate band will appear for each.

Isoelectric focusing is a procedure used to determine the isoelectric point (pI) of a protein. A pH gradient is established by allowing a mixture of low molecular weight organic acids and bases to distribute themselves in an electric field generated across the gel. When a protein mixture is applied, each protein migrates until it reaches the pH that matches its pI. Proteins with different isoelectric points are thus distributed differently throughout the gel.

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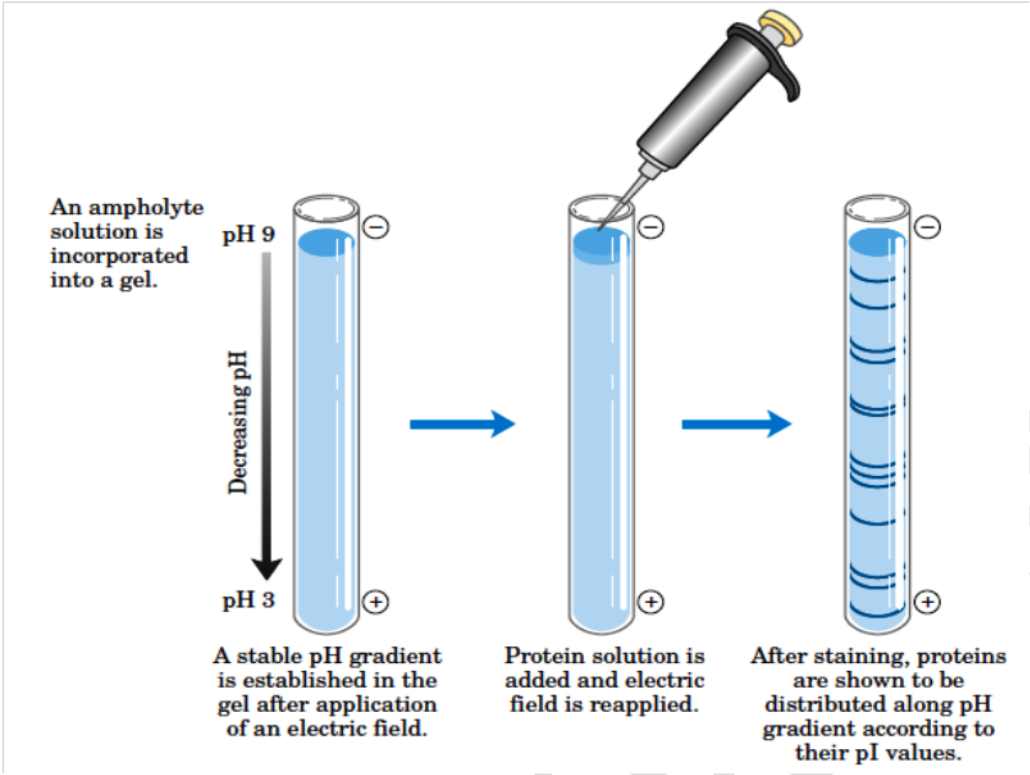
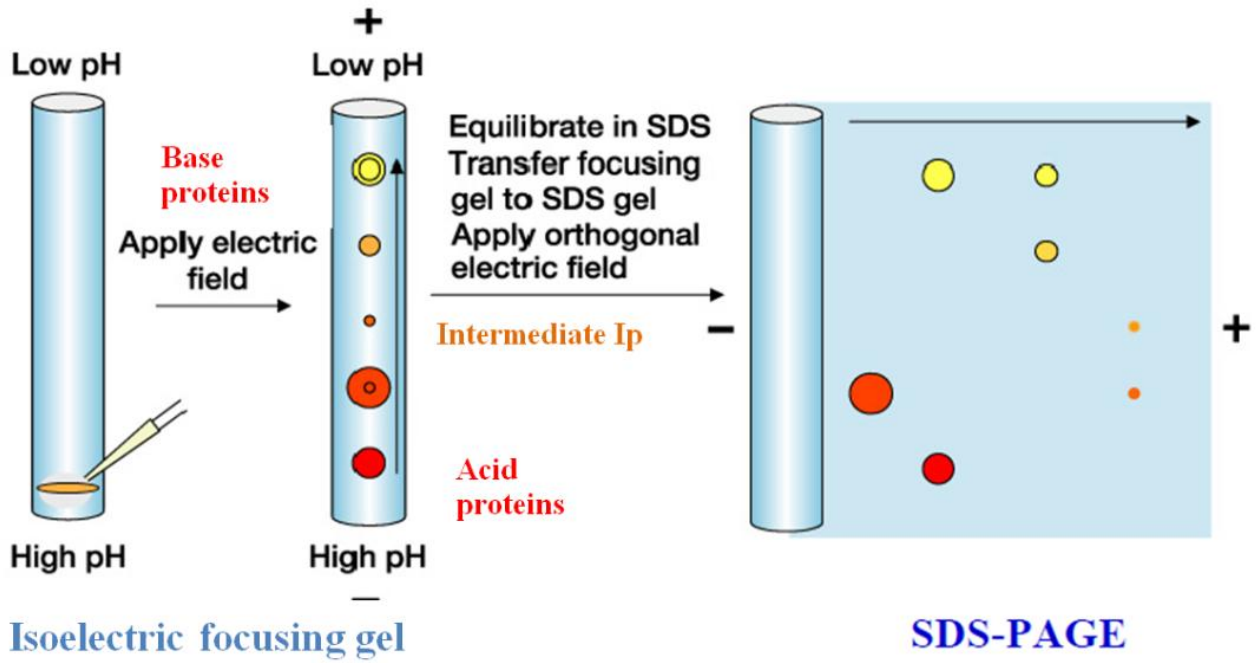


FIGURE:Isoelectric focusing.

This technique separates proteins according to their isoelectric points. A stable pH gradient is established in the gel by the addition of appropriate ampholytes. A protein mixture is placed in a well on the gel. With an applied electric field, proteins enter the gel and migrate until each reaches a pH equivalent to its pI. Remember that when pH = pI, the net charge of a protein is zero.



Combining isoelectric focusing and SDS electrophoresis sequentially in a process called **two-dimensional electrophoresis** permits the resolution of complex mixtures of proteins. This is a more sensitive analytical method than either electrophoretic method alone. Two-dimensional electrophoresis separates proteins of identical molecular weight that differ in pI, or proteins with similar pI values but different molecular weights.

Why purify proteins?

Technicians and scientists in biotechnology must face the challenge of protein purification regularly. In a research environment, proteins must be purified in order to determine their structure and study their biochemical properties. In industrial settings, proteins are purified on a larger scale in order to be sold as products such as drugs, vaccines, diagnostic tools or food additives.

Separation methods are based on protein properties.

Because of the differences in amino acid composition and sequence, and the possible presence of non-protein groups, each protein has different chemical characteristics that make it unique. These characteristics include size (molecular weight), charge, solubility, hydrophobicity and biological affinity. Differences in these characteristics are the basis for separation methods such as filtration, salt precipitation, and chromatographic procedures. A wide variety of separation methods are available.

The ideal purification strategy has the following goals: maximum recovery of the target protein; minimal loss of biological activity; and maximum removal of contaminating proteins, as well as low cost.

Analysis of purity of the protein during the various steps of purification

In order to purify a protein, it is essential to have a specific assay, a method for detecting and measuring the target protein. The assay makes it possible to determine the yield, which is how much of the target protein has been purified. In rare cases, the target protein can be detected easily because it is colored (myoglobin) or fluorescent (Green Fluorescent Protein) but most proteins are colorless. If the protein is an enzyme, such as ß-galactosidase, it may be assayed by its ability to catalyze a reaction, provided that there is a way to monitor the formation of product, or the disappearance of substrate.

After each separation step is performed, an aliquot of the product mixture is set aside for analysis of purity and yield. As the purification process proceeds, the specific activity should increase because contaminants are being removed, thereby decreasing the total weight of all proteins in the sample. This indicates greater purity due to fewer contaminating proteins. Maximum specific activity will be obtained when the protein is purified to homogeneity; in other words, there are no other proteins present other than the target protein. Losses of yield are inevitable as the product mixture is manipulated - the goal is to minimize this loss.

The purification table

The results of activity and protein assays, from a protein purification, are typically summarized in a so called purification table as give in the below example.

Table 1. Summary of the purification of β-glucosidase from *Cellulomonas biazotea* NIAB442 grown on Sigmacell-100.

Treatment	Volume (ml)	Units/ml*	Total units	Protein (mg/ml)	Total protein	Specific activity (units/mg)	% Yield	Purification factor
Crude extract	520	0.59	307	1.16	603	0.50	100	1
Dialysis	580	0.38	219	0.7	406	0.54	71	1.08
$(NH_4)_2SO_4$ precipitation	13	5.13	66.7	1.50	19.5	3.42	22	6.33
Anion-exchange chromatography	12	4.86	58.32	0.56	6.72	8.67	19	17.0

*One β-glucosidase unit is defined as 1 μmole of *p*-nitrophenol released/ml/min from *p*-nitrophenyl-β-D-glucopyranoside in sodium acetate buffer at pH 5.5 and 37 °C.

Now here in the above example

Total units = (Units / ml) X (Total volume in ml) i.e.,

307 = (0.59) X (520)

Total protein = (Protein mg/ml) X (Volume in ml) i.e.,

603 = (1.16) X (520)

Specific activity = (Units in ml) / (Protein mg/ml) i.e.,

0.50 = 0.59 / 1.16

Purification fold or factor = Specific activity (Units in mg) in the present step i.e., after Purification

---

Specific activity (Units in mg) in the previous step i.e., before Purification

1.08 = 0.54 / 0.50

So by definition

- ***Volume*** **(ml)** this refers to the measured total solution volume at the particular stage in the isolation.

• ***Total protein*** **(mg)** - the primary measurement is of protein concentration, i.e. mg /ml, which is obtained using a protein assay. Multiplying the protein concentration by the total volume gives the
total protein (i.e. mg/ml x ml = mg).

• ***Total activity*** **(units)** the activity, in units / ml, is obtained from an activity assay. Multiplying the activity by the total volume gives the total activity (i.e. units/ml x ml = units).
• *Specific-activity*(units/mg) - the specific activity is obtained by dividing the total activity by the total protein. Alternatively, the activity (units/ml) can be divided by the protein concentration (mg/ml), in which case the ìmlîs cancel out, leaving units/mg.

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• ***Purification*** **(fold)** Fold refers to the number of multiples of a starting value. In this case it refers to the increase in the specific activity, i.e. the purification is obtained by dividing the specific activity at any stage by the specific activity of the original homogenate. The purification "per step" can also be obtained by dividing the specific activity after that step by the specific activity of the material before that step.

• ***Yield*** **(%)** - the yield is based on the recovery of the activity after each step. The activity of the original homogenate is arbitrarily set at 100%. The yield (%) is calculated from the total activity (units) at each step divided by the total activity (units) in the homogenate, multiplied by 100. The yield can also be calculated on a "per step" basis by dividing the total activity after that step by the total activity before that step and multiplying by 100.

The efficiency of a step - is calculated as:-

$$\text{Purification (for that step)} \times \frac{\%\,\text{yield (for that step)}}{100}$$

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BIOINFORMATICS

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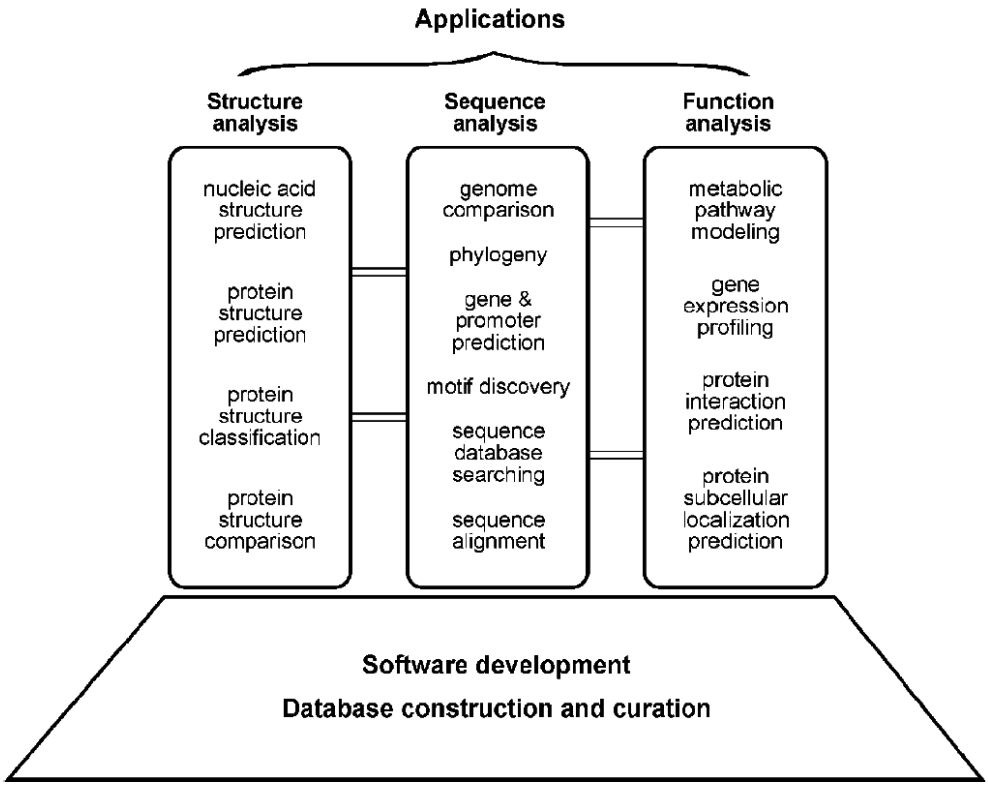


Figure 1.1: Overview of various subfields of bioinformatics. Biocomputing tool development is at the foundation of all bioinformatics analysis. The applications of the tools fall into three areas: sequence analysis, structure analysis, and function analysis. There are intrinsic connections between different areas of analyses represented by bars between the boxes.

What is a Database?

A *database* is a computerized archive used to store and organize data in such a way that information can be retrieved easily via a variety of search criteria. Databases are composed of computer hardware and software for data management.

Based on their contents,

Primary databases, secondary databases, and specialized databases.

Primary databases contain

original biological data , Redundant

archives of raw sequence or structural data submitted by scientific community

GenBank, EMBL and DDBJ are primary sequence databases and Protein Data Bank (PDB) – structural database are examples of primary databases

Secondary databases contain

Computationally processed or manually curated information, based on original information from primary databases

Non-redundant

Multiple entries of the same sequence in primary databases are merged to create a single sequence in the secondary database with extensive annotation derived from all available information on the sequence. The sequence and all the information about it are manually curated.

NCBI RefSeq database is a secondary database that is a collection of curated, non-redundant, well-annotated sequences including genomic DNA, transcripts, and proteins

Table 5.2: Publicly Available Secondary and Specialized Database

Database	**Comments (with URLs)**
Universal Protein Resource Knowledgebase (UniProtKB)	The UniProt Knowledgebase (UniProtKB) is the central repository for the collection of sequence and functional information on proteins with accurate, consistent, and rich annotation. UniProtKB is the product of UniProt, which is an international consortium between the European Bioinformatics Institute (EBI), the Swiss Institute of Bioinformatics (SIB) and the Protein Information Resource (PIR) at the Georgetown University Medical Center. In 2002, EBI, SIB, and PIR started collaboration to create a single high-quality database of protein sequence and function, by unifying the Swiss-Prot, TrEMBL, and PIR-PSO database. Before this collaboration, EMBL-EBI maintained TrEMBL, SIB maintained Swiss-Prot, and PIR maintained the Protein Sequence Database (PIR-PSD). These data sets conexisted with different protein-sequence coverage and annotation priorities[26,27]. (www.uniprot.org) UniProtKB has two sections: UniProt/Swiss-Prot and UniProt/TrEMBL. UniProt/Swiss-Prot contains sequences that are manually annotated, compared, and verified (curated) based on information from literature and curator-evaluated computational analysis. UniProt/TrEMBL (TrEMBL = translated EMBL) contains computationally annotated, unreviewed sequences. TrEMBL sequences are eventually manually curated to become part of Swiss-Prot and removed from TrEMBL. Before becoming part of UniProt, PIR-PSD was the oldest annotated

	and curated protein-sequence database, established in 1984 as a successor to the original National Biomedical Research Foundation (NBRF) Protein Sequence Database. It was developed over a 20 year period by the late Margret Dayhoff and published as the "Atlas of Protein Sequence and Structure" from 1965 to 1978. The link to PIR-PSD is http://pir.georgetown.edu/[28].
Worldwide Protein Data Bank (wwPDB)	Experimentally determined structures of proteins, and complex assemblies. wwPDB is a publicly available archive of macromolecular structural data[29]. (http://www.wwpdb.org/)
Structural Classification of Proteins (SCOP) database	The SCOP database aims to provide a detailed and comprehensive description of the structural and evolutionary relationships between all proteins whose structure is known, including all entries in the PDB, Proteins are classified into families (clear evolutionarily relationship; this generally means that pairwise residue identities between the proteins are 30% and greater), superfamilies (probable common evolutionary origin), and folds (major structural similarity)[30] (http://scop.mrc-lmb.cam.ac.uk/scop/)
Class, Architecture, Topology, Homology (CATH) database	CATH is a manually curated classification of protein domain structures. Each protein is chopped into structural domains and assigned into homologous superfamilies (groups of domains that are related by evolution). This classification procedure uses a combination of automated and manual techniques, which include computational algorithms, empirical and statistical evidence, literature review, and expert analysis[31] (http://www.cathdb.info/)
PROSITE database	This consists of a large collection of biologically meaningful signature patterns or profiles. These signatures are not easily revealed by standard sequence alignment. Each signature can be linked to useful biological information on the protein family, domain, or functional site. Therefore, the database can be used to rapidly and reliably identify which known family of protein (in any) the new sequence belongs to. The PROSITE database uses two kinds of signatures, patterns and generalized profiles, to identify conserved regions[32]

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	(http://prosite.expasy.org/)
PRINTS database	This is a compendium of protein fingerprints; a fingerprint is a group of conserved motifs used to characterize a protein family[33] (http://www.bioinf.man.ac.uk/dbbrowser/PRINTS/index.php)
Protein Family (Pfam) database	Pfam is a comprehensive database of protein families; members of a family share significant similarity, thereby suggesting homology. Pfam allows the analys is sequence data in order to search for related proteins in the database based on domains. Domains are regions of the protein, which in different combinations can determine the protein's function. Thus, proteins can be viewed as built from a specific combination of domains. Pfam contains two types of families: high-quality manually curated Pfam-A families and automatically generated Pfam-B families. Pfam uses multiple sequence alignments and hidden Markov models (HMM)[34] (http://www.sanger.ac.uk/resources/databases/pfam.html)
InterPro database	InterPro integrates various predictive protein signatures from diverse source repositories, such as (Gene3D, PANTHER, Pfam, PIRSF, PRINTS, ProDom, PROSITE, SMART, SUPERFAMILY, and TIGRF AMs. Protein signatures from various databases are integrated into InterPro manually. Curators combine signatures representing the same protein family, domain, or site into single database entries, and, where possible, trace biological relationships between the constituent signatures[35] (http://www.ebi.ac.uk/interpro/)
Biological General Repository for Interaction Datasets **(BioGRID)**	The BioGRID database is an online repository of interactions in which data are curated from both high-throughput data sets and individual focused studies, as derived from over 40,000 publications in the primary literature. The current compilation (as of July, 2013) has more than 700,000 raw protein and manually annotated genetic interactions from major model organisms. All BioGRID interaction records are directly mapped to experimental evidence in the supporting publication[36]

	(http://thebiogrid.org/)
Molecular Interaction database (MINT)	MINT is a public repository for protein-protein interactions reported in peer-reviewed journals. It focuses on experimentally verified protein-protein interactions mined from the scientific literature by expert curators. Currently it contains over 240,000 interaction data captured from over 4750 publications[37,38] (http://mint.bio.uniroma2.it/mint/)
Munich Information System for Protein Sequences (MIPS) database	The MIPS mammalian protein-protein interaction database is a resource of high-quality experimental protein-interaction data. The content is based on published experimental evidence that has been processed by human expert curators. MIPS also contains large-scale secondary data of protein similarities, currently containing 38 million non-redundant protein sequences [39, 40] (http://mips.helmholtz-muenchen.de/proj/ppi/)
IntAct	IntAct is a freely available, open source molecular interaction database populated by data either curated from the literature or from direct data depositions. As of September 2011, IntAct contained approximately 275,000 curated binary interaction evidence records from over 5000 publications. The IntAct database also captures protein-small molecule (including phospholipids), protein-nucleic acid, and protein-gene locus interactions[41] (http://www.ebi.ac.uk/intact/)
Structural Database of Allergenic Proteins (SDAP)	SDAP is a web server that integrates a database of allergenic proteins with various computational tools that can assist structural biology studies related to allergens, including predicting the IgE-binding potential of food proteins. This database allows bioinformatics analysis as recommended by the *Codex Alimentarius* and UN Food and Agriculture Organization (FAO)/World Health Organization (WHO) Expert Committee on potential allergenicity of foods derivated through modern biotechnology[a] (http://fermi.utmb.edu/SDAP/)
AllergenOnline/	AllergenOnline provides access to a peer-reviewed allergen list and

FARRP database (FARRP = Food Allergy Research and Resource Program at the University of Nebraska-Lincoln)	sequence searchable database intended for the identification of proteins, including food proteins, that may present a potential risk of allergenic cross-reactivity. The objective is to identify proteins that may require additional tests, such as serum IgE binding, basophil histamine release, or in vivo challenge to evaluate potential cross-reactivity. (http://www.allergenonline.org/)
Allermatch database	The Allermatch database allows the comparison of a protein sequence with sequences of allergenic proteins in the database, in order to predict whether the protein being evaluated can be allergenic. This database allows bioinformatics analysis as recommended by the *Codex Alimentarius* and FAO/WHO Expert Committee on potential allergenicity of foods derived through modern biotechnology[42] (http://www.allermatch.org/)
Online Mendelian Inheritance in Man (OMIM) database	OMIM is a comprehensive compendium of human genes and genetic-disease-associated phenotypes. The full-text referenced overviews in OMIM contain information on all known Mendelian disorders and over 12,000 genes[b] (http://www.ncbi.nlm.nih.gov/omim/ and http://omim.org/)
ArrayExpress database	A public database of microarray gene-expression data at the EBI. It accepts data generated by sequencing or array-based technologies and currently contains data from almost a million assays, from over 30,000 experiments. Experiments are submitted directly to ArrayExpress or are imported from the NCBI GEO database[43]. ArrayExpress uses the minimum information about a microarray experiment (MIAME) annotation standard[c]. (http://www.ebi.ac.uk/arrayexpress/)
Gene Expression Omnibus (GEO) database	The GEO is a public repository that archives and freely distributes MIAME-compliant microarray data, next-generation sequencing data, and other forms of high-throughout functional genomic data submitted by the scientific community. It is one of three international functional genomics public data repositories, alongside ArrayExpress at the EBI and the DDBJ Omics Archive[44,45]

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	(http://www.ncbi.nlm.nih.gov/geo/)
Array Track database	A public database of microarray gene-expression data at the US Food and Drug Administration. ArrayTrack provides an integrated solution for managing, analyzing, and interpreting microarray gene-expression data and experimental parameters associated with pharmacogenomics or toxicogenomic studies-that is, studies on the effects of drugs or other chemicals on gene expression. ArrayTrack supports MIAME-compliant data[46] (http://www.fda.gov/ScienceResearch/BioinformaticsTools/Arraytrack/default.htm)
Comparative Toxicogenomic database (CTD)	This is a public database of information built on curated data from the scientific literature about interactions between environmental chemicals and gene products and their relationships to diseases. As of 2013, CTD contains over 15 million toxicogenomic relationships. A user can look up specific literature-based information about genes, gene products, and toxicants of interest and their interactions[47] (http://ctdbase.org/)
Chemical Effects in Biological Systems (CEBS) database	The CEBS database has been developed by the National Center for Toxicogenomics within the National Institute of Environmental Health Science (NIEHS). CEBS integrates data obtained using 'omics technologies (transcriptomics, proteomics, metabolomics) as well as from traditional toxicology studies. Thus, CEBS combines the molecular genetic data with traditional clinical chemistry and histopathology data. This combination allows researchers to fully capture information on dose response, time response, and environmental-stress-induced gene expression. The database captures information from multiple species, such as humans, rats, mice, and *Caenorhabdit is elegans*[48] (http://www.niehs.nih.gov/research/resources/databases/cebs/index.cfm)
DrugMatrix database	DrugMatrix is a toxicogenomic and molecular toxicology database and informatics system developed by the National Toxicology Program (NTP). It contains data from standard toxicological experiments along with large-scale gene-expression data from various organs and tissues. DrugMatrix contains toxicogenomic profiles for 638 different

	compounds that include approved drugs, withdrawn drugs, and industrial and environmental toxicants[d] (https://ntp.niehs.nih.gov/drugmatrix/index.html)
FlyBase database	FlyBase is the leading database and web portal for genetic and genomic information focusing on *Drosophila melanogaster*, but also including data on other *Drosophila* species and related drosophilids. The current content of FlyBase comprises > 200,000 references, including > 87,000 research papers from > 2400 different journals, with publication dates ranging from the seventeenth century through to the present day[49, 50] (http://flybase.org/)
NCBI database	Collection of various databases. This is separately discussed below, in Section 5.6.1 (http://www.ncbi.nlm.nih.gov/)

The BLAT (BLAST-like Alignment Tool) was written by Jim Kent at UCSC. BLAT is used to map the input sequence to the genome - that is, to identify the location of a sequence in the genome. Therefore, BLAT works with the genomic context in memory, but it works by alignment-based similarity search

BLAT works for both DNA and proteins.

BLAT is different from BLAST because, unlike BLAST, BLAT does not search the sequences from GenBank/EMBL-Bank/DDBJ; rather, BLAT uses an index derived from the genome assembly and it consists of all non-overlapping 11-mers except the heavily repeated sequences. For proteins, BLAT uses 4-mers.

SEQUENCE ANALYSIS

Sequence identity means the same residues being present at corresponding positions in two sequences being compared. For proteins, it means the same amino acids; for nucleic acids, it means the same bases.

Sequence similarity means similar residues being present at corresponding positions in the two sequences being compared. For nucleic acids, sequence similarity and sequence identity are the same. However, for proteins, sequence similarity involves amino acids with similar physicochemical and functional properties.

For example,

Substitution of lysine and arginine by one another will be regarded as similar substitution because both are positively charged hydrophilic amino acids.

Substitution of aspartic acid and glutamic acid

Substitution of asparagine by aspartic acid and substitution of glutamine by glutamic acid, or vice versa, are also regarded as similar substitutions.
Substitution of isoleucine, leucine, and valine by one another will be regarded as similar substitutions because they have similar aliphatic hydrophobic side chains.

Substitution of serine and threonine by one another is also regarded as similar substitution.

Similar substitutions are also referred to as conservative substitutions. A conservative amino acid substitution is not expected to disrupt the structural/functional attributes of the protein.

Sequence homology is an evolutionary term. Sequences are called homologous if they have a common evolutionary origin - that is, if they are derived from a common ancestral sequence. So, sequences are either homologous or not homologous and there is no quantitation of homology.

Homologous – genes or proteins derived from a common ancestor, and have not changed much, probably because of functional constraints; hence, they are homologous.

Homologous genes in different species performing the same function are called **orthologs**.

Two proteins can be considered homologous despite low similarity if one or more of the following conditions are met:
(1) the similarity extends over a long stretch of sequence and is statistically significant;
(2) despite low sequence similarity, the same pattern of identical and similar amino acid residues is seen in multiple sequences; or
(3) the pattern of sequence similarity reflects the similarity between experimentally determined structures of the respective proteins, or at least corresponds to the known key elements of one such structure

Phenylalanine and tryptophan have very hydrophobic aromatic side chains, whereas tyrosine and histidine are less so. The latter two can often be found in positions that are somewhere between buried and exposed. The hydrophobic aromatic amino acids can sometimes substitute for aliphatic residues of a similar size, for example phenylalanine to leucine, but not tryptophan to valine.

Sequence identity and sequence similarity can be calculated based on the proportion of identical and similar amino acids, respectively:

% Identity (PID) = (# of identical amino acids / # of total amino acids) × 100; (6.1)

% Similarity = {(# of identical amino acids + # of similar substitutions) / # of total amino acids} × 100 (6.2)

In the above formulae, the denominator (# of total amino acids) can vary. For example, the denominator could be (1) the length of the shortest sequence, (2) the length of the longest sequence, (3) the mean length of the sequences, (4) the length of the aligned region (aligned positions excluding overhangs), etc. Therefore, PID is a rough measure and can be influenced by how it is calculated. However, because of the simplicity of calculation, PID is widely used.[3]

different. In a protein sequence alignment, *sequence identity* refers to the percentage of matches of the same amino acid residues between two aligned sequences. *Similarity* refers to the percentage of aligned residues that have similar physicochemical characteristics and can be more readily substituted for each other.

There are two ways to calculate the sequence similarity / identity. One involves the use of the overall sequence lengths of both sequences; the other normalizes by the size of the shorter sequence. The first method uses the following formula:

$$S = [(L_s \times 2) / (L_a + L_b)] \times 100 \quad \text{(Eq. 3.1)}$$

where S is the percentage sequence similarity, L_s is the number of aligned residues with similar characteristics, and L_a and L_b are the total lengths of each individual sequence. The sequence identity (*I%*) can be calculated in a similar fashion:

$$I = [(L_i \times 2) / (L_a + L_b)] \times 100 \quad \text{(Eq. 3.2)}$$

where L_i is the number of aligned identical residues.

The second method of calculation is to derive the percentage of identical / similar residues over the full length of the smaller sequences using the formula:

$$I(S)\% = L_{i(s)} / L_a\% \quad \text{(Eq. 3.3)}$$

where L_a is the length of the shorter of the two sequences.

Sequence alignment

The purpose of alignment is to identify regions of similarity that may have structural, functional, and evolutionary consequences

Some widely used online pairwise alignment tools use local alignment strategy (**Smith-Waterman algorithm)** and are shown in Table 6.1.

Table 6.1: Online Pairwise Alignment Tools Using the Smith – Waterman Algorithm

Online Tool	**URL**
PIR Research	http://pir.georgetown.edu/pirwww/search/pairwise.shtml[5]
NCBI specialized	bl2seq resource; look for the Align link on the NCBI BLAST

BLAST	home page under Specialized BLAST
SIM	http://web.expasy.org/sim/
LALIGN*	http://www.ch.embnet.org/software/LALIGN_form.html

The LALIGN program is William Pearson's, and it implements the algorithm of X. Huang and W. Miller.[6]

Multiple sequence alignments are useful in identifying conserved sequence segments across the sequences being aligned.

Such **conserved regions across multiple sequences usually indicate an evolutionary relationship**. For an unknown protein, for example, such conserved sequence segments identified through multiple alignment can be used in conjunction with other information to predict functionally important and evolutionarily conserved motifs within the proteins

MSA is also needed for construction of phylogenetic trees.

Most widely used MSA tool is ClustalW or Clustal Omega

The T-Coffee, CBRC (Computational Biology Research Center at the National Institute of Advanced Industrial Science and Technology, Japan) MAFFT, and EMBL-EBI MUSCLE all use ClustalW, so the output format is similar. NCBI COBALT has a very different output format.

In order to perform multiple alignment, the ClustalW algorithm goes through a number of steps, as follows:

it calculates all possible pairwise alignments of the input sequences;
computes the score of each alignment, where the score reflects the distance between the two sequences;
creates a dendrogram (guide tree) based on the matrix of the distance;

and uses the dendrogram as the basis to perform multiple alignment, where closely related pairs of sequences are aligned first.

Table 6.2: Online Multiple Alignment Tools

Online Tool	**URL**
COBALT (NCBI)	http://www.ncbi.nlm.nih.gov/tools/cobalt/cobalt.cgi?link_loc = BlastHomeLink[7]
ClustalW (DDBJ)	http://clustalw.ddbj.nig.ac.jp/inde.php?lang=en[8, 9]

MAFFT (CBRC)	http://mafft.cbrc.jp/alignment/server/[10]
MUSCLE (EMBL-EBI)	http://www.ebi.ac.uk/Tools/msa/muscle/[11]
T-Coffee	http://www.tcoffee.org/Projects/tcoffee/. Then click any of the server links on this page, such as http://www.tcoffee.org/ and from there the type of alignment program needed for analysis[12]

ALIGNMENT ALGORITHMS, GAPS, AND GAP PENALTIES

An algorithm is a step-by-step procedure that utilizes a finite number of instructions for automated reasoning and the calculation of a function.

The algorithm that drives **global alignment** is the **Needleman- Wunsch algorithm**, and the algorithm that drives **local alignment** is the **Smith-Waterman algorithm**.

Both these algorithms are examples of dynamic programming.

Dynamic programming is a method for solving complex problems by breaking them down into simpler sub problems.
In the case of sequence alignment, dynamic programming involves setting up a two-dimensional matrix in which one sequence is listed vertically and the other sequence is listed horizontally; then calculating the scores, one row at a time. For example, a match can be given a 1, a mismatch a 0, and a gap a 21. A 100% perfect alignment will produce a diagonal straight line (with a negative slope) spanning from the top left to bottom right. If the alignment is not perfect, gaps are introduced in the matrix. For the sequence represented horizontally, gaps are introduced vertically, and for the sequence represented vertically, gaps are introduced horizontally, and the alignment is determined by a traceback step.

The basic sequence alignment method is the dot matrix or dot plot method. In this method, two sequences being compared are written in the vertical and horizontal axes of the matrix. Then each residue is scanned and each match is given a dot; mismatches are left blank. When enough dots are lined up, they are connected.

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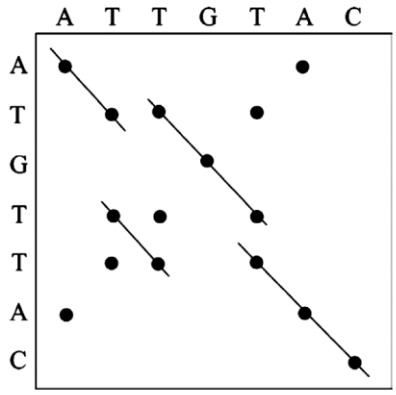


Figure 6.8 Comparison of two sequences using dot matrix or dot plot.

In both global and local alignment, the final output is given an alignment score. **Gaps have to be introduced to improve the alignment. The reason gaps are introduced** is because one of the sequences may have **gained or lost sequence characteristics** (insertion deletion) during evolution that did not happen with the other sequence.

However, the **number of gaps is kept to a minimum to keep the alignment meaningful**; otherwise an artificially **high alignment score can be obtained even when the two sequences are not related.** The gap penalty value is subtracted from the gross alignment score to obtain the final alignment score (alignment score and scoring matrix are discussed in the next section). The insertion of no more than 1 gap per 20 amino acid residues is ideal but that is not possible in most cases. **For each gap opened, a gap-opening penalty** value is assigned, and **for each gap extended, a gapextension penalty value is assigned.** A **gap-opening penalty is always much higher than a gap-extension penalty**. Often, a default value of -10 for a gapopening penalty and -1 for a gap-extension penalty are used. However, these values can be different and can also be adjusted by the user. This type of differential penalty for gap opening and gap extension is called affine gap penalty.

There are other types of gap penalties, such as constant gap penalty, linear gap penalty, and proportional gap penalty, but for all practical purposes affine gap penalty is the most relevant for sequence alignment. **Affine gap penalty is calculated as follows**:

$$G_t = G_o + G_e \times L_n, \quad (6.3)$$

where G_t = total gap penalty, G_o = gap – opening penalty, G_e = gap – extension penalty, and L_n = length of the extension gaps. For any given block of gaps, L_n = # of total gaps – 1, because the first gap is the opening, the rest in the block are extensions.

A raw alignment score can be calculate based on the following simple formula:

$$S = \sum_i + \sum_m - G_t, \quad (6.4)$$

where S = raw score, $\sum_i$ = total score for identities, $\sum_m$ = total score for mismatches, and G_t = total gap penalty.

For both nucleic acids and proteins, the **alignment score is calculated using a scoring matrix**. **A scoring matrix is a set of values representing the likelihood of one residue being substituted by another during sequence divergence through evolution**.

This is why the scoring matrix is also known as the substitution matrix.

A scoring matrix for comparing DNA sequences can be simple because there are only four nucleotides and the mutation frequencies are assumed to be equal (the Jukes and Cantor assumption). A high positive score (e.g. 5) is assigned for a match and a low negative score (e.g. 24) for a mismatch, thus creating a simple model.

However, the **frequency of transition mutations** (purine replaced by purine or pyrimidine replaced by pyrimidine) **is higher than transversion mutations** (purine replaced by pyrimidine or vice versa). To deal with this differential mutation frequency, sophisticated statistical models have been developed by Kimura and others. For generating a DNA sequence-alignment score, the simple scoring matrix is still used, such as the NUC4.2 and NUC4.4 DNA scoring matrices. These matrices can be obtained from the NCBI (ftp://ftp.ncbi.nih.gov/blast/matrices/).

Scoring matrices for amino-acid substitutions are more complex, reflecting the similarity of physicochemical properties, as well as the likelihood of one amino acid being substituted by another at a particular position in homologous proteins. The scoring matrices for proteins are 20 X 20 matrices. **Two well-known types of scoring matrices for proteins are PAM and BLOSUM.**

PAM (point accepted mutation—that is, accepted point mutation—also called percent accepted mutation) matrices were first **developed by Margaret Dayhoff** and colleagues in 1978 and hence are **also known as Dayhoff PAM matrices**.

A PAM represents a substitution of one amino acid by another that has been fixed by natural selection because either it does not alter the protein function or it is beneficial to the organism.

In a PAM1 matrix, which is the original PAM matrix generated, a PAM unit is an evolutionary time over which 1% of the amino acids in a sequence are expected to undergo accepted mutations, resulting in 1% sequence divergence.

The PAM1 matrix was built by aligning closely related protein sequences (71 protein families) that had at least 85% sequence identity.

Subsequently, in order to deal with protein sequences that are more diverged and distantly related, other PAM matrices, such as PAM100 and PAM250, were generated. These later PAM matrices were generated by multiplying the PAM1 matrix by itself hundreds of times. For example, the **PAM250 matrix can be obtained by multiplying the PAM1 matrix by itself 250 times over.** Figure 6.9 shows the PAM250 substitution matrix. The values in the matrix are log odds scores

PAM 250 Matrix

Ala	A	2																	
Arg	R	-2	6																
Asn	N	0	0	2															
Asp	D	0	-1	2	4														
Cys	C	-2	-4	-4	-5	12													
Gln	Q	0	1	1	2	-5	4												
Glu	E	0	-1	1	3	-5	2	4											
Gly	G	1	-3	0	1	-3	-1	0	5										
His	H	-1	2	2	1	-3	3	1	-2	6									
Ile	I	-1	-2	-2	-2	-2	-2	-2	-3	-2	5								
Leu	L	-2	-3	-3	-4	-6	-2	-3	-4	-2	2	6							
Lys	K	-1	3	1	0	-5	1	0	-2	0	-2	-3	5						
Met	M	-1	0	-2	-3	-5	-1	-2	-3	-2	2	4	0	6					
Phe	F	-3	-4	-3	-6	-4	-5	-5	-5	-2	1	2	-5	0	9				
Pro	P	1	0	0	-1	-3	0	-1	0	0	-2	-3	-1	-2	-5	6			
Ser	S	1	0	1	0	0	-1	0	1	-1	-1	-3	0	-2	-3	1	2		
Thr	T	1	-1	0	0	-2	-1	0	0	-1	0	-2	0	-1	-3	0	1	3	
Trp	W	-6	2	-4	-7	-8	-5	-7	-7	-3	-5	-2	-3	-4	0	-6	-2	-5	17
Tyr	Y	-3	-4	-2	-4	0	-4	-4	-5	0	-1	-1	-4	-2	7	-5	-3	-3	0 10
Val	V	0	-2	-2	-2	-2	-2	-2	-1	-2	4	2	-2	2	-1	-1	-1	0	-6 -2 4
		A	R	N	D	C	Q	E	G	H	I	L	K	M	F	P	S	T	W Y V

Figure 6.9 A PAM250 substitution matrix made by writing amino acids in alphabetical order.

BLOSUM will be referred to as BLOSUM matrix here. BLOSUM (blocks substitution matrices) scoring matrices were proposed by Steven Henikoff and Jorja Henikoff in 1992. BLOSUM represents an alternative set of scoring matrices, which are widely used in sequence alignment algorithms.

Like PAM, BLOSUM matrices are also log-odds matrices.

BLOSUM matrices were developed based on multiple alignment of 500 groups of related protein sequences, which yielded.2000 blocks of conserved amino-acid patterns. Blocks are ungapped multiple sequence alignments corresponding to the most conserved regions of the proteins involved. Henikoff and Henikoff used their BLOCKS database of trusted alignments. In each multiple alignment, the sequences showing similar % identity were clustered into groups and averaged. Using these groups, the substitution frequencies for all pairs of amino acids were calculated and the matrix was developed.

Different BLOSUM matrices differ in the % sequence identity used in clustering. Therefore, **BLOSUM62** means that the **sequences used to create this matrix have approximately 62% identity.** Substitution frequencies weigh more heavily by protein sequences having less than 62% identity. **Therefore, BLOSUM62 is useful for aligning and scoring proteins that show less than 62% identity.** Shown below is an example of an ungapped multiple alignment. The conserved amino acids are shaded for identification.

Henikoff and Henikoff tested the performance of hierarchical multiple alignment of three serine proteases using **BLOSUM45, BLOSUM62, BLOSUM80, PAM120, PAM160, and PAM250** matrices. **All BLOSUM matrices performed better than PAM matrices**; the number of residues misaligned was three to five times lower when BLOSUM matrices were used compared to PAM matrices. **BLOSUM62 performed slightly better than BLOSUM45 and BLOSUM80**.

BLOSUM 62 matrix

	A	R	N	D	C	Q	E	G	H	I	L	K	M	F	P	S	T	W	Y	V
Ala A	4																			
Arg R	-1	5																		
Asn N	-2	0	6																	
Asp D	-2	-2	1	6																
Cys C	0	-3	-3	-3	9															
Gln Q	-1	1	0	0	-3	5														
Glu E	-1	0	0	2	-4	2	5													
Gly G	0	-2	0	-1	-3	-2	-2	6												
His H	-2	0	1	-1	-3	0	0	-2	8											
Ile I	-1	-3	-3	-3	-1	-3	-3	-4	-3	4										
Leu L	-1	-2	-3	-4	-1	-2	-3	-4	-3	2	4									
Lys K	-1	2	0	-1	-3	1	1	-2	-1	-3	-2	5								
Met M	-1	-1	-2	-3	-1	0	-2	-3	-2	1	2	-1	5							
Phe F	-2	-3	-3	-3	-2	-3	-3	-3	-1	0	0	-3	0	6						
Pro P	-1	-2	-2	-1	-3	-1	-1	-2	-2	-3	-3	-1	-2	-4	7					
Ser S	1	-1	1	0	-1	0	0	0	-1	-2	-2	0	-1	-2	-1	4				
Thr T	0	-1	0	-1	-1	-1	-1	-2	-2	-1	-1	-1	-1	-2	-1	1	5			
Trp W	-3	-3	-4	-4	-2	-2	-3	-2	-2	-3	-2	-3	-1	1	-4	-3	-2	11		
Tyr Y	-2	-2	-2	-3	-2	-1	-2	-3	2	-1	-1	-2	-1	3	-3	-2	-2	2	7	
Val V	0	-3	-3	-3	-1	-2	-2	-3	-3	3	1	-2	1	-1	-2	-2	0	-3	-1	4

Figure 6.10 BLOSUM62 substitution matrix made by writing the amino acide in alphabetical order.

The following shows the PAM_BLOSUM matrix equivalence, and their preferred use:

PAM 100 ≈ BLOSUM 90 (for less divergent proteins)
PAM 120 ≈ BLOSUM 80
PAM 160 ≈ BLOSUM 62 (for most other proteins)
PAM 200 ≈ BLOSUM 52
PAM 250 ≈ BLOSUM 45 (for more divergent proteins)

In general, BLOSUM matrices are widely used for detecting local alignments. BLOSUM62 is the most frequently used matrix for detecting the majority of weak protein similarities, and BLOSUM45 is very suitable for detecting long and weak alignments. While aligning unknown sequences, if one wants to use the most appropriate matrix based on how similar the sequences are, one has to first try multiple matrices and then use the one that gives the highest ungapped alignment score.

Statistical Significance of Sequence Alignment

6.7.3.1 P-Value

The P-value of an alignment represents the probability of obtaining a score $\geq S$ by chance. For example, if the P-value is 10^{-5}, it means that the probability of obtaining an alignment with a score $\geq S$ is 1 out of 10^5. Thus, different alignments can be compared based on their P-values. The P-value ranges from 0 to 1; the closer it is to 0, the better is the alignment.

6.7.3.2 Z-Score

In the statistical sense, Z is the distance between S and the mean of scores obtained using randomized sequences.
Z > 20: two sequences are definitely homologous (family)
Z between 10 and 20: two sequences most likely homologous (Family / Superfamily)

Z between 6 and 8: two sequences are less likely to be homologous
Z < 6: not significant.

6.7.3.3 E-Value

This is particularly relevant in relation to sequence-similarity searching using BLAST and FASTA, which are discussed later in this chapter. The **E-value** is the **expectation** value that indicates the number of alignments with a score ≥ S that one can expect to find by chance in a database of size N. Hence, the E-value is dependent on the database size N. Hence, the E-value is dependent on the database size and the query length. The closer the E-value to 0, the better is the alignment. For E < 1e – 2 (= $1 \times 10^{-2} = 0.01$), P ≈ E. *The E-value is the most widely used measure for estimating the quality of sequence alignment – that is, the extent of sequence similarity.*

The typical threshold for the E-value when judging homology, particularly using BLAST, is E ≤ 1e – 5 (= 1×10^{-5}), and the lower the value, the better it is. For BLAST (both nucleotide and protein), the default E-value is set at 10 in the **Expect threshold** box under **Algorithm parameters** (lower left corner of the BLAST home page). This means that 10 matches are expected to be found merely by chance, according to the stochastic model of Karlin and Altschul (1990).[19]

6.8.2.3 Suggested BLAST E-Value Cut-off

For nucleic-acid-based search, the suggested threshold (minimum significant hit) for the E-value is ≤ 1e-6 (= 10^{-6}), and a sequence identity of ≥ 70%. For protein-based search, the suggested threshold for the E-value ≤ le-4 (= 10^{-4}), with a sequence identity of ≥ 35%[g]. However, typically for protein-based homology search the threshold used is E ≤ 1e-5 (= 10^{-5}), and the lower it is, the better. For example, an E-value of le-25 (= 10^{-25}) will indicate a clear homology.

6.7.3.4 Bit Score

The **bit score** (S') is a normalized raw score expressed in *bits*; it is an estimate of the search space one has to search through – that is, the number of sequence pairs one has to score – before one can come across a raw alignment score ≥ S, by chance.

For example, a bit score of 30 means that, on average, one has to score 2^{30} (= 1 billion) sequence pairs before one will come across a score ≥ S, by chance. Usually, good alignments produce a bit score > 50. *It should be emphasized that the bit score is dependent on sequence length, and short sequences may not produce high bit scores despite very high identity.*

To summarize the utility of the statistical estimates of sequence alignment in simple terms, **the better the alignment (e.g. homologous sequences), the lower the P-values and E-values, and the higher the Z-values and bit scores.**

Megablast, Blastn, and Discontinuous Megablast

Megablast is optimized for highly similar sequences. It efficiently finds long alignments between highly similar (>95%) sequences, and thus is the best tool to find the identical match to the query sequence. The default **word size is 28 and the lowest word size is 16**.

Blastn is optimized for somewhat similar sequences. The reason blastn is more sensitive than megablast is because it uses a shorter default **word size (11)**

blastn is better than megablast at finding alignments to related nucleotide sequences from other organisms. Reducing the word size from 11 (default) to 7 (lowest) increases the sensitivity of search—that is, increases the number of positive hits.

Discontinuous megablast is optimized for more dissimilar or diverged sequences especially sequences from different organisms. Instead of using the exact word match as seed for an alignment extension, discontinuous megablast uses a noncontiguous word within a longer window of template

6.8.3.1 Searching for Distantly Related Proteins – PSI-BLAST

Many homologous proteins have similar three-dimensional structure, but in pairwise alignment they may not show significant sequence similarity. Therefore, regular protein BLAST (blastp) is not useful in identifying these proteins. Position-Specific Iterative BLAST (PSI-BLAST) is designed to detect weak relationships between the query sequence and other sequences in the database that are not necessarily detectable by standard BLAST searches. When a new genome is

6.8.3.2 Searching for Pattern Hit—PHI-BLAST

Many proteins contain signature sequences (motifs) that are characteristics of a protein family. These signature sequences are part of important structural or functional domains. Pattern-hit-initiate (PHI)-BLAST is designed to search the database for proteins that are significantly related to the query sequence and also contain a pattern. In other words, PHI-BLAST searches for significantly similar sequences to both a query sequence and a signature. This dual requirement is supposed to reduce the number of database hits that contain the pattern but are likely to have no true homology to the query.

6.8.5.1 Comparison of BLAST and FASTA

BLAST and FASTA are both heuristic algorithms that perform database searches to find sequences related to a query sequence. However, there are some differences between the two:

1. BLAST begins a search by looking for matches that include exact matches and conservative substitutions; FASTA begins a search by looking at exact matches.

2. BLAST scans a larger window size that FASTA; hence, FASTA may produce better coverage for homologs.

3. BLAST may produce multiple best-scoring alignments (also called **high-scoring segment pairs or HSPs**) from the same sequence; FASTA returns only one alignment from one sequence.

4. BLAST automatically masks low-complexity regions; FASTA does not employ such automatic masking. Therefore, if the query sequence has non-unique segments such as repeats, compositionally biased segments, etc., FASTA search may return alignments with artificially high scores.

5. For a given sequence search, the BLAST output is larger than that of FASTA.

6. For a given sequence search, BLAST is faster than FASTA.

Table 2.1: Major Biological Databases Available Via the World Wide Web

Database and Retrieval System	**Brief Summary of Content**	**URL**
AceDB	Genome database for *Caenorhabditis elegans*	www.acedb.org
DDBJ	Primary nucleotide sequence database in Japan	www.ddbj.nig.ac.jp
EMBL	Primary nucleotide sequence database in Europe	www.ebi.ac.uk/embl/index.html
Entrez	NCBI portal for a variety of biological databases	www.ncbi.nlm.nih.gov/gquery/gquery.fcgi
ExPASY	Proteomics database	http://us.expasy.org/
FlyBase	A database of the *Drosophila* genome	http://flybase.bio.indiana.edu/
FSSP	Protein secondary structures	www.bioinfo.biocenter.helsinki.fi:8080/dali/index.html

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GenBank	Primary nucleotide sequence database in NCBI	www.ncbi.nlm.nih.gov/Genbank
HIV databases	HIV sequence data and related immunologic information	www.hiv.lanl.gov/content/index
Microarray gene expression database	DNA microarray data and analysis tools	www.ebi.ac.uk/microarray
OMIM	Genetic information of human diseases	www.ncbi.nlm.nih.gov/entres/query.fcgi?db=OMIM
PIR	Annotated protein sequences	http://pir.georgetown.edu/pirwww/pirhome3.shtml
PubMed	Biomedical literature information	www.ncbi.nlm.nih.gov/PubMed
Ribosomal database project	Ribosomal RNA sequences and phylogenetic trees derived from the sequences	http://rdp.cme.msu.edu/html
SRS	General sequence retrieval system	http://srs6.ebi.ac.uk
SWISS-Prot	Curated protein sequence database	www.ebi.ac.uk/swissprot/access.html
TAIR	Arabidopsis information database	www.arabidopsis.org

Table 8.5: Some Online Tools for the Analysis of Possible Secondary Structure of a Protein

Online Tool	**Comments and URL**
APSSP	http://imtech.res.in/raghava/apssp/*

CFSSP (Chou-Fasman[16] Secondary-Structure Prediction)	http://www.biogem.org/tool/chou-fasman/*
GOR IV	GOR IV[17]; GOR I, the original GOR[18] (http://npsa-pbil.ibcp.fr/cgi-bin/npsa_automat.pl?page=npsa_gor4.html)*[19]
HMMSTR	HMM-based[20,21] (http://www.bioinfo.rpi.edu/bystrc/hmmstr/server.php)
JPred 3	Combines the analysis from multiple prediction algorithms, such as DSC, JNET, PHD, and PREDATOR[22] (http://www.compbio.dundee.ac.uk/www.jpred/)
NPS@ (Network Protein Sequence Analysis	This site contains links to a number of prediction tools including GOR and PHD. However, GOR and PHD are mentioned here separately as well. Pay attention to those that were developed in the late 1990s. Compare the output from these tools[*19] (http://npsa-pbil.ibcp.fr/cgi-bin/npsa_automat.pl?page=/NPSA/npsa_server.html)
PHD	Neural-network-based[23-25] (http://npsa-pbil.ibcp.fr/cgi-bin/npsa_automat.pl?page=/NPSA/npsa_phd.html)
PredictProtein	Meta-server that combines the analysis from multiple prediction algorithms such as Jpred, PHD, PROF, and PSIPRED. It is a good secondary-structure prediction program[**] (https://www.predictprotein.org/)
PROTEUS 2	Combination of HMM-and neural-network-based prediction[26] (http://wishart.biology.ualberta.ca/proteus2/)
PSIPRED	Combination of homology modeling and neural-network-based

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	prediction. It is a good secondary-structure prediction program[22] (http://bioinf.cs.ucl.ac.uk/psipred/)
Quick2D	Provides an overview of secondary-structure features like α – helices, extended β – sheets, coiled coils, transmembrane helices, and disordered regions. Predictions by PSIPRED, JNET, Prof(Rost), Prof (Ouali), Coils, MemSAT2, HMMTOP, DISOPRED2 and VSL2^{++} (http://toolkit.tuebingen.mpg.de/quick2_d)
SCRATCH Protein Predictor	The SCRATCH software suite includes predictors for a number of parameters, such as secondary structure, relative solvent accessibility, disordered regions, domains, individual residue contracts, teritiary structure and more[28] (http://scracth –proteomics.ics.uci.edu/index.html)
SSPro 4.0	Bidirectional recurrent neural network (BRNN)-based[29, 30] (http://scratch.proteomics.ics.uci.edu/sspro4.html)
SYMPRED	SYMPRED can be run using any combination of the following programs: PHD, PROF, SSPro201, YASPIN, JNet and PSIPRED. The consensus of the outputs is derived through dynamic programming to achieve a higher level of prediction accuracy[31] (http://wwwibi.vu.nl/programs/sympredwww/)
SOPMA	An improved self-optimized prediction method (SOPM)[32] (http://npsa-pbit.ibcp.fr/cgi-bin/npsa_automat.pl?page=npsa_sopma.html)
YASPIN	Neural-network-based[33] (http://www.ibi.vu.nl/programs/yaspinwww/)

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SYNTENY

A chromosomal segment that has been inherited from the common ancestor during evolution without a major rearrangement of the order of genes is called a **syntenic block** (or synteny block). Syntenic blocks contain specific non-repetitive genomic markers that are in the same order and orientation in the genomes being compared. These genomic markers could be protein-coding genes, RNA-coding genes, noncoding sequences, pseudogenes, etc., and are called **syntenic anchors** (or synteny anchors)

Charting the blocks of conserved synteny creates a **synteny map**, which shows the large-scale evolutionary relationships between genomes that are related through a common ancestor, but have diverged during evolution.

Table 7.1: Some Online Tools for Prediction of Promoters, Cis-Regulatory Elements, Transcription Start and Initiation Sites, and the ORF

Online Analysis Tool	Comments and URL
BPROM	**Bacterial promoter prediction.** A SoftBerry utility that predicts putative transcription start positions of bacterial genes regulated by sigma 70 promoters. The prediction accuracy is about 80%; the specificity is also about 80% when tested on equal numbers of promoter and non-promoter sequences. It uses the signal and content information of the sequence (e.g. consensus sequence). BPROM should be run on a region between two neighboring ORFs located on the same strand, or on a sequence upstream from an ORF (most promoters are located within 150 bp upstream of the ORF). BPROM should not be used for whole genomes, to avoid the many false positives (http://linux1.softberry.com/berry.phtml?topic=bprom&group=programs&subgroup=gfindb)
Virtual Footprint	**Prokaryotic promoter prediction.** Virtual Footprint is a software suite for analyzing transcription-factor-binding sites in whole bacterial genomes and their underlying regulatory networks. The results is a list of potential binding sites and corresponding genes defining the whole regulon. There are two types of analysis: analysis of a whole prokaryotic genome with one regulator pattern, and analysis of a promoter region with several regulator patterns[28] (http://www.prodoric.de/vip/vfp_promoter.php)

BDGP (Berkeley *Drosophila* Genome Project)	**Prokaryotic and eukaryotic promoter prediction.** Neural network promoter prediction (NNPP)-based. NNPP is method that consists mainly of two recognition features for predicting eukaryotic promoters; one for recognizing the TATA-box and one for recognizing the initiator element. Both features are combined into one output unit, which gives output scores between 0 and 1. The default score is set at 0.8. The prediction accuracy for prokaryotic promoters is greater than that for eukaryotic promoters[29] (http://www.fruitfly.org/seq_tools/promoter.html)
FindTerm	**Rho-independent-terminator prediction in the bacterial genome.** A SoftBerry utility that predicts terminators in the bacterial genome. The search utilizes certain known features of bacterial terminators, such as T-rich regions, possible combinations of spacer lengths, all hairpins etc., and the result output shows all putative terminators (http://linux1.softberry.com/berry.phtml?topic=findterm&group =programs&subgroup=gfindb)
Promoter 2.0	**Vertebrate pol II transcription start site (TSS) prediction.** The program builds on principles that are common to neural networks and genetic algoritms[30] (http://www.cbs.dtu.dk/servies/Promoter/)
Tfsitescan	**Eukaryotic promoter sequence and putative transcription-factor-binding site prediction.** Works best with sequences of ~ 500 nt. The output is in graphic display and shows expectation sores for the putative binding sites[a] (http://www.ifti.org/cgi-bin/ifti/Tfsitescan.pl)
SoftBerry Search for promoters / functional motifs	**SoftBerry utility providing a suite of prediction tools for promoter/ functional motif prediction.** For example: 1. Plant promoter prediction (**TSSP**) 2. Human pol II promoter prediction (**TSSG and TSSW**) 3. Human promoter prediction (**EPROM**) 4. Promoter prediction using orthologous sequences in eukaryotic

	genome (**PromH(G) and PromH(W)**) 5. Regulatory motif prediction (**Nsite**) (http://linux1.softberry.com/berry.phtml?topic=index&group=programs&subgroup=promoter)
WWW Signal Scan	**Eukaryotic transcriptional elements prediction** based on scoring homologies of published *cis-regulatory* transcriptional signal sequences (e.g. in TFD, TRANSFAC databases) in the input sequences[b,31] (http://www-bimas.cit.nih.gov/molbio/proscan/)
Human Core-Promoter Finder	**Transcription start site (TSS) prediction in human core-promoters.** The input genomic DNA sequence should be longer than 240 bp and less than 2001 bp. The functional core-promoter is assumed to span between – 60 and + 40 nt with respect to the TSS (+ 1). The program is able to localize a TSS to a 100-bp interval ~60% of the time[c]. (http://rulai.cshl.org/tools/genefinder/CPROMOTER/human.htm)
EP3 (Easy Promoter Prediction Program)	**Eukaryotic core promoter prediction.** Performs very well in identifying regions in human genes that are associated with transcription initiation. EP3 uses universal properties of the promoter to detect those regions in a whole-genome context[33] (downloadable) (http://bioinformatics.psb.ugent.be/webtools/ep[3])
Eponine	**Transcription start site prediction in mammalian genomic sequence.** A probabilistic method with good specificity and excellent positional accuracy. Eponine is estimate to detect > 50% of transcription start sites, with ~70% specificity[34] (downloadable from Sanger Center) (http://www.sanger.ac.uk/resources/software/eponine/)
Footprinter	**Prediction of regulatory elements in DNA sequences based on phylogenetic footprinting.** Phylogenetic footprinting method identifies regions of DNA that are highly conserved across a set of orthologous sequences[35] (downloadable from the University of Washington (Motif Discovery link) (http://bio.cs.washington.edu/software)

ORF Finder	**Open reading frame (ORF) prediction.** A very user-friendly ORF finder on the web. It is a graphical analysis tool that finds all ORFs in the input sequence, using the standard or alternative genetic codes. The putative ORFs are displayed in six frames, three sense and three antisense[d] (http://www.ncbi.nlm.nih.gov/gorf/gorf.html)
NetStart 1.0	**Translation initiation site prediction.** NetStart produces neural network predictions of translation start sites in vertebrate and *Arabidopsis thaliana* nucleotide sequences. The program has been trained on cDNA-like sequences; therefore, it shows better performance for cDNAs and ESTs. It has not been tested on genomic data[36] (http://www.cbs.dtu.dk/services/NetStart/)
ATGPr	**Translation initiation site prediction.** ATGpr can be used to predict whether an initiation codon for cases where there are multiple ATG condons. The method uses linear discriminant analysis, and has been tested on a non-redundant data set of 660 sequences[37] (http://atgpr.dbscls.jp/)

Table 7.2: Some Online Tools for RNA Secondary-Structure Prediction

Online Analysis Tool	Comments and URL
RNAfold	RNAfold predicts secondary strcutures of single-stranded RNA or DNA sequences based on the classic minimum-free-energy algorithm of Zuker and Stiegler[42] as well as the partition-function algorithm of McCaskill.[43] Current limits are 10,000 nt for minimum-free-energy-only predictions and 7500 nt for partition-function calculations. The server function can be tested using the sample sequence provided[44] (http://rna.tbi.univie.ac.at/cgi-bin/RNAfold.cgi)
RNAsoft	RNAsofts is a collection of online services for the computational prediction and design of RNA/DNA structures based on a standard free-energy model.[45] The underlying algorithms have been designed and implemented by members of the Bioinformatics, Empirical and Theoretical Algorithmics (BETA) Lab at the

	Department of Computer Science of the University of British Columbia (http://www.rnasoft.ca/)
CONTRAfold	CONTRAfold is a novel secondary-structure prediction method based on conditional log-linear models (CLLMs), a flexible class of probabilistic models with high prediction accuracy[46] (http://contra.stanford.edu/contrafold/server.html)
RNAstructure	RNAstructure uses several secondary-structure prediction algorithms, including thermodynamic and partition-function algorithms. It is a complete package for RNA and DNA secondary-structure prediction and analysis. It can take different types of experiment mapping data to constrain or restrain structure prediction[47] (http://rna.urmc.rochester.edu/RNAstructureWeb/)
IPKnot	IPKnot performs integer-programming (IP)-based prediction of RNA pseudoknots. IPknot can also predict the consensus secondary structure when a multiple alignment of RNA sequences is given[48] (http://rna.naist.jp/jpknot/)
CYLOFOLD	RNA secondary-structure (including pseudoknot) prediction tool. Some examples of RNA sequences are provided that can be used to perform a test run. The bracket notation output is in brackets and hyphens instead of brackets and dots* (http://cylofold.abcc.ncifcrf.gov/)
CentroidHomfold and CentroidFold	CentroidHomfold predicts the secondary structure of an input RNA sequence by employing automatically collected homologous sequences of the target[49,50] CentroidFold uses the CONTRAfold model as the default setting to calculate base-pairing probabilities, and predicts RNA secondary structure using a γ - centroid estimator. Currently, the

	input sequence should be less than or equal to 200 bases[51] (http://www.ncrna.org/)
PknotsRG	pknotsRG is a tool for prediction RNA secondary structures, including the class of simple recursive pseudoknots. It uses the thermodynamic energy model extended by some pseudoknot-specific values.[52] The program on the BiBiserv is limited to sequences of length up to 800 bases (http://bibiserv.techfak.uni-bielefeld.de/pknotsrg/submission.html) pknotsRG will be discontinued and replaced by pKiss in the near future (http://bibiserv2.cebitec.uni-bielefeld.de/pkiss)

Table 7.3: IUPAC Codes for Nucleotides

A = adenine	**T = thymine**	**G = guanine**	**C = cytosine**		
R = A/G	Y = C/T	S = G/C	W = A/T	K = G/T	M = A/C
B = C/G/T	D = A/G/T	H = A/C/T	V = A/C/G	N = any base	

Table 8.2: Some Online Analysis Tools for Prediction of Post-Translational Protein Modifications, Protein Sorting, Localization Signals.

Online Tool	**URL**
CBS Prediction Servers (Center for Biological Sequence Analysis, Technical University of Denmark DTU)	http://www.cbs.dtu.dk/services/*

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PSORT (Protein Sorting)	http://psort.hgc.jp/+
Gene Infinity	http://www.geneinfinity.org/sp/sp_proteinptmodifs.html++

Table 8.7: Some Online Tools for Transmembrane – Helix Prediction

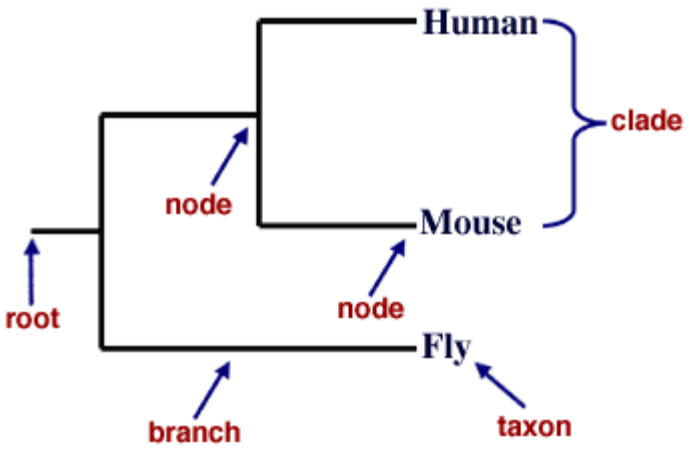
Online Tool	**Comments and URL**
TMHMM	Hidden-Markov-model-based[46] (http://www.cbs.dtu.dk/services/TMHMM/)
RHYTHM	Utilizes the structural information from ever-growing data sets and evolutionary information from conserved-sequence patterns in a representative data set of membrane proteins[47] (http://proteinformatics.charite.de/rhythm/)
Phobius	Hidden-Markov-model-based[48] (http://phobius.sbc.su.se/)
OCTOPUS	Artificial-neural-network-based[49] (http://octopus.cbr.su.se/)

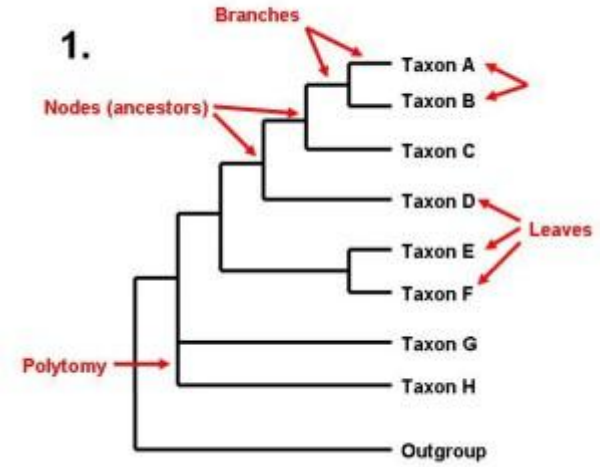
Phylogeny refers to the evolutionary history of species. **Phylogenetics** is the study of phylogenies—that is, the study of the evolutionary relationships of species. **Phylogenetic analysis** is the means of estimating the evolutionary relationships.

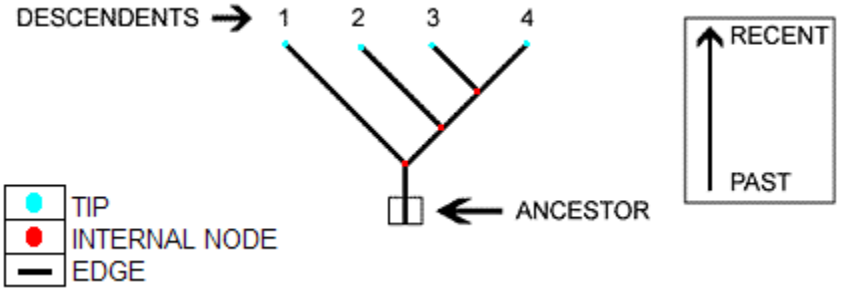
The evolutionary relationship obtained from phylogenetic analysis is usually depicted as branching, treelike diagram—the **phylogenetic tree**.

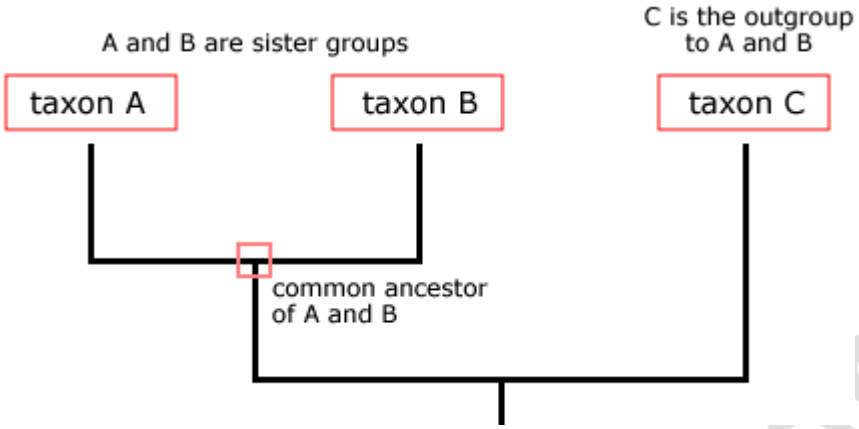
A phylogenetic tree or evolutionary tree is a diagrammatic representation of the evolutionary relationships among various taxa (Figure 9.1 A-D). It is a branching diagram composed of **nodes and branches**. The branching pattern of a tree is called the topology of the tree.

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Human
clade
node
Mouse
root
node
Fly
branch
taxon


1.
Branches
Taxon A
Taxon B
Nodes (ancestors)
Taxon C
Taxon D
Leaves
Taxon E
Taxon F
Taxon G
Polytomy
Taxon H
Outgroup


DESCENDENTS
1
2
3
4
RECENT
PAST
ANCESTOR
TIP
INTERNAL NODE
EDGE



The **nodes represent taxonomic units, such as species (or higher taxa), populations, genes, or proteins**.
A **branch is called an edge**, and **represents the time estimate of the evolutionary relationships among the taxonomic units**.

One branch can connect only two nodes.

In a phylogenetic tree, **the terminal nodes represent the operational taxonomic units (OTUs) or leaves (Fig B) . The OTUs are the actual objects**—such as the species, populations, or gene or protein sequences—being compared, **whereas the internal nodes represent hypothetical taxonomic units (HTUs)**. **An HTU is an inferred unit and it represents the last common ancestor (LCA) to the nodes arising from this point**.

Descendants (taxa) that split from the same node form sister groups, and a taxon that falls outside the cladea is called an outgroup.

For example, in Figure 9.1 B, **T2 and T3 are sister groups**, and **T1 is an outgroup to T2 and T3**.

Phylogenetic trees can be scaled or unscaled. In a scaled tree, the branch length is proportional to the amount of evolutionary divergence (e.g. the number of nucleotide substitutions) that has occurred along that branch. In an unscaled tree, the branch length is not proportional to the amount of evolutionary divergence, but usually the actual number is indicated somewhere on the branch.

Phylogenetic trees can be rooted (Figure 9.1 A and B) or unrooted (Figure 9.1 C). **A rooted tree has a node (the root) from which the rest of the tree diverges. This root is frequently referred to as the last universal common ancestor (LUCA) (FIG B),** from which the other taxonomic groups have descended and diverged over time. **In molecular phylogenetics, the LUCA and LCA are represented by DNA or protein sequences. Obtaining a rooted tree is ideal,** but **most phylogenetic-tree reconstruction algorithms produce unrooted trees**

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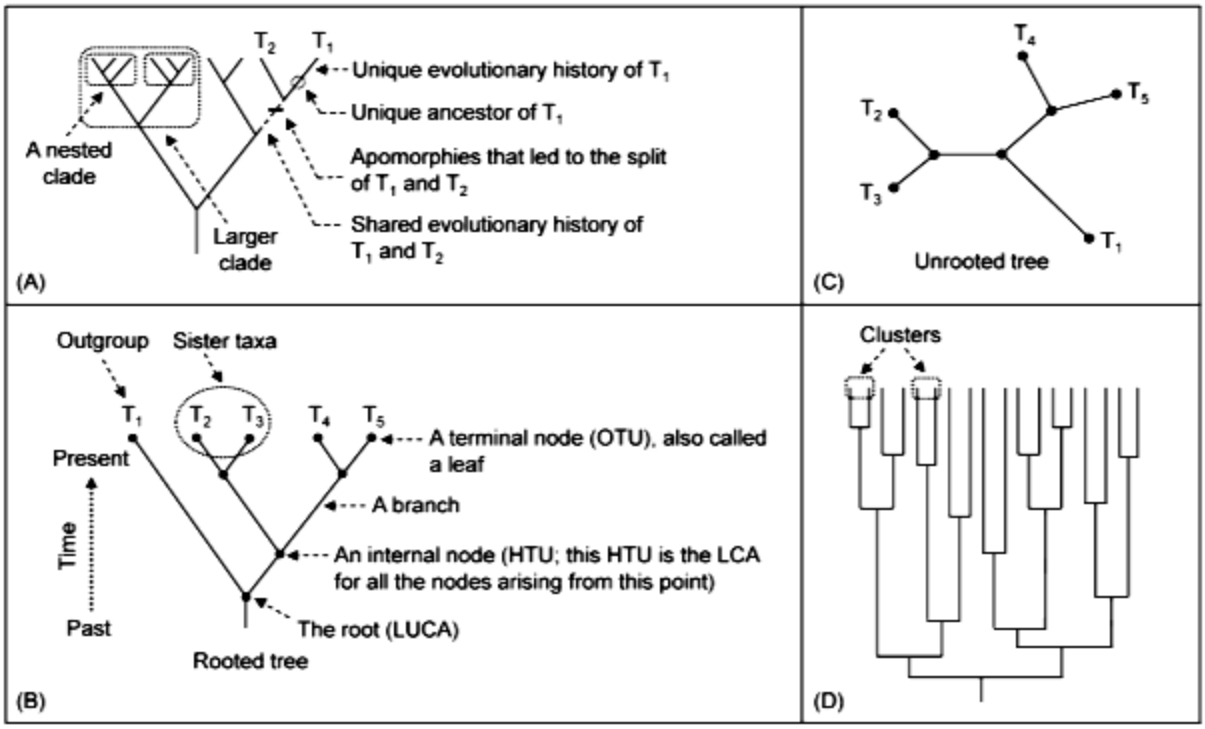


Figure 9.1: Different forms of presentation of the phylogenetic tree. The phylogenetic tree in D is a dendrogram derived from hierarchical clustering (see text). A, B, and D show rooted trees, while C shows an unrooted tree. taxa that share specific derived characters are grouped into clades. (A) Smaller clades located within a larger clade are called nested clades. (B) The terminal nodes represent the operational taxonomic units, also called "leaves"; each terminal node could be a taxon (species or higher taxa), or a gene or protein sequence. The internal nodes represent hypothetical taxonomic units. An HTU represents the last common ancestor to the nodes arising from this point. Two descendants that split from the same node are called sister groups and a taxon that falls outside the clade is called an outgroup. Rooted trees have a node from which the rest of the tree diverges, frequently called the last universal common ancestor (LUCA).

[a]: Taxa that share specific derived characters are grouped more closely together than those who do not. The groups are called clades; each clade consists of an ancestor and all of its descendants.

Number of Rooted and Unrooted Trees

Number of unrooted trees for n taxa $N_u = (2n - 5)*(2n - 7)* \ldots *3*1 = (2n - 5)!/[2^{n-3}*(n - 3)!]$

Number of rooted trees for n tax $N_r = (2n - 3)*(2n - 5)*(2n - 7)* \ldots *3*1 = (2n - 3)!/[2^{n-2}*(n - 2)!]$

Note that the number of unrooted trees for n sequences is equal for the number of rooted trees for (n – 1) sequences.

Number of Taxa	Number of unrooted trees	Number of rooted trees
3	1	3
4	3	15
5	15	105
6	105	945
7	945	10395
8	10395	135135

Phylogenetics also provides the basis for comparative genomics, which is a more recent term that came into existence in the age of genomics.

In mathematics, a factorial is a function applied to natural numbers greater than zero. The symbol for the factorial function is an exclamation mark after a number, like this: 2! The usual definition of "n factorial" is "n! = n * (n – 1) * (n – 2) *…* 2 * 1," where 'n' is a positive integer.

Molecular Dynamics and Simulations

Molecualr Dynamics is a computational method which describes equilibrium and dynamics properties of a biological system. Molecular dynamics (MD) simulations represent the computer approach to statistical mechanics. As a counterpart to experiment, MD simulations are used to estimate equilibrium and dynamic properties of complex systems that cannot be calculated analytically. Representing the exciting interface between theory and experiment, MD simulations occupy a venerable position at the crossroads of mathematics, biology, chemistry, physics, and computer science. The static view of a biomolecule, as obtained from X-ray crystallography for example - while extremely valuable - is still insufficient for understanding a wide range of biological activity. It only provides an average, frozen view of a complex system. Certainly, molecules are live entities, with their constituent atoms continuously interacting among themselves and with their environment. Their dynarnic motions can explain the wide range of thermally-accessible states of a system and thereby connect structure and function. By following the dynarnics of a molecular system in space and time, we can obtain a rich amount of information concerning structural and dynarnic properties.

The molecular dynamics method was first introduced by Alder and Wainwright in the late 1950's (Alder and Wainwright, 1957,1959) to study the interactions of hard spheres. Many important insights concerning the behavior of simple liquids emerged from their studies.

The next major advance was in 1964, when Rahman carried out the first simulation using a realistic potential for liquid argon (Rahman, 1964). The first molecular dynamics simulation of a realistic system was done by Rahman and Stillinger in their simulation of liquid water in 1974 (Stillinger and Rahman, 1974). The first protein simulations appeared in 1977 with the simulation of the bovine pancreatic trypsin inhibitor (BPTI) (McCammon, et al, 1977).

Today in the literature, one routinely finds molecular dynamics simulations of solvated proteins, protein-DNA complexes as well as lipid systems addressing a variety of issues including the thermodynamics of ligand binding and the folding of small proteins. The number of simulation techniques has greatly expanded; there exist now many specialized techniques for particular problems, including mixed quantum mechanical classical simulations, that are being employed to study enzymatic reactions in the context of the full protein. Molecular dynamics simulation techniques are widely used in experimental procedures such as Xray crystallography and NMR structure determination.

Furthermore their use is no longer reserved to experts and many experimentalists use computer simulations nowadays as a tool to analyze or interpret their measurements whenever they are too complex to be described by simple analytical models. This is particularly true in the case of neutron scattering, as in this case the experimental observables correlate directly with the properties obtained in MD simulations and the spatial and time scales that can be measured match very well those amenable to computation. Such features, together with the availability of user-friendly and reliable software to perform this kind of calculation (e.g. Charmm , NAMD , Amber , Gromacs , Gromos , DL_POLY, etc.) and to visualize and analyse their output (VMD, gOpenMol , nMoldyn, . . .), enable that today MD simulations are routinely used by many neutron users.

Even if the available software makes quite easy for a novice user to perform a MD simulation without needing lengthy training, it is advisable to acquire a minimum knowledge on the principles of the technique, as well as on the meaning of the terms and choices that one may encounter when using any MD program. The goal of the present chapter is precisely to give to the reader such basic knowledge. But if we want to study a particular system by means of a MD simulation, the requirement of a good program to perform the computation is a necessary but not sufficient condition. In addition we need a good model to represent the interatomic forces acting between the atoms composing the system. Ideally this can be done from first principles, solving the electronic structure for a particular configuration of the nuclei, and then calculating the resulting forces on each atom. Since the pioneering work of Car and Parrinello the development of *ab initio* MD (AIMD) simulations has grown steadily and today the use of the density functional theory (DFT) allows to treat systems of a reasonable size (several hundreds of atoms) and to achieve time scales of the order of hundreds of ps, so this is the preferred solution to deal with many problems of interest. However quite often the spatial and/or time-scales needed are prohibitively expensive for such ab initio methods. In such cases we are obliged to use a higher level of approximation and make recourse to empirical force field (FF) based methods. They allow to simulate systems containing hundreds of thousands of atoms during times of several nanoseconds or even microseconds. On the other hand the quality of a force field needs to be assessed experimentally. And here again neutrons play a particularly important role because

while many different experimental results can be used to validate some of the FF parameters, the complementarity mentioned above makes neutron scattering a particularly useful tool in this validation procedure.

Statistical mechanics is the branch of physical sciences that studies macroscopic systems from a molecular point of view. The goal is to understand and to predict macroscopic phenomena from the properties of individual molecules making up the system. The system could range from a collection of solvent molecules to a solvated protein-DNA complex. In order to connect the macroscopic system to the microscopic system, time independent statistical averages are often introduced. We start this discussion by introducing a few definitions.

Definitions

The thermodynamic state of a system is usually defined by a small set of parameters, for example, the temperature, T, the pressure, P, and the number of particles, N. Other thermodynamic properties may be derived from the equations of state and other fundamental thermodynamic equations.

The **mechanical or microscopic state** of a system is defined by the atomic positions, q, and momenta, p; these can also be considered as coordinates in a multidimensional space called phase space. For a system of N particles, this space has 6N dimensions. A single point in phase space, denoted by Γ, describes the state of the system. An ensemble is a collection of points in phase space satisfying the conditions of a particular thermodynamic state. A molecular dynamics simulations generates a sequence of points in phase space as a function of time; these points belong to the same ensemble, and they correspond to the different conformations of the system and their respective momenta. Several different ensembles are described below.

An **ensemble** is a collection of all possible systems which have different microscopic states but have an identical macroscopic or thermodynamic state. There exist different ensembles with different characteristics.

Microcanonical ensemble (NVE) : The thermodynamic state characterized by a fixed number of atoms, N, a fixed volume, V, and a fixed energy, E. This corresponds to an isolated system.
Canonical Ensemble (NVT): This is a collection of all systems whose thermodynamic state is characterized by a fixed number of atoms, N, a fixed volume, V, and a fixed temperature, T.

IsobaricIsothermal Ensemble (NPT): This ensemble is characterized by a fixed number of atoms, N, a fixed pressure, P, and a fixed temperature, T. Grand canonical Ensemble (μVT): The thermodynamic state for this ensemble is characterized by a fixed chemical potential, μ, a fixed volume, V, and a fixed temperature, T.

FORCE FIELDS

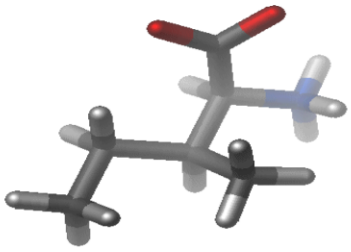
A **force field** is a mathematical expression describing the dependence of the energy of a system on the coordinates of its particles. It consists of an analytical form of the interatomic potential energy, and a set of parameters entering into this form. The parameters are typically obtained either from ab initio or semi-empirical quantum mechanical calculations or by fitting to experimental data such as neutron, X-ray and electron diffraction, NMR, infrared, Raman and neutron spectroscopy, etc. Molecules are simply defined as a set of atoms that is held together by simple elastic (harmonic) forces and the FF replaces the true potential with a simplified model valid in the region being simulated. Ideally it must be simple enough to be evaluated quickly, but sufficiently detailed to reproduce the properties of interest of the system studied. There are many force fields available in the literature, having different degrees of complexity, and oriented to treat different kinds of systems.

The Effective molecular energy can be expressed as a sum of potentials derived simple physical forces

$$V = E_{bonded} + E_{nonbonded}$$

$$E_{bonded} = E_{bond-strech} + E_{angle-bend} + E_{rotate-along-bond}$$

$$E_{nonbonded} = E_{van-der-waals} + E_{electrostatic}$$



However a typical expression for a FF may look like this:

$$U = \sum_{bonds} \frac{1}{2} k_b \left(r - r_O\right) + \sum_{angles} \frac{1}{2} k_a \left(\theta - \theta_O\right)^2 + \sum_{torsions} \frac{V_n}{2} \left[1 + \cos\left(n\phi - \delta\right)\right]$$

$$+ \sum_{improper} V_{imp} + \sum_{LJ} 4 \in_{ij} \left(\frac{\sigma_{ij}^{12}}{r_{ij}^{12}} - \frac{\sigma_{ij}^{6}}{r_{ij}^{6}} \right) + \sum_{elec} \frac{q_i q_j}{r_{ij}}$$

where the first four terms refer to intramolecular or local contributions to the total energy (bond stretching, angle bending, and dihedral and improper torsions), and the last two terms serve to describe the repulsive and Van der Waals interactions (in this case by means of a 12-6 Lennard-Jones potential) and the Coulombic interactions.

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Molecular Dynamics

An MD simulation is a technique to produce a dynamical trajectory for a system composed of N particles by integrating Newton's equations of motion. We need a set of initial conditions (positions and velocities of each particle), a good model to represent the forces acting between the particles (either from electronic structure calculations or using the empirical force fields presented in the previous section), and to define the boundary conditions to be employed. Then we need to solve the classical equation of motion:

$$m_i \frac{d^2 r_i}{dt^2} = f_i = -\frac{\partial}{\partial r_i} U\left(r_1, r_2, ..., r_N\right)$$

where U(r1, r2, . . . , rN) is the potential energy depending on the coordinates of the N particles. This is a system of N coupled second order non linear differential equations that cannot be solved exactly, so equation above has to be solved numerically step by step using an appropriate integration algorithm.

SIMULATION

Molecular dynamics (MD) simulation, first developed in the late 70s, has advanced from simulating several hundreds of atoms to systems with biological relevance, including entire proteins in solution with explicit solvent representations, membrane embedded proteins, or large macromolecular complexes like nucleosomes or ribosomes. Simulation of systems having <50,000–100,000 atoms are now routine, and simulations of approximately 500,000 atoms are common when the appropriate computer facilities are available. This remarkable improvement is in large part a consequence of the use of high performance computing (HPC), and the simplicity of the basic MD algorithm

We can consider that there are four different steps in a MD simulation:

1. **Setting up the system**: This stage includes most of the aspects mentioned in the preceding sections. We have to create a reasonable configuration of the system we are interested in, choose a valid force field for this system, select a cut-off radius and the method to treat the electrostatic interactions if the system contains partial charges, choose an integration algorithm (if the MD code we use allows it) and a correct time step, and select the working ensemble (NVE, NVT or NPT). At this point we have to be sure that the size of the system is enough to explore the physics of interest.

2. **Equilibration**: Normally the initial configuration will not be representative of the conditions we want to explore. For example, if it comes from a previous MD simulation we may want to simulate the system at a different temperature. If the initial positions were determined randomly we may have some atoms that are too close or a lack of short-range correlations which is not real. On the other hand, if we start from an ordered configuration and we are interested in the liquid, we need to melt the system and guarantee that the long-range order has completely disappeared. This step usually requires the use of a NPT simulation in order to allow the system to achieve the equilibrium density corresponding to the desired pressure and temperature. However the choice of the thermostat and barostat is not critical, as this part of the simulation

will not be used to compute any property. It is important to ensure that we have achieved an equilibrated state, so during this phase we should follow the pressure, the density and the different energy components, which at equilibrium should fluctuate around some average value without showing any drift. If we have started from a lattice configuration and we are melting it, we can also check the evolution of a translational and/or rotational order parameter. Finally we can verify that some standard properties such as the radial distribution function or the mean square displacement are independent of the segment of the trajectory employed to compute them, but we have to be aware that this will be the case only when the segment is long enough so that the ergodic hypothesis holds.

3. **Production**: Once we have attained an equilibrium situation at the desired temperature and pressure, we can perform the production run. It may be convenient to work in the NVE ensemble to avoid any influence from the thermostat or the barostat, as we saw before. Nevertheless in very long simulations the drift in temperature originated mainly from truncation effects can be embarrassing, so we may need to apply any of the thermostats described above. The pressure should oscillate around the average value achieved during the equilibration, so normally there is no need to use any barostat. Thus we can work with a fixed cell and avoid the inconvenience of dealing with volume fluctuations during the analysis of the trajectory. A clear exception here is in the case where we are interested in simulating a phase transition. We also have to decide how long to run the simulation. We need to run it for a time much longer than the relaxation time of the property of interest. But the simulation time should also be sufficiently long to ensure the validity of the ergodic hypothesis, i.e. that the time average of a given property coincides with the ensemble average. While this condition is true in most cases (such as when simulating a standard liquid or crystal), some systems such as glasses are not ergodic, so the results will depend on the history of the system. Similarly, if we study a large biomolecule that can adopt different conformations, the simulation time should be enough to allow the macromolecule to explore all the possible configurations. Otherwise the results obtained (e.g. atomic displacements) may correspond to one of the global conformations of the macromolecule, but not to the real situation.

4. **Analysis**: Finally the simulated trajectory must be analysed to extract the desired properties. We have access to the atomic positions, velocities and even forces as a function of time, so any statistical mechanical property that can be expressed in terms of those variables can be computed. In the following section we present some examples of the different properties that can be computed in a MD simulation, while a more detailed account can be found in this collection.

Introduction to Molecular dynamics of Proteins and Nucleic acids

The study of the macromolecular structure is a key point in the understanding of biology. Biological function is based on molecular interactions, and these are a consequence of macromolecular structures. Since initial structure determinations in the 50s, both in the protein and in the nucleic acid worlds, the increase in the knowledge of how macromolecular structures are built has been continuous. At present, protein data bank (PDB)1 holds more than 110,000 entries, including more than 100,000 proteins, 2,800 nucleic acids, alone or forming complexes, and approximately 20,000 small molecules complexed to macromolecules. Molecular

recognition rules as defined by such structural knowledge powers the understanding of basic biological phenomena, like enzyme mechanisms and regulation, transport across membranes, the building of large structures like ribosomes, or viral capsids, or how DNA is read and transcription is controlled. The study and prediction of protein–protein interaction networks is one of the growing fields in modern systems biology. On a more practical note, protein three-dimensional (3D) structures are the basis for structure-based drug design. The simple visual analysis of 3D structures of protein or nucleic acids, as obtained from the experiments, has driven large number of successful studies in biochemistry. However, despite their enormous utility, structures stored at the PDB provide only a partial view of 3D structure. Both protein and nucleic acids are flexible entities, and dynamics can play a key role in their functionality. Proteins undergo significant conformational changes while performing their function. As a rule, any complex made by any protein implies some structural rearrangement. This can be easily checked just by comparing a series of PDB entries that just differ in a small ligand bound to a given protein.

Binding to ligands (allosteric regulators), or simply protein concentration, or crowding, may switch stabilities among conformations and trigger the shape transition. Additionally, some features of protein function can be understood only when dynamic properties are taken into account. For instance, diffusion of small substrates through heme-dependent enzyme molecules requires the transient appearance of channels in the protein structure. Also, cavities have transient phenomena that in some cases can only be revealed or analyzed following its dynamics.

In the case of nucleic acids, conformational changes are even more complex. Standard B-DNA has a relatively simple structure in comparison with protein or complex RNAs; however, it is an extremely plastic molecule that undergoes large conformational changes to adapt to its interaction partners. Binding of transcription factors to DNA, for example, is not only dependent on DNA sequence recognition, but also a direct consequence of the ability of the DNA molecule to adapt to the protein surface.

Career Avenues

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